UNIVERSITY OF PISA

DEPARTMENT OF INFORMATION ENGINEERING

Large-Scale and Multi-Structured Databases

WineAdvisor

Work group:

Valentina Bertei

Marta Bertelà

Francesco Tarchi

ACADEMIC YEAR 2024/2025

Index

Chapter 1

Application Manual

This guide explores the features and vision behind WineAdvisor, a web platform developed during the Large-Scale and Multi-Structured Databases course. WineAdvisor is designed for wine lovers, offering a platform to discover, review and share wines from all over the world. Users can search for wines based on various criteria, such as region, grape variety and price. The application also features a personalized recommendation system and social features such as following other users and exploring their favorites. WineAdvisor aims at enhancing the wine experience through detailed statistics, tailored suggestions and wine rankings.

The project code is available at the following link.
`https://github.com/francetarchi/LargeProject`

The application APIs are available through Postman at the following link.
`https://www.postman.com/wineadvisor-9550/wineadvisor/overview`

1.1 Application Highlights

WineAdvisor integrates a range of features designed to provide users with a rich, personalized, and socially engaging wine experience. Below we list the key functionalities that characterize the application:

- **Advanced Search Capabilities:** Users can search for wines based on a variety of criteria, including price range, country and region of origin, grape variety and wine type. This allows for highly targeted and meaningful exploration.
- **User Reviews and Favorites:** Each user has the ability to rate and review wines, as well as curate a personal list of favorites, which can be accessed and updated at any time.
- **Personalized Recommendations:** The application offers wine suggestions based on user preferences, previous reviews, and newly released labels, helping

users discover products aligned with their tastes.

- **Wine Statistics and Trends:** WineAdvisor presents insights such as review trends over specific time periods, the most appreciated wines in a given area, and top-rated wines in terms of quality-to-price ratio.
- **Social Interaction:** Users can follow others and be followed in return, gaining access to their reviews and favorite selections. This fosters a community-driven approach to wine discovery.
- **Business Accounts:** Wine producers can register as companies, enabling them to manage their wine catalogs directly by uploading, editing, or removing products.

1.2 User Manual

When accessing the site, users are presented with options to log in or register.
To fully access and utilize the features of the application, users are required to create an account and log in. Without registering, users will not be able to interact with the platform.

Figure 1.1: Log-in/Register page mock-up

The application supports three main types of users: **Basic Users**, **Winery Users**, and **Administrators**. **Basic Users** represent regular customers who can search for wines, write reviews, interact with other users, and perform additional actions described in detail in the dedicated user manual chapter. **Winery Users** correspond to wine businesses and have access to a different set of functionalities, such as managing their own wine catalog by adding, editing, or removing products. Some features available to Basic Users are not accessible to Winery Users, and vice

versa, reflecting the distinct roles they hold within the platform. Finally, **Administrators** (**Admins** from now on) have full access to almost all functionalities and are responsible for maintaining the integrity and performance of the platform.

1.2.1 Basic User

After registering and logging in, Basic Users gain access to the core functionalities of WineAdvisor. They are welcomed with a personalized homepage, featuring a dynamic feed populated by posts and reviews from the users they follow.

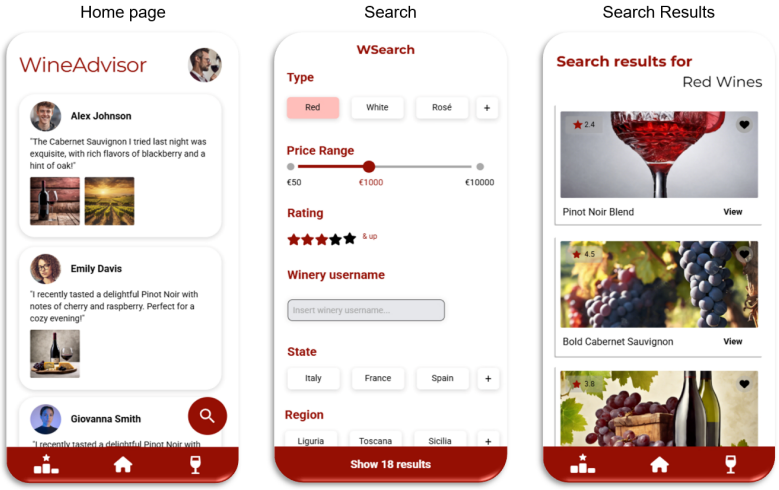


Figure 1.2: Basic user main pages mock-ups

Basic Users can:

- Search for wines using various filters;
- Explore wine suggestions based on their preferences, review history, and new releases;
- Write and post reviews about the wines they have tasted;
- Add wines to their list of favorites for easy access;
- Like or dislike reviews posted by other users;
- Gift (suggest) wines to other users;
- Search for other users or wineries profiles and view their public activities;
- Follow or unfollow users and wineries to personalize their feed and discover new wines.

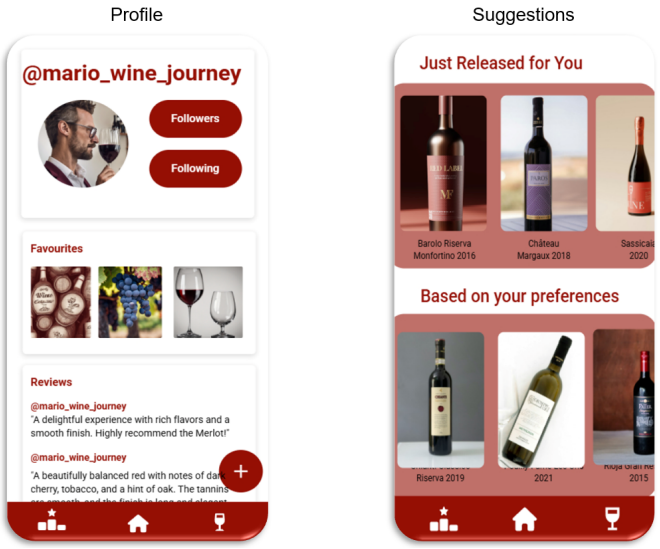


Figure 1.3: Basic user main pages mock-ups

1.2.2 Winery User

Winery Users represent wine-producing businesses that manage their own digital presence on the platform. After creating a winery account and logging in, these users gain access to a dedicated interface that enables them to curate and update information about their products and brand.

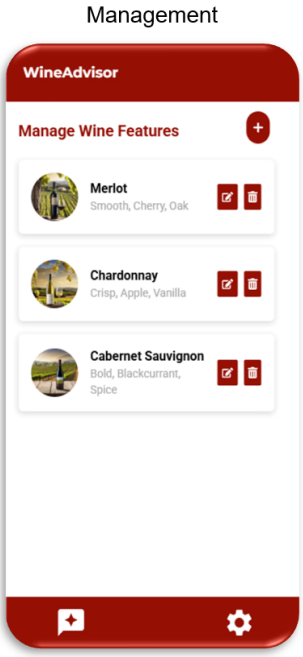


Figure 1.4: Winery user main page mock-up

Winery Users can:

- Upload and manage their wine catalog, including the ability to add, edit, or remove wines;
- Specify different vintages for each wine and update related information;
- Create a gallery of images showcasing their winery, to enhance visibility and brand identity;
- View reviews and feedback posted by basic users about their wines.

Unlike Basic Users, Winery Users do not have access to social features such as writing reviews or following other users. Their role is focused on product and brand management within the WineAdvisor system.

1.2.3 Administrators

Administrators hold the highest level of privileges within the application. They are entrusted with complete control over the database and its resources. They can create, delete and edit almost any data. This level of access allows them to oversee the correct functioning of the system, perform maintenance operations, and ensure that the platform remains secure, up-to-date, and aligned with its intended goals.

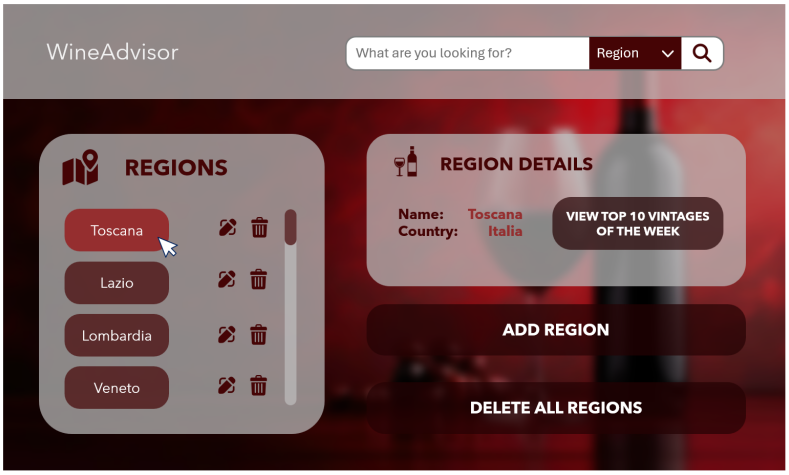


Figure 1.5: Admin dashboard mock-up

Chapter 2

Design Overview

In this chapter, we take a closer look at how the system is built, focusing on its main components and how they work together. Understanding this structure helps to make sense of how the application works and how it can grow or be improved in the future.

2.1 Main Actors

The WineAdvisor system involves four main types of actors, each playing a distinct role within the platform.

- **Unlogged Users**: users who have not yet authenticated within the application. When the application is opened, users are considered unlogged by default. In this state, their access is limited just to sign up/log in themselves, avoiding their access to any type of content in the system.
- **Basic Users**: "regular" users who interact with the application by searching for wines, posting reviews, exploring suggestions, and engaging with other users through a social feed. They represent the core audience of the platform.
- **Winery Users**: users who upload, manage, and promote their own wine catalog (they are the wine producers). They can add new wines and vintages, modify product details, and showcase their winery through images and descriptions. Unlike *Basic Users*, they cannot write reviews or engage in social features as regular users do.
- **Administrators** (**Admins**): users who have full control over the system. They can access and manage almost all resources available. Their role ensures the proper functioning and maintenance of the entire platform. As *Winery Users*, they cannot write reviews or engage in social features as regular users do.

2.2 Requirements

2.2.1 Functional Requirements

This section outlines the main functionalities that each type of user can access within the WineAdvisor platform. These requirements are essential to ensure the system supports the intended user experience and behaviors.

Unlogged Users

Unlogged users should be able to:

- Register and create an account to become a *Basic* or *Winery User*.
- Log in with existing credentials in order to access the platform as a *Basic User* or *Winery User* or *Admin*.

Basic Users

After registering and logging in, *Basic Users* should be able to:

- Search for wines by name, region, grape variety, price, and much more.
- Add wines to their list of favorites.
- Explore wine recommendations based on their tastes and past activity.
- Review wines and assign a rating and optional description.
- Like or dislike reviews from other users.
- Follow other users and view their profiles and reviews.
- Access a personalized homepage/feed with updates from followed users.

Winery Users

After registering and logging in, *Winery Users* should be able to:

- Manage their company profile and update winery information.
- Upload new wines and vintages, including all relevant details.
- Edit or remove wines they own from the catalog.
- Add an image gallery of their winery.
- View reviews and statistics related to their wines.
- Search for other wines or users for reference or comparison.

Note: *Winery Users* cannot post reviews or interact socially like *Basic Users*.

Admins

After registering and logging in, *Admins* should be able to:

- Monitor, edit, and remove almost any content within the system.
- Manage platform statistics.
- Maintain data integrity and oversee the general health of the system.

2.2.2 Non-functional Requirements

These requirements ensure the system quality in terms of performance, reliability, maintainability, and security. While not directly describing user-facing features, they are fundamental for the stability and long-term success of the platform.

- **Web-Based Architecture:** The application is implemented as a web service, providing RESTful APIs to enable future integration with graphical interfaces.
- **Fault Tolerance:** The architecture minimizes the issue of the single point of failure in order to enhance the system's robustness. Failures in non-critical components should not compromise core functionalities.
- **High Availability:** The system is designed to be accessible with minimal downtime, even if data presented is slightly outdated.
- **Password Security:** User passwords are securely stored using the *BCrypt* hashing algorithm, which incorporates salting and an adaptive workload factor to protect against brute-force attacks and ensure strong encryption over time.
- **Modular and Object-Oriented Design:** Code is developed using an object-oriented paradigm, promoting separation of concerns, code re-usability, and ease of maintenance.
- **Future-Proofing:** The system is designed with extensibility in mind, allowing the addition of new roles, features, and data sources without requiring big changes.

2.3 Use Case Diagram

This section presents the *Use Case Diagram* that models the interactions between the main actors of the WineAdvisor system and the functionalities they can access. The diagram helps to visualize the system from a user-centric perspective, highlighting the roles of each actor and their respective operations within the platform.

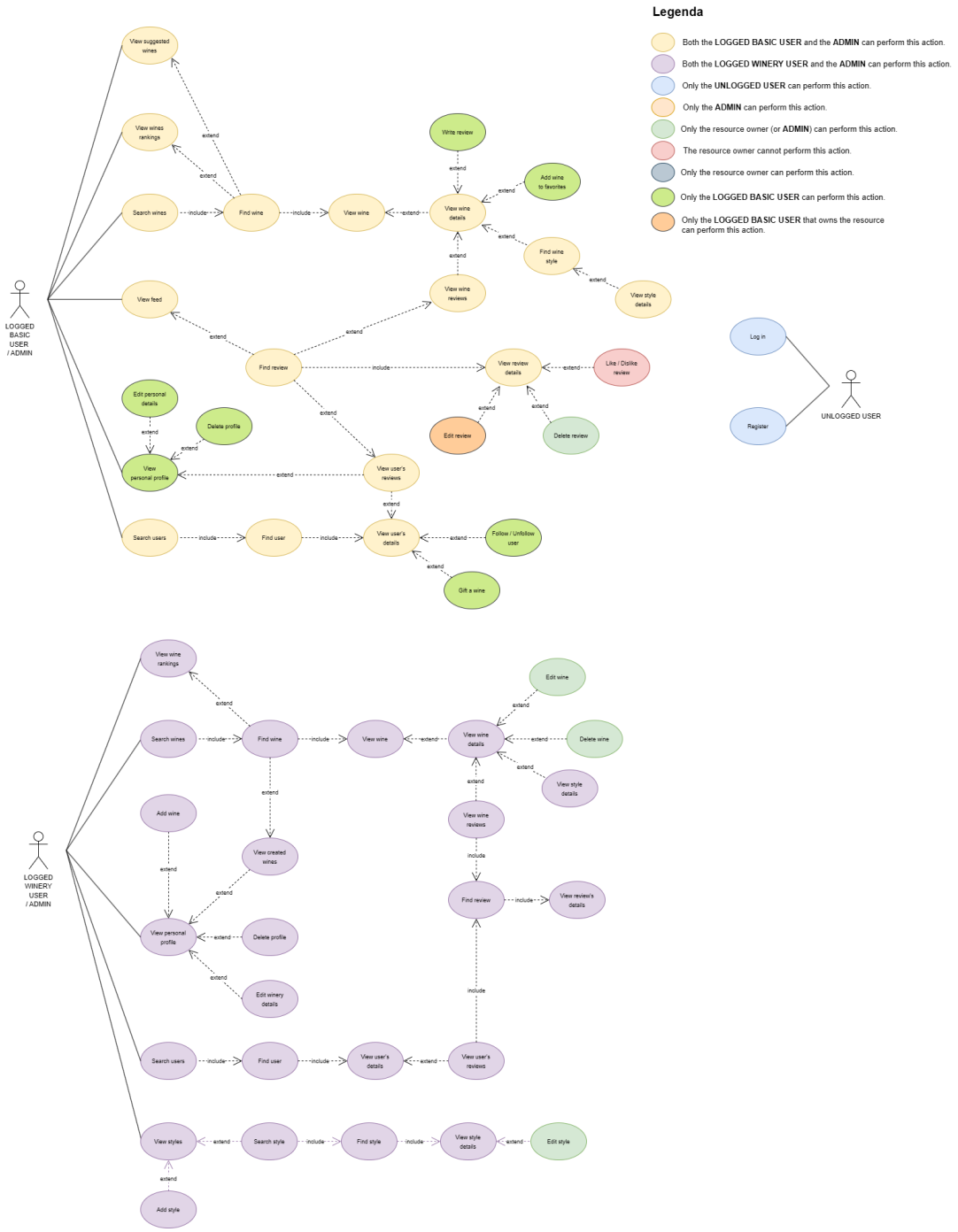


Figure 2.1: WineAdvisor use case diagram (part 1)

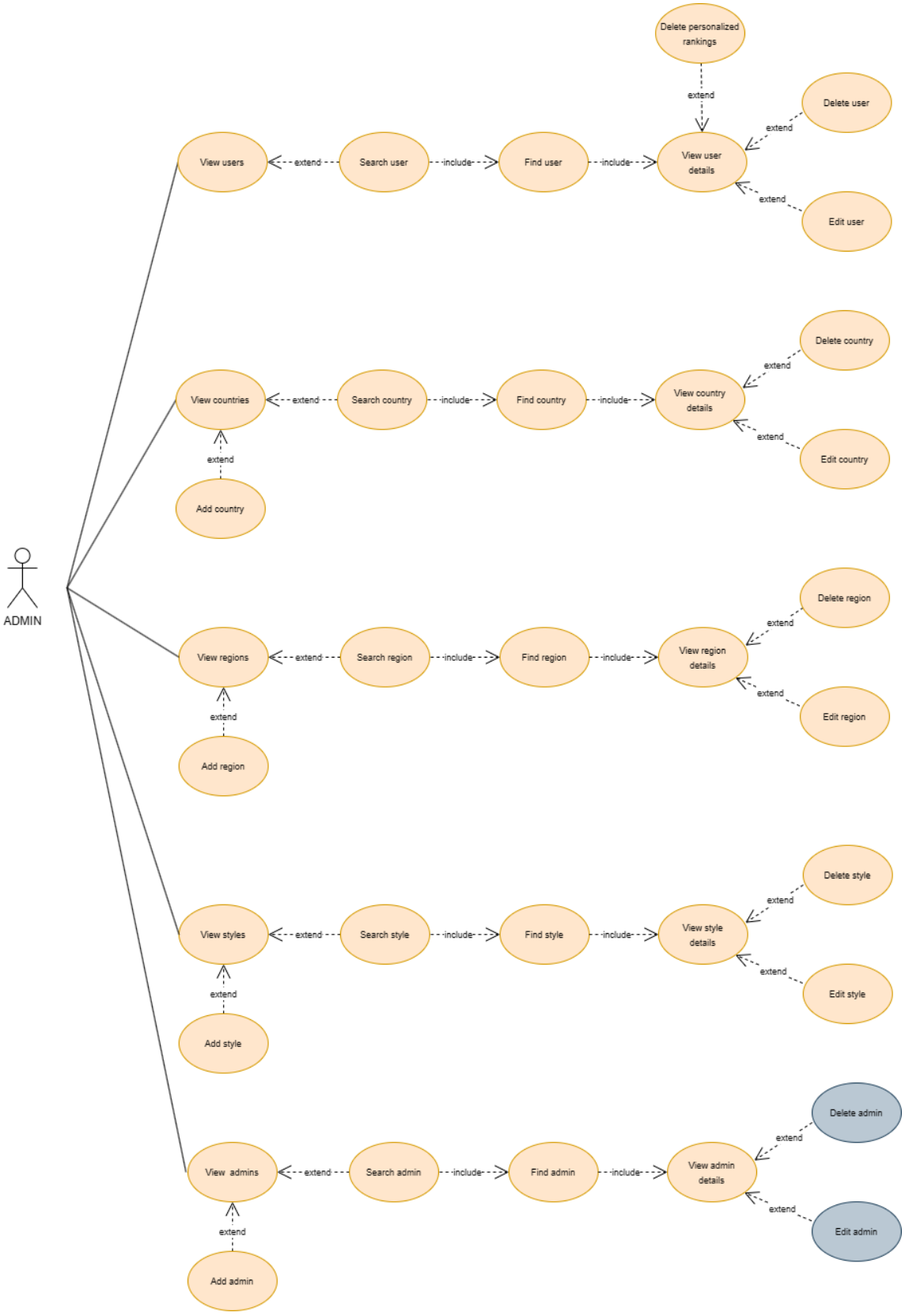


Figure 2.2: WineAdvisor use case diagram (part 2)

2.4 UML Class Diagram

This section presents the *Class Diagram* that models the main entities of our system and their relative properties (data that characterizes each entity).

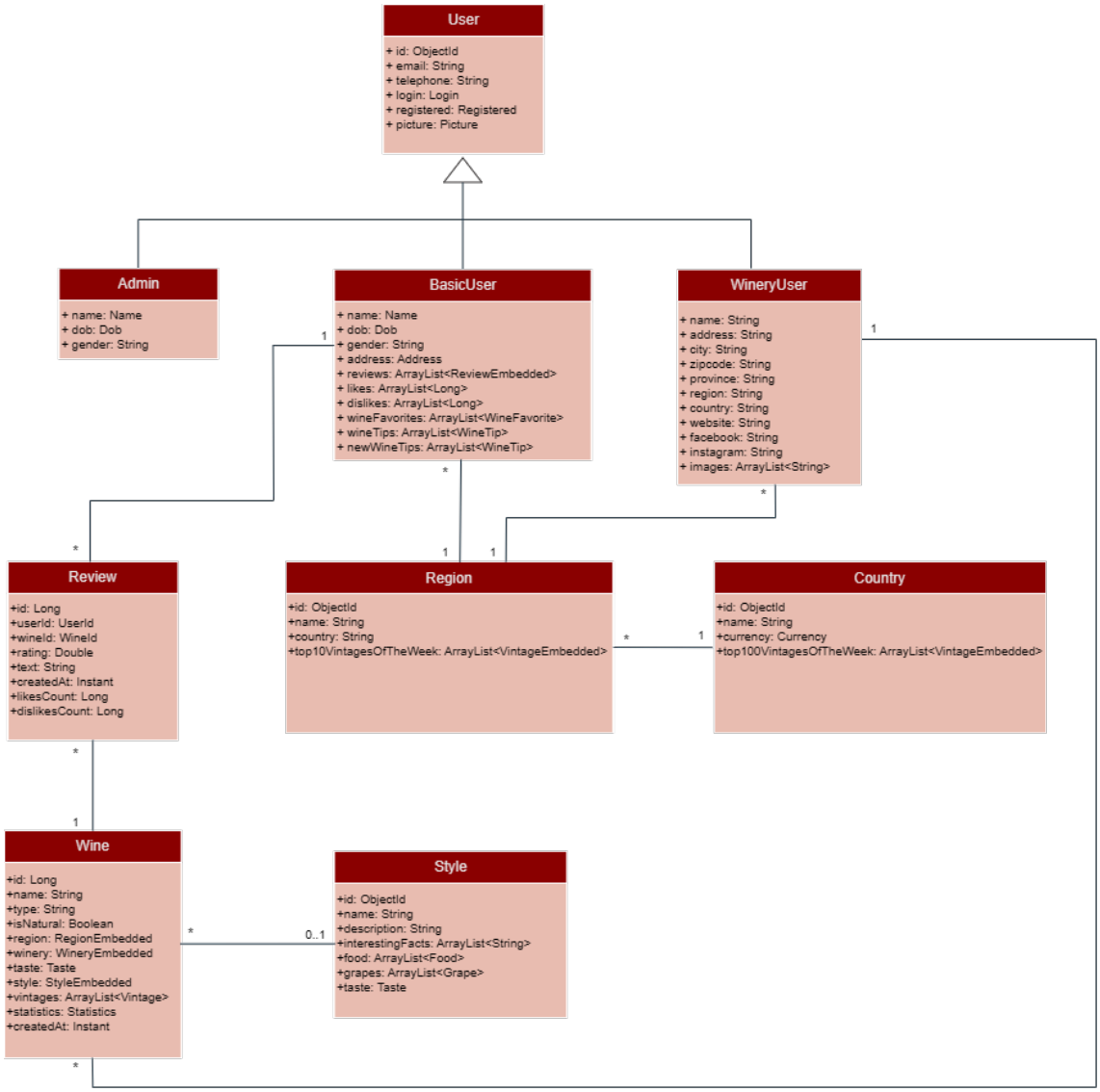


Figure 2.3: WineAdvisor class diagram

Chapter 3

Data Modeling and Structure

3.1 Dataset Creation

In accordance with the principle of *data variety*, our dataset was assembled from a combination of multiple structured sources, public repositories, and scraped content. To ensure a diverse and representative dataset, we relied on multiple sources and techniques, all oriented toward maximizing data realism. The core of our dataset was built using data retrieved from three main sources: **Vivino.com**, **Bereilvino.it**, and **RandomUser.me**. Each of these sources served a specific purpose within the dataset creation pipeline and contributed unique information.

3.1.1 Web APIs

Vivino API

We used Vivino's public (but unofficial) API to retrieve structured data related to wines and their vintages. The API queries returned arrays of matches, where each match corresponded to a specific vintage of a wine. Within each vintage object, we accessed keys such as `price`, `prices`, and a nested `wine` object, which contained all relevant information about the wine itself. Statistical attributes such as average rating and number of reviews were provided per vintage, not per wine.

Data per wine provided by this API was huge, since that it also includes unnecessary data to our application (such as prizes or awards) and a large amount of repeated data.

We made a considerable pre-processing work on this data to let it be useful to our goals.

RandomUser API

To simulate a realistic user base for our application, we generated synthetic user data through the `randomuser.me` API. This allowed us to populate our system with diverse user profiles while avoiding any privacy concerns or the need for real user data.

3.1.2 Scraping

Bereilvino.it

Since no public API was available, we resorted to web scraping to extract wine information from `bereilvino.it`, a website for Italian wines and wineries. The scraped data included details about wine producers, such as names, regions, addresses and contact info. This information was semi-structured and required a pre-processing work before being integrated into our dataset with data coming from the other sources.

3.1.3 Building Documents

Reviews

Our review dataset was obtained primarily from Vivino. Each review was associated with a specific vintage, not directly with the wine itself. To build consistent review documents, we extracted key fields such as rating (on a scale from 0 to 5), review text, timestamp, and the corresponding user and vintage identifiers.

The Vivino's reviews had been used to generate other reviews (following that style), in order to have a bigger and more realistic review dataset.

Users

User profiles were generated using two main strategies. A portion of users was created by leveraging the `randomuser.me` API, which provided realistic personal information such as names, surnames, emails, birth dates, and nationalities. For the remaining users, we employed the *Faker* Python library to synthetically generate similar fields. We then linked the users from Vivino (the owners of the reviews we had taken) respectively to our generated users. The users' surplus had been fulfilled with the generated reviews we specified in the previous sub-paragraph.

To ensure secure authentication, passwords were hashed using the *BCrypt* algorithm.

Wineries

Wineries were extracted from `bereilvino.it` using web scraping techniques. Each winery document included essential attributes such as the name, region, and country. Since wineries are also a specific type of account within the system, login credentials and access-related fields (e.g., email, password) were synthetically generated using the `Faker` Python library.

Passwords were hashed using the `BCrypt` algorithm to ensure secure authentication.

Wines

Wine documents were built by combining information from Vivino. Each document contained fields such as name, type, style, taste, winery reference, and a nested list of vintages. Each vintage stored year-specific data. This structure ensured that both general wine attributes and vintage-specific data could be efficiently queried and displayed.

Styles

Wine styles were originally embedded within the wines documents. To improve data normalization and re-usability, these styles were extracted and modeled as a separate entity. The styles documents include the style name, description, and typical characteristics associated with each style. This separation allows for easier querying and classification of wines by their style attributes across the dataset.

Regions and Countries

The regions and countries documents were manually curated to ensure accuracy and consistency across the dataset. These entities provide standardized geographic classifications used to organize wineries and wines. Each region document includes the region name and its associated country, while each country document contains essential information such as the country name and currency.

3.2 Scripts and functions for dataset assembly

To prepare a consistent and enriched dataset, we developed a suite of Python scripts performing data cleaning, transformation, and integration tasks across multiple source files. Key operations included:

- Populating flavor profiles and image links, including generating a set of random images saved in the database, referenced by an `image` field where needed.
- Standardizing review documents by renaming `note` to `review_text`, removing redundant fields such as `language`, and adding `wine_id` to reviews for better linkage.
- Reorganizing statistical data by moving wine-related statistics from individual vintages to the parent wine document.
- Cleaning style information by removing unnecessary fields.
- Merging and de-duplicating wineries data, assigning unique IDs, and updating winery references in wines accordingly.
- Embedding user data (username and profile picture) directly into wine reviews to reduce relational queries.

- Transforming review relations by replacing `user_id` and `wine_id` fields with embedded documents containing relevant user and wine details.
- Augmenting user profiles with samples of their recent reviews.
- Associating the three most recent reviews with each vintage, instead of the wine as a whole.
- Normalizing geographic data by replacing sub-regions with official Italian regions in wine documents.
- Replacing numeric `type_id` fields in wines with descriptive strings (e.g., "rosso", "bianco").
- Separating fortified or aromatized wines into a distinct category.
- Moving the `image` field from the wine level to each vintage document.
- Cleaning country data by removing unnecessary fields.
- Standardizing user fields: renaming `phone` to `telephone` and `state` to `region`, removing unused fields like `timezone` and `coordinates`.
- Localizing user data to Italian profiles with assistance from GPT-generated formatting.
- Removing redundant country information from individual wine documents.
- Proposing the introduction of `Region` and `Country` entities within the graph database to enable geographically contextualized recommendations for users and wineries.
- Adding `likes` and `dislikes` arrays to each user to efficiently manage user reactions to reviews and avoid duplication of likes or dislikes.

This comprehensive pre-processing pipeline ensured data consistency, enriched document contents, and optimized the dataset for both querying and analytical use cases.

3.3 Databases

Our application relies on **MongoDB** and **Neo4j**. In this chapter, we delve into the rationale behind this selection, the methods used to store data, and the thoughtful techniques we applied.

3.3.1 Volume Considerations

Based on an estimated user base of approximately 5,000 users, we assume that 20% are *Daily Active Users (DAU)*, and an additional 10% access the platform occasionally. With an average of about 90 actions per active user per day, this results in an estimated 138.000 user actions daily. The following table provides a breakdown of the main user interactions, along with the estimated number of daily occurrences and the corresponding operations performed on MongoDB and Neo4j.

Note: The *update* and *delete* operations (e.g. deleting an account, updating user info, updating winery images) and operations done by the admins are not considered in the analysis because we suppose they are really rare actions.

Note: The percentage in the following table are to be intended on the total of the *DAU*.

Action	% Users	Daily Occurrences	Mongo Ops	Neo4j Ops	Description
Winery add	~1%	~0	1	1	Registering as new *winery* user
Winery search	30%	5,000	1	0	Searching for wineries
Winery page	25%	7,500	1	0	Viewing winery details
Wine add	~5%	~1,000	1	0	Adding a new wine
Wine search	40%	10,000	1	0	Searching for wines
Wine page	50%	15,000	1	0	Viewing wine details
Check rankings	20%	6,000	1	0	Checking wine rankings
Review write	10%	1,500	2	1	Creating a review + updating stats
Review like	50%	15,000	1	0	Liking/disliking a review
User add	~5%	~1,000	1	1	Registering as new *basic user*
User search	40%	10,000	1	0	Searching for users
User page	50%	15,000	1	0	Viewing another user details
User reviews	30%	5,000	1	0	Viewing all user's reviews
Check profile	30%	5,000	1	0	Checking your own wine tips
Follow	10%	1,500	1	1	Following/Unfollowing
Feed	60%	17,000	0	2	Viewing feed + follow tips
Totals			**~100,000**	**~38,000**	

Table 3.1: Estimated Daily Requests per Action

3.3.2 MongoDB

Although MongoDB was a required technology for the project, we would have chosen it regardless due to the advantages offered by its document-oriented approach. The ability to embed related information within a single document allows us to avoid complex joins, leading to simpler queries and better runtime performance. Additionally, MongoDB's horizontal scalability makes it a solid choice for applications

expected to grow over time, thanks to efficient indexing and a robust aggregation pipeline for processing large datasets.

3.3.2.1 MongoDB Collections

These are the following MongoDB collections used in our application:

- `Reviews`
- `Users`
- `Wines`
- `Wineries`
- `Admins`
- `Regions`
- `Countries`
- `Styles`
- `Counters`
- `TopVintagesByOurQopType`
- `TopVintagesByQopType`
- `TopVintagesRatingsType`
- `TopWineriesRatings`
- `TopWinesRatingsType`

Reviews Collection

Here is an example of a document stored in this collection:


```
{
    "_id": 1,
    "user_id": {
        "username": "silverelephant535",
        "thumbnail":
        "https://randomuser.me/api/portraits/thumb/men/98.jpg"
    },
    "wine_id": {
        "id": 542,
        "name": "Oracolo Veneto Rosso",
        "year": 2000,
        "image": "https://i.postimg.cc/76rp4DV8/wine19.png"
    },
    "rating": 4.0,
    "text": "Un vino ricco e strutturato, con un equilibrio
    perfetto tra tannini e frutta matura, ideale con piatti
    di carne alla griglia.",
    "created_at": {
        "$date": "2024-12-10T17:26:49.316Z"
    },
    "likes_count": 180,
    "dislikes_count": 268
}
```

Users Collection

Here is an example of a document stored in this collection:


```
{
    "_id": {
        "$oid": "6839380fcb18a722a38c545b"
    },
    "gender": "female",
    "name": {
        "title": "Miss.",
        "first": "Michela",
        "last": "Gianetti"
    },
    "address": {
        "street": {
            "number": "49",
            "name": "Viale Gustavo"
        },
        "city": "Gorizia",
        "region": "Friuli-Venezia Giulia",
        "country": "Italia",
        "postcode": "34170"
    },
    "email": "michela.gianetti@example.com",
    "login": {
        "username": "crazyduck840",
        "password":
        "$2b$12$.iJSeJyL1M7.m7Vg/QR/Se1Px33jt...AIbixgNEvb3n8xqkq"
    },
    "dob": {
        "date": { "$date": "1981-04-28T03:32:29.035Z" },
        "age": 43
    },
    "registered": {
        "date": { "$date": "2020-02-14T19:45:21.187Z" },
        "age": 4
    },
    "telephone": "+39 327 9756577",
    "picture": {
        "large": "https://randomuser.me/api/portraits/women/95.jpg",
        "medium":
        "https://randomuser.me/api/portraits/med/women/95.jpg",
        "thumbnail":
```

```
        "https://randomuser.me/api/portraits/thumb/women/95.jpg"
    },
    "reviews": [
        {
            "review_id": 16381,
            "user_id": {
                "username": "crazyduck840",
                "thumbnail":
                "https://randomuser.me/api/.../women/95.jpg"
            },
            "wine_id": {
                "id": 4461,
                "name": "Ovello Barbaresco",
                "year": 2020,
                "image": "https://i.postimg.cc/.../wine20.png"
            },
            "rating": 4.6,
            "text": "Superbo e umile.",
            "created_at": { "$date": "2023-10-30T10:22:47.000Z" },
            "likes_count": 90,
            "dislikes_count": 399
        },
        {
            "review_id": 3602,
            "user_id": {
                "username": "crazyduck840",
                "thumbnail":
                "https://randomuser.me/api/.../women/95.jpg"
            },
            "wine_id": {
                "id": 201,
                "name": "Pinot Nero",
                "year": 2021,
                "image": "https://i.postimg.cc/.../wine19.png"
            },
            "rating": 4.0,
            "text": "Ciliegie, fragole, more mature.",
            "created_at": { "$date": "2022-04-15T21:37:06.000Z" },
            "likes_count": 1,
            "dislikes_count": 303
        },
        {
            "review_id": 8323,
```

```
            "user_id": {
                "username": "crazyduck840",
                "thumbnail":
                "https://randomuser.me/api/.../women/95.jpg"
            },
            "wine_id": {
                "id": 1775,
                "name": "Faius",
                "year": 2021,
                "image": "https://i.postimg.cc/.../wine12.png"
            },
            "rating": 4.0,
            "text": "Gran vino, con finale persistente.",
            "created_at": { "$date": "2021-08-21T18:55:57.000Z" },
            "likes_count": 1,
            "dislikes_count": 17
        }
    ],
    "likes": [ 21, 25, 31, 64, 68, 85, 90 ],
    "dislikes": [ 8, 19, 107, 117, 128, 143 ],
    "wine_favorites": [],
    "wine_tips": [],
    "new_wine_tips": []
}
```


As shown, each user document includes their **three most recent reviews**. Such list is updated in real time when a new review is submitted or an existing one is deleted: newly submitted reviews are immediately added to the top of the list, and deleted ones are instantly removed. The latter is done without inserting a third review in real time retrieving it from the *review* collection, in order to have a lighter operation with shorter response times. This "consistency" operation is scheduled to run once a day. This approach enables us to display a summary of a user's reviews directly on their profile page, whether viewed by themselves or others, without needing to query a separate collection. In this way, users can quickly get an overview of someone else posts.

The **arrays** `likes` and `dislikes` contain the ID of reviews which the user liked or disliked, respectively.

The **array** `wine_favorites` lists the user's favorite wines: this information is the starting point (in combination with the ratings to wines the user provided in his reviews) for generating personalized recommendations.

The **arrays** `wine_tips` and `new_wine_tips` keep trace of the recommendations.

Wines Collection

Here is an example of a document stored in this collection:


```
{
    "_id": 1,
    "name": "Grandi Annate Sangiovese",
    "type": "rosso",
    "is_natural": false,
    "region": {
        "name": "Toscana",
        "country": {
            "name": "Italia",
            "currency": {
                "code": "EUR",
                "name": "Euro",
                "prefix": "€"
            }
        }
    },
    "winery": {
        "name": "Avignonesi",
        "username": "avignonesi2572",
        "thumbnail":
        "https://i.postimg.cc/LX6F6wtH/winery112.jpg"
    },
    "taste": {
        "acidity": 3.7327719,
        "fizziness": null,
        "intensity": 3.4893339,
        "sweetness": 1.5484701,
        "tannin": 3.5251772
    },
    "style": {
        "name": "Vino Nobile di Montepulciano",
        "description": "You may not be familiar with Vino
        Nobile di Montepulciano, but you should be. Despite
        its long name, it makes delicious wines filled
        with preserved red cherry flavors. ...",
        "interesting_facts": [
            "Vino Nobile di Montepulciano was one of the
            original four DOCGs created in 1980.",
            "By law, the wines must be aged for 2 years to
            be called Vino Nobile and 3 years to qualify as
```

```
            Riserva.",
            "The area of Montepulciano is warmer than the
            rest of Tuscany.",
            "Vino Nobile di Montepulciano must contain at
            least 70% Prugnolo Gentile (aka Sangiovese)."
        ],
        "food": [
            {
                "name": "Manzo",
                "image":
                "https://i.postimg.cc/mDbRbCNG/manzo.jpg"
            },
            {
                "name": "Cacciagione (cervo, camoscio)",
                "image":
                "https://i.postimg.cc/Vshyh3Tw/cacciagione.jpg"
            }
        ],
        "grapes": [
            {
                "name": "Sangiovese",
                "wines_count": 125094
            },
            {
                "name": "Canaiolo nero",
                "wines_count": 4879
            }
        ]
    },
    "vintages": [
        {
            "year": 2012,
            "price": 80,
            "statistics": {
                "ratings_count": 726,
                "ratings_average": 4.2
            },
            "image":
            "https://i.postimg.cc/hPHLGGHY/wine01.png",
            "reviews": [
                {
                    "review_id": 3002,
                    "user_id": {
```

```
                    "username": "bigpeacock714",
                    "thumbnail":
                    "https://randomuser.me/.../91.jpg"
                },
                "wine_id": {
                    "id": 1,
                    "name": "Grandi Annate Sangiovese",
                    "year": 2012,
                    "image":
                    "https://i.postimg.cc/.../wine01.png"
                },
                "rating": 5.0,
                "text": "Veramente un vino sorprendente!",
                "created_at": {
                    "$date": "2021-08-11T08:53:24.000Z"
                },
                "likes_count": 13,
                "dislikes_count": 199
            }
        ],
        "created_at": {
            "$date": "2014-04-18T21:06:31.839Z"
        }
    },
    {
        "year": 2015,
        "price": 80,
        "statistics": {
            "ratings_count": 579,
            "ratings_average": 4.3
        },
        "image": "https://i.postimg.cc/.../wine01.png",
        "reviews": [
            {
                "review_id": 3003,
                "user_id": {
                    "username": "beautifulkoala533",
                    "thumbnail":
                    "https://randomuser.me/.../46.jpg"
                },
                "wine_id": {
                    "id": 1,
                    "name": "Grandi Annate Sangiovese",
```

```
                    "year": 2015,
                    "image":
                    "https://i.postimg.cc/.../wine01.png"
                },
                "rating": 4.0,
                "text": "Da provare.",
                "created_at": {
                    "$date": "2024-03-11T21:40:29.000Z"
                },
                "likes_count": 13,
                "dislikes_count": 474
            }
        ],
        "created_at": {
            "$date": "2018-03-15T01:47:19.197Z"
        }
    },
    {
        "year": 2016,
        "price": 79,
        "statistics": {
            "ratings_count": 199,
            "ratings_average": 4.3
        },
        "image": "https://i.postimg.cc/.../wine01.png",
        "reviews": [
            {
                "review_id": 3001,
                "user_id": {
                    "username": "smalllion777",
                    "thumbnail":
                    "https://randomuser.me/.../75.jpg"
                },
                "wine_id": {
                    "id": 1,
                    "name": "Grandi Annate Sangiovese",
                    "year": 2016,
                    "image":
                    "https://i.postimg.cc/.../wine01.png"
                },
                "rating": 4.3,
                "text": "Tannic, but bellissimooo",
                "created_at": {
```

```
                        "$date": "2023-04-20T13:12:19.000Z"
                    },
                    "likes_count": 2,
                    "dislikes_count": 167
                }
            ],
            "created_at": {
                "$date": "2023-08-04T03:04:32.912Z"
            }
        }
    ],
    "statistics": {
        "ratings_count": 4140,
        "ratings_average": 4.3
    },
    "created_at": {
        "$date": "2014-04-18T21:01:31.839Z"
    }
}
```


Similarly to the approach used for the *users* collection, for each vintage we store the **three most recent reviews**. Updates to such list are done in the exact same way we stated for the *users* collection. This design choice avoids the need to access a separate collection when displaying the wine page, enabling faster retrieval of last reviews.

Wineries Collection

Here is an example of a document stored in this collection:


```
{
"_id": {
    "$oid": "6888350fcb18a722a38c545b"
},
"name": "ARPEPE",
"address": "Viale Europa, 30",
"city": "Mantova",
"zipcode": "46100",
"province": "Mantova",
"region": "Lombardia",
"country": "Italia",
"telephone": "+39 336 1157764",
"email": "info@arpepe.com",
"website": "http://www.arpepe.com/",
"facebook": "https://www.facebook.com/arpepe",
"instagram": "https://www.instagram.com/arpepe/",
"images": [
  "https://i.postimg.cc/KYhkCGjh/winery092.jpg",
  "https://i.postimg.cc/CKbPVx4G/winery181.jpg",
  "https://i.postimg.cc/VvZzwRFb/winery007.jpg",
  "https://i.postimg.cc/gjVBqkxc/winery060.jpg",
  "https://i.postimg.cc/DZjzfnvZ/winery106.jpg",
  "https://i.postimg.cc/fRppNFQN/winery171.jpg",
  "https://i.postimg.cc/vmG8WcDT/winery113.jpg",
  "https://i.postimg.cc/9QgVqJ38/winery152.jpg"
],
"login": {
  "username": "arpepe3749",
  "password": "$2b$12$ixjEU5Bo6I...ENzPLOqPoqCMFb7MGq"
},
"registered": {
  "date": { "$date": "2020-03-17T16:38:26.9322Z" },
  "age": 5
},
"picture": {
  "large": "https://i.postimg.cc/W3btL9tZ/winery112.jpg",
  "medium": "https://i.postimg.cc/VNv2xnm6/winery112.jpg",
  "thumbnail": "https://i.postimg.cc/LX6F6wtH/winery112.jpg"
}
}
```

Admins Collection

Here is an example of a document stored in this collection:


```
{
"_id": {
    "$oid": "68363507cb18a722a38c545c"
},
"gender": "male",
"name": {
  "title": "Mr.",
  "first": "Mario",
  "last": "Rossi"
},
"email": "mario.rossi@example.com",
"telephone": "+39 392 3928397",
"login": {
  "username": "admin",
  "password": "$2b$12$5M1iObzaPCAu...dgno2grW38XpKrSI2"
},
"registered": {
  "date": {
    "$date": "2020-03-24T14:30:15.123Z"
  },
  "age": 5
},
"dob": {
  "date": {
    "$date": "2000-03-24T14:30:15.123Z"
  },
  "age": 25
},
"picture": {
  "large": "https://randomuser.me/.../men/6.jpg",
  "medium": "https://randomuser.me/.../med/men/6.jpg",
  "thumbnail": "https://randomuser.me/.../thumb/men/6.jpg"
}
}
```

Regions Collection

Here is an example of a document stored in this collection:

```
{
    "_id": {
        "$oid": "6236350fcb18a722a38c545b"
    },
    "name": "Valle d'Aosta",
    "country": "Italia",
    "top_10_vintages_of_the_week": []
}
```

The **array** `top_10_vintages_of_the_week` is updated dynamically based on aggregation pipelines, which will be discussed in more detail later.

Countries Collection

Here is an example of a document stored in this collection:

```
{
    "_id": {
        "$oid": "6839350fcb18a722a38c545b"
    },
    "name": "Italia",
    "currency": {
        "code": "EUR",
        "name": "Euro",
        "prefix": "€"
    },
    "top_100_vintages_of_the_week": []
}
```

The **array** `top_100_vintages_of_the_week` is updated dynamically based on aggregation pipelines, which will be discussed in more detail later.

Styles Collection

Here is an example of a document stored in this collection:


```
{
    "_id": {
        "$oid": "6836350fcb18a722a38c585b"
    },
    "name": "Vino Nobile di Montepulciano",
    "description": "You may not be familiar with Vino Nobile
    di Montepulciano, but you should be. Despite its long name,
    it makes delicious wines filled with preserved red
    cherry flavors. ...",
    "interesting_facts": [
        "Vino Nobile di Montepulciano was one of the original
        four DOCGs created in 1980.",
        "By law, the wines must be aged for 2 years to be
        called Vino Nobile and 3 years to qualify as Riserva.",
        "The area of Montepulciano is warmer than the rest
        of Tuscany.",
        "Vino Nobile di Montepulciano must contain at least
        70% Prugnolo Gentile (aka Sangiovese)."
    ],
    "food": [
        { "name": "Manzo", "image": "https://manzo.jpg"},
        {
            "name": "Cacciagione (cervo, camoscio)",
            "image": "https://cacciagione.jpg"
        }
    ],
    "grapes": [
        { "name": "Sangiovese", "wines_count": 125094},
        { "name": "Canaiolo nero", "wines_count": 4879 }
    ],
    "taste": {
        "acidity": 4.0,
        "fizziness": null,
        "intensity": 3.5,
        "sweetness": 1.0,
        "tannin": 4.0
    }
}
```

Counters Collection

Here is an example of a document stored in this collection:

```
{
    "_id": "reviews",
    "seq": 17466
},
{
    "_id": "wines",
    "seq": 4824
}
```

In the *counters* collection, we store the current sequential ID for managing the auto-incremented identifiers of the *reviews* and *wines* collections.

TopWineriesRatings Collection

Here is an example of a document stored in this collection:

```
{
  "_id": {
    "$oid": "6836350fcb18a722a38c4cec"
  },
  "winery_username": "le_vigne_di_san_pietro7924",
  "thumbnail": "https://i.postimg.cc/26cRqGGn/winery105.jpg",
  "winery": "Le Vigne di San Pietro",
  "region": "Veneto",
  "country": "Italia",
  "ratings_average": 5,
  "ratings_count": 3
}
```

This collection contains the result of an *aggregation pipeline* that identifies the **top wineries** based on the average ratings of their own wines. As with other similar structures, the purpose is to store precomputed results for quick access. The logic and methodology behind this aggregation will be detailed in the *Analytics section.*

TopVintagesByOurQopType Collection

Here is an example of a document stored in this collection:


```
{
    "_id": {
        "$oid": "6846350fcb18a722a38c545b"
    },
    "type": "bianco",
    "vintages": [
        {
            "wine": "Lutum (Trebbiano d'Abruzzo)",
            "winery": "Tenuta Antonini",
            "year": 2023,
            "image": "https://i.postimg.cc/Sx82sf40/wine08.png",
            "ratings_count": 3,
            "quality": 5,
            "price": 14,
            "points": 100
        },
        {
            "wine": "Punggl Pinot Grigio",
            "winery": "Nals Margreid",
            "year": 2022,
            "image": "https://i.postimg.cc/fTyYwdkZ/wine04.png",
            "ratings_count": 6,
            "quality": 4.6,
            "price": 19.5,
            "points": 84.2
        },
        ...
    ]
}
```


This collection stores the result of an *aggregation pipeline*, whose output is saved here for efficiency and fast access. It contains, for each wine `type`, a list of the **top 1000 vintages**, based on our formula for quality/price. Further details about the logic behind aggregations will be discussed in the *Analytics section*.

TopVintagesByQopType Collection

Here is an example of a document stored in this collection:


```
{
    "_id": {
        "$oid": "6836350fcb18a722a38c545b"
    },
    "type": "bianco",
    "vintages": [
        {
            "wine": "Chardonnay",
            "winery": "Barbanera",
            "year": 2023,
            "image": "https://i.postimg.cc/MHCZ2gvn/wine16.png",
            "ratings_count": 3,
            "quality": 4.166666666666667,
            "price": 6.4,
            "points": 100
        },
        {
            "wine": "Catarratto - Pinot Grigio",
            "winery": "Costallegra",
            "year": 2022,
            "image": "https://i.postimg.cc/MKBKbZFV/wine18.png",
            "ratings_count": 1,
            "quality": 3.4,
            "price": 6.5,
            "points": 80.3
        },
        ...
    ]
}
```


This collection stores the result of an *aggregation pipeline*, whose output is saved here for efficiency and fast access. It contains, for each wine `type`, a list of the **top 1000 vintages**, based on the basic formula for quality/price. Further details about the logic behind aggregations will be discussed in the *Analytics section*.

TopVintagesRatingsType Collection

Here is an example of a document stored in this collection:


```
{
    "_id": {
        "$oid": "68363510cb18a722a38c5462"
    },
    "type": "rosato",
    "vintages": [
        {
           "wine": "Rosa dei Frati ... Classico Rosé",
           "winery": "Cà dei Frati",
           "year": 2023,
           "price": 12.4,
           "image": "https://i.postimg.cc/bYC2Fvg5/wine06.png",
           "ratings_average": 4.9,
           "ratings_count": 1
        },
        {
           "wine": "Furía di Calafuria",
           "winery": "Tormaresca",
           "year": 2022,
           "price": 24.7,
           "image": "https://i.postimg.cc/4491DM2Z/wine21.png",
           "ratings_average": 4.8,
           "ratings_count": 2
        },
        ...
    ]
}
```


This collection stores the result of an *aggregation pipeline*, whose output is saved here for efficiency and fast access. It contains, for each wine `type`, a list of the **top 1000 vintages**, based on the average ratings of vintages. Further details about the logic behind aggregations will be discussed in the *Analytics section*.

TopWinesRatingsType Collection

Here is an example of a document stored in this collection:


```
{
  "_id": {
    "$oid": "68363510cb18a722a38c5469"
  },
  "type": "rosato",
  "wines": [
        {
          "wine": "Diamante Rosé",
          "winery": "Comincioli",
          "image": "https://i.postimg.cc/fTyYwdkZ/wine04.png",
          "prices_average": 20.98,
          "ratings_average": 4.675,
          "ratings_count": 4
        },
        {
          "wine": "Rosato",
          "winery": "Casa Grande della Quercia",
          "image": "https://i.postimg.cc/XJNpsMC3/wine09.png",
          "prices_average": 14.95,
          "ratings_average": 4.666666666666667,
          "ratings_count": 3
        },
        {
          "wine": "Morso Rosa Susumaniello Biologico",
          "winery": "Tenuta Viglione",
          "image": "https://i.postimg.cc/c4Pt05VM/wine07.png",
          "prices_average": 12.9,
          "ratings_average": 4.525,
          "ratings_count": 4
        },
        ...
    ]
}
```


This collection contains the result of an *aggregation pipeline*, whose output is saved here for efficiency and fast access. It contains, for each wine `type`, a list of the **top 1000 wines**, based on the average ratings of wines. Further details about the logic behind aggregations will be discussed in the *Analytics section*.

3.3.3 Neo4j

Given the social nature of some features in our application, we adopted Neo4j to efficiently manage and explore user interactions. What follows is an overview of the graph's key elements and how they connect.

Justification of Neo4j Adoption

Neo4j was adopted to support the **social features** of our application, such as *follow/unfollow* operations, generating *profile suggestions* (other users to follow), and building a *personalized feed* for each user. These features are inherently graph-oriented, making a graph database like Neo4j a natural and efficient choice.

To avoid redundancies and maintain a clean separation of concerns between MongoDB and Neo4j, we chose to model user relationships **exclusively in Neo4j**. This design allows us to efficiently traverse connections and generate recommendations based on graph queries, without duplicating relationship data across both databases.

Moreover, since our service does not require real-time updates of **social metrics** (e.g., number of followers or liked reviews), we embraced an *eventual consistency* approach for this kind of data: derived statistics and aggregated counters are updated in MongoDB periodically, while real-time relationship logic is handled in Neo4j. This trade-off enables us to prioritize **performance and scalability**, while still meeting the responsiveness required by our features.

3.3.3.1 Entities

User

The *User* node in Neo4j is a simplified version of the one stored in MongoDB. Since the graph database is used to support social features only, we decided to store just the essential information required for these functionalities: username, name, surname, and thumbnail.

Review

In our Neo4j graph, we decided to store only the latest three *Review* nodes for each user. Each review node contains only a subset of information, which is sufficient for generating personalized feeds and account suggestions: the review ID, the wine ID, name, image and year and the creation date of the review, as well as the review rating and text. This design avoids unnecessary redundancy and ensures a lighter and faster traversal during graph queries.

Winery

The *Winery* node represents a wine producer and is a simplified version of the entity stored in MongoDB. For the purposes of the social features implemented in Neo4j, we only store the username, name and thumbnail attributes.

3.3.3.2 Relationships

We defined two main relationship types: `FOLLOWS` and `WROTE`, each with specific domains and characteristics.

Followings

The `FOLLOWS` relationship is used to represent following connections between nodes in the graph, primarily between *users*.

$$(\texttt{u1:User}) - [\texttt{f:FOLLOWS}] - > (\texttt{u2:User})$$

This relationship is *unidirectional*: if a user `u1` follows another user `u2`, the reverse does not automatically hold. In case of mutual following, two distinct `FOLLOWS` edges will exist. We opted for this design to ensure faster traversal performance in Neo4j. Traversing only outgoing relationships is generally more efficient than checking both incoming and outgoing ones along with an additional property filter.

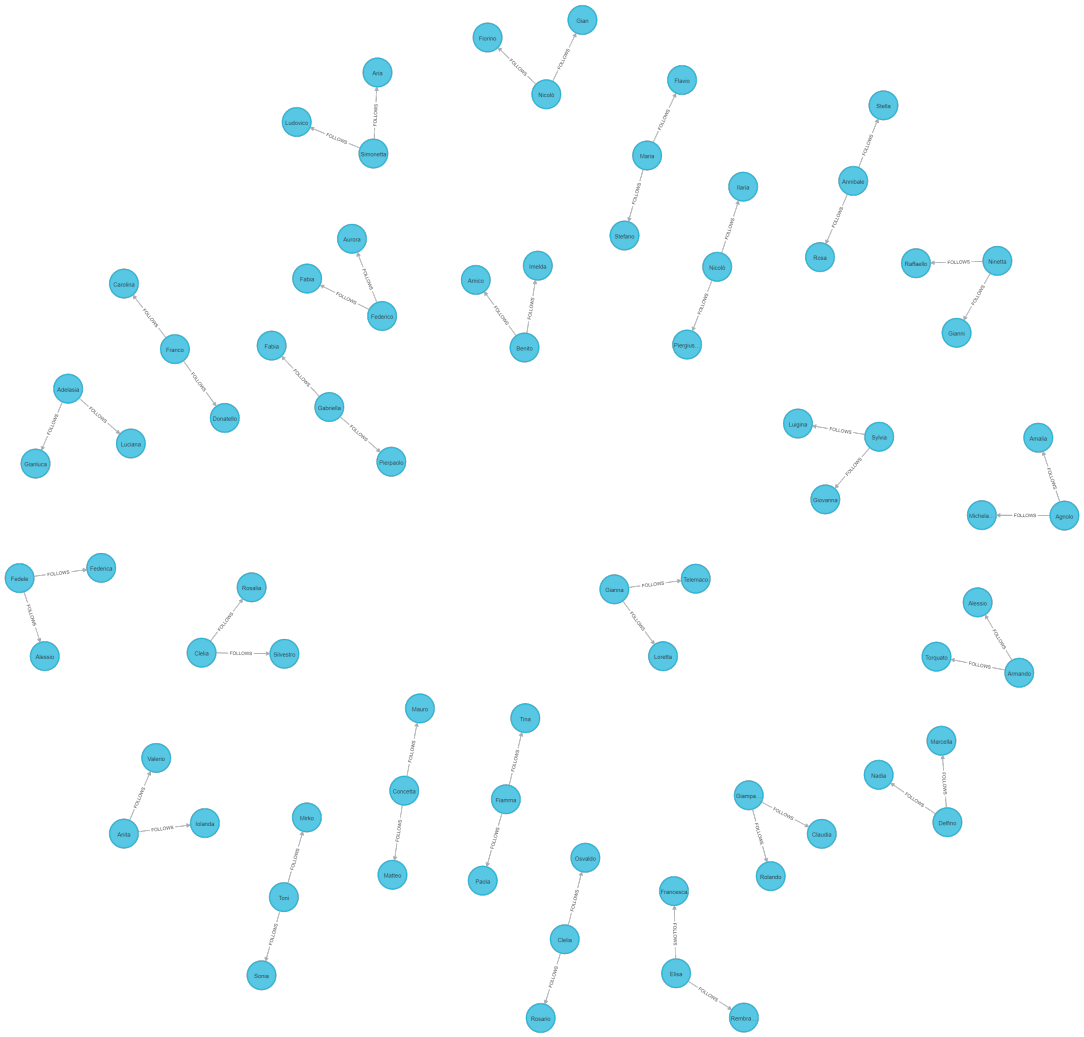


Figure 3.1: User nodes linked with `FOLLOWS` relationships.

Additionally, this relationship is also used between a *user* and a *winery*.

$$(\texttt{u:User}) - [\texttt{f:FOLLOWS}] - > (\texttt{w:Winery})$$

This *uni-directionality* is enforced by design: winery accounts cannot follow anyone, as they are intended only as followable entities. Wineries are included in the graph primarily to enable indirect recommendation strategies. For instance, users may receive winery suggestions based on which wineries are followed by people they follow.

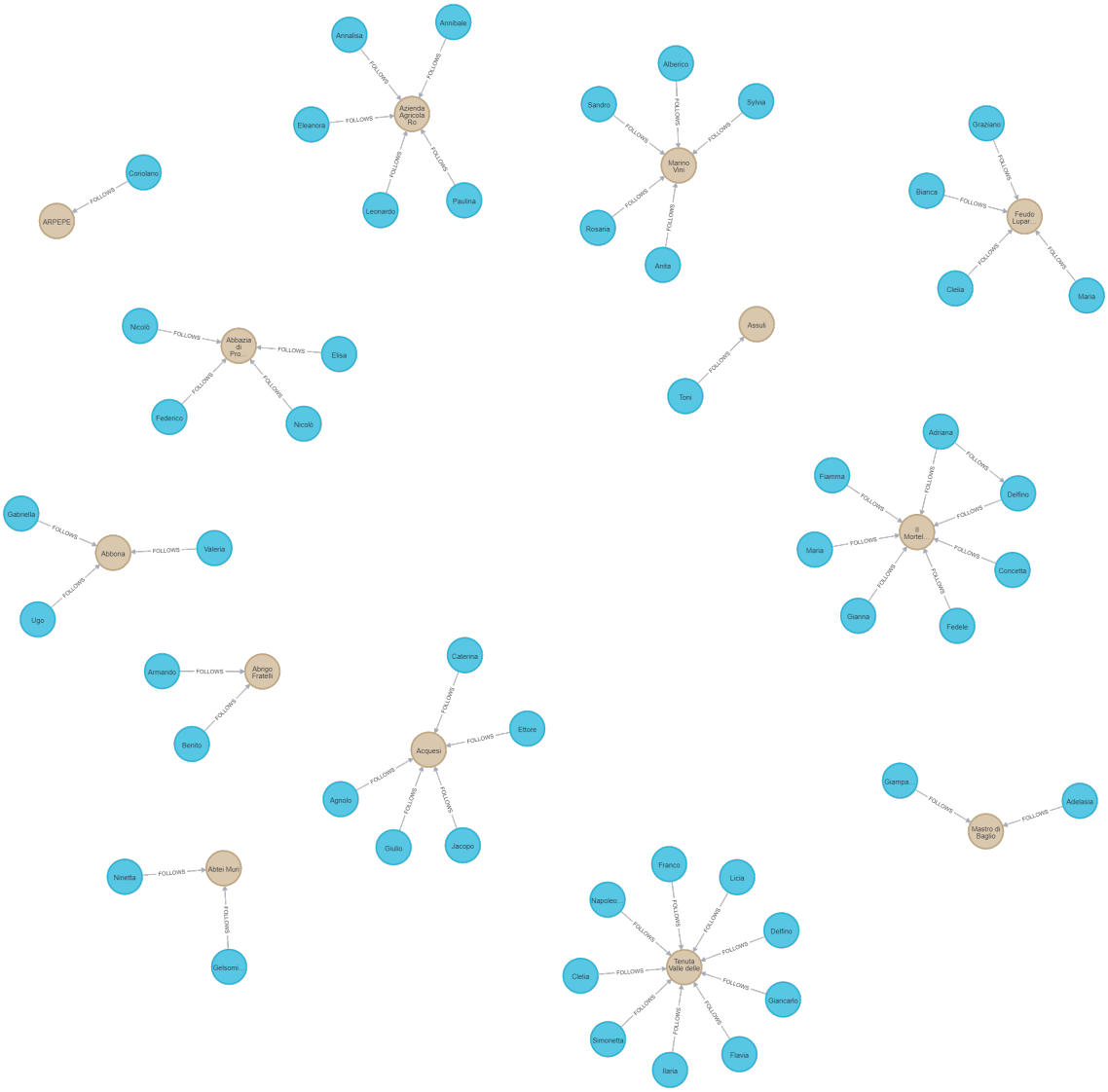


Figure 3.2: User and Winery nodes linked with `FOLLOWS` relationships.

Review ownership

The `WROTE` relationship connects a *User* node to a *Review* node, representing the action of writing a review.

$$(\texttt{u:User}) - [\texttt{w:WROTE}] - > (\texttt{r:Review})$$

This relationship allows us to retrieve all most recent reviews authored by a given user, and it is crucial in order to build the user's personal feed. Since each review is written by exactly one user, the `WROTE` relationship is *unidirectional* and *unique* for each review node.

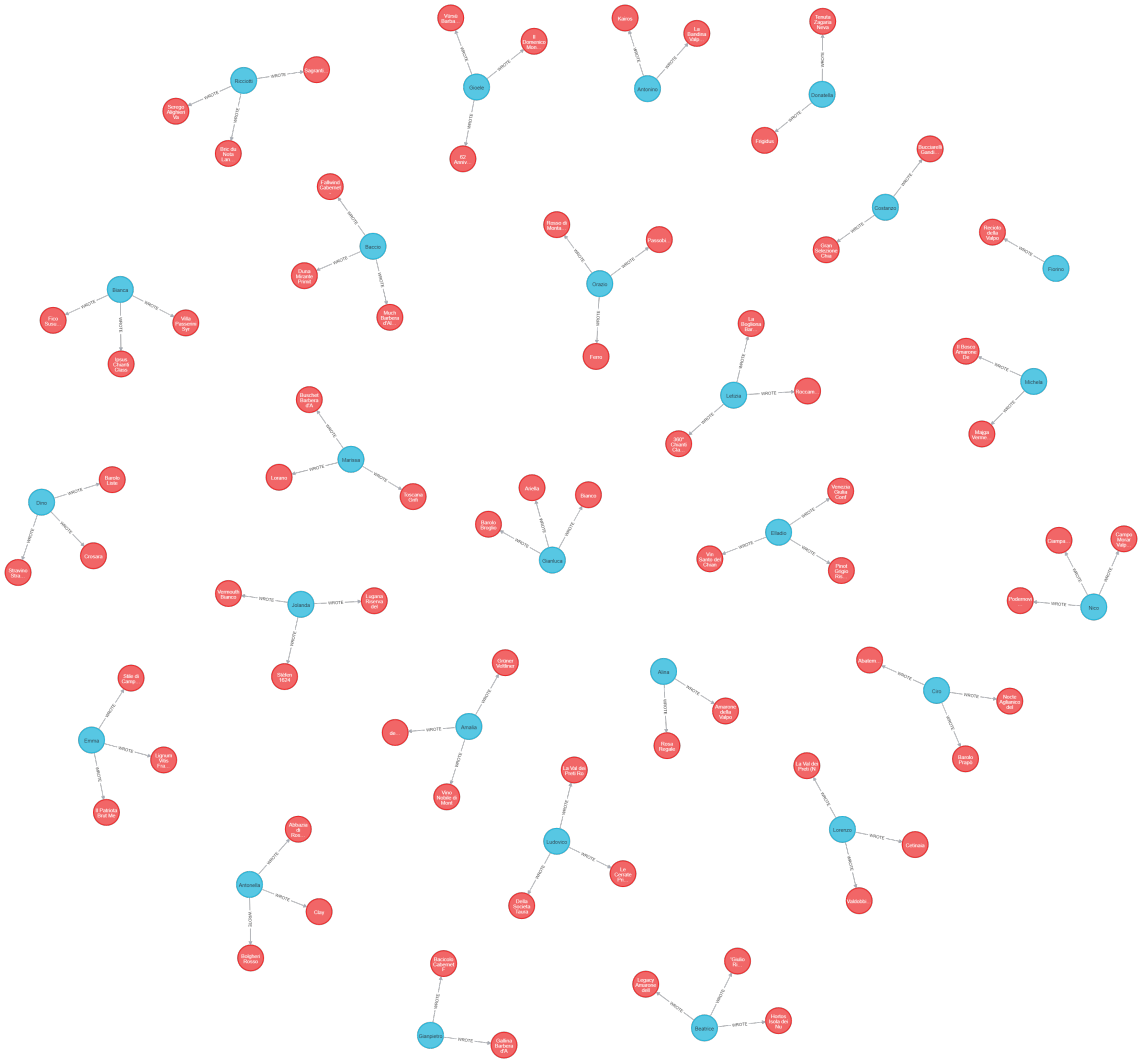


Figure 3.3: User and Review nodes linked with `WROTE` relationships.

Chapter 4

Implementation

4.1 Implementation Frameworks

For the development of our application, we chose to adopt the **Spring Boot** framework as the fundamental backbone. To facilitate the development and testing of our REST APIs, we made extensive use of **Swagger** and **Postman**.

4.1.1 Spring Boot

This framework greatly simplifies the configuration and management of the entire application infrastructure, providing an embedded web server that allows us to rapidly build and deploy our backend services without the need for complex external setup. This feature alone has accelerated our development process by enabling a streamlined workflow and reducing boilerplate code.

In addition, we leveraged the `@RestController` annotation, which has proven crucial in designing RESTful APIs. This approach allowed us to map HTTP requests directly to JAVA methods in a clean and efficient manner, facilitating clear separation between the backend logic and the HTTP protocol details. The use of `@RestController` also helped us maintain consistency across the API endpoints, making the system easier to maintain and extend.

Moreover, Spring Boot's powerful integration with **Spring Security** provided robust mechanisms to secure our application. This includes authentication and authorization which are essential to safeguard sensitive user data and implement role-based access controls seamlessly. By using Spring Security within the Spring Boot ecosystem, we ensured a scalable and secure environment that meets the requirements of our web application.

4.1.2 Swagger UI

Swagger was integrated directly into the application via the **SpringDoc library**, providing a clear, interactive documentation of all available endpoints. This greatly improved the debugging and validation process by allowing real-time interaction

with the backend services. The annotation `@Schema` has been used in all *DTO* classes to provide pre-written parametres and request bodies on both Swagger and Postman.

4.1.3 Postman

Postman has been used in the last part of the application development, basically after having added security roles and access to our application: you need to be authenticated on Swagger to send requests to server and this prevents from testing the APIs which need a non-authenticated request.

Postman allows to choose the authentication protocol for each request. Moreover, it gives a visually better interface to interact with the server.

4.2 Project Structure and Packages Organization

In our project, the code is arranged to promote modularity and simplify understanding. Different packages have been created, each dedicated to a distinct aspect of the application functionalities.

4.2.1 `com.wineadvisor.wineadvisor.config`

This package contains the configuration classes for the application, including security settings, database connections, and other environment-specific parameters.

- `SecurityConfig`: Configures the application security settings, defining authentication rules, route permissions, and stateless session management. It uses Spring Security with basic authentication and disables *CSRF* to simplify testing with Postman.

- `RepositoryConfig`: Sets up the `MongoTemplate` for interacting with MongoDB, disabling the automatic inclusion of the `_class` field in documents, thus keeping the stored data cleaner.

```java
@Configuration
@RequiredArgsConstructor
@EnableMethodSecurity(securedEnabled = true, jsr250Enabled = true)
public class SecurityConfig {
    ////////////////////////////////
    ////////// VARIABLES //////////
    ////////////////////////////////
    private final AuthenticationService authenticationService;



    ////////////////////////////////
    ///////// PUBLIC METHODS ///////
    ////////////////////////////////

    ////////////////////////////////
    //// Authentication settings ////

    // Imposta il provider di autenticazione per l'autenticazione degli utenti
    @Bean
    public DaoAuthenticationProvider authenticationProvider() {
        DaoAuthenticationProvider authProvider = new DaoAuthenticationProvider();
        authProvider.setUserDetailsService(authenticationService);
        authProvider.setPasswordEncoder(PasswordDTO.passwordEncoder());
        return authProvider;
    }

    // Imposta il gestore di autenticazione per l'autenticazione degli utenti
    @Bean
    public AuthenticationManager authenticationManager(AuthenticationConfiguration authConfig) throws Exception {
        return authConfig.getAuthenticationManager();
    }

    // Configura la catena di filtri di sicurezza per gestire le richieste HTTP
    @Bean
    public SecurityFilterChain securityFilterChain(HttpSecurity http) throws Exception {
        http
            .csrf(csrf -> csrf.disable())   // Disabilito CSRF (utile per testare con Postman)
            .sessionManagement(session -> session.sessionCreationPolicy(SessionCreationPolicy.STATELESS))   // Nessuna sessione
            .authorizeHttpRequests(auth -> auth
                .requestMatchers("/api/authentication/login").anonymous()       // Permetto l'accesso a /api/authentication/** solamente ad utenti non autenticati (endpoint di login)
                .requestMatchers(HttpMethod.POST, "/api/users").anonymous()     // Permetto l'accesso a /api/users solamente ad utenti non autenticati per le richieste di tipo POST (endpoint di registrazione di un utente)
                .requestMatchers(HttpMethod.POST, "/api/wineries").anonymous()  // Permetto l'accesso a /api/wineries solamente ad utenti non autenticati per le richieste di tipo POST (endpoint di registrazione di un'azienda vinicola)
                .requestMatchers("/api/admins/**").hasRole("ADMIN")     // Permetto l'accesso a /api/admins solamente ad utenti con ruolo ADMIN (endpoint di gestione degli admin)
                .requestMatchers("/api/regions/**").hasRole("ADMIN")    // Permetto l'accesso a /api/countries solamente ad utenti con ruolo ADMIN (endpoint di gestione delle regioni)
                .requestMatchers("/api/countries/**").hasRole("ADMIN")  // Permetto l'accesso a /api/countries solamente ad utenti con ruolo ADMIN (endpoint di gestione delle nazioni)
                .anyRequest().authenticated()   // Tutte le altre richieste richiedono autenticazione (senza specificare il ruolo)
            )
            .authenticationProvider(authenticationProvider())
            .httpBasic(httpBasic -> httpBasic.realmName("WineAdvisor"));    // Configuro l'autenticazione Basic per Postman

        return http.build();
    }

    //// END of auth. settings ////
    ////////////////////////////////
}
```

Figure 4.1: `SecurityConfig`

4.2.2 `com.wineadvisor.wineadvisor.controller`

The controller package manages the HTTP requests.

- `AdminController`: Handles CRUD operations for administrator management, including CRUD operations and updates to username/password. Access is restricted to admins using security annotations.
- `AnalyticsController`: Handles CRUD operations to provide statistics about top wines, vintages, and wineries based on criteria like QOP, rating, and price. Also includes cleanup operations for monthly charts, accessible only by admins.
- `AuthenticationController`: Exposes an endpoint to authenticate users using Spring Security authentication manager. Returns a message indicating successful login along with the user's role.
- `CountryController`: Handles CRUD operations for countries. Allows paginated retrieval, renaming, and deletion of countries with appropriate validations.
- `RegionController`: Handles creation, retrieval, update, and deletion of regions. It supports filtering by country and pagination for fetching region data.
- `ReviewController`: Handles creation, retrieval, update, and deletion of reviews.
- `StyleController`: Handles creation, retrieval, update, and deletion of wine styles. Admins and wineries can create and manage styles through secured endpoints.
- `UserController`: Manages user-related operations such as account creation, profile updates, search, password changes, and interactions like likes, dislikes, and favorites. Secured with role-based access control.
- `WineController`: Manages wine-related endpoints, allowing wineries to create, update, and delete wines and vintages. Provides search and retrieval of wines with filtering and pagination, with role-based access control.
- `WineryController`: Handles winery management operations such as creating, updating, and deleting winery profiles. Supports search by winery name and includes security checks for authenticated updates and deletions.

```
@PutMapping("/{username}")
@Secured({"ROLE_ADMIN", "ROLE_USER" })
@PreAuthorize("#username == authentication.principal.username or hasRole('ROLE_ADMIN')")
public ResponseEntity<?> updateUser(
        @NotBlank(message = "Username cannot be blank.") @PathVariable String username,
        @NotNull(message = "User update info cannot be null.") @Valid @RequestBody UpdateUserDTO updateUserDTO) {
    return ResponseEntity.status(HttpStatus.OK).body(userService.updateUser(username, updateUserDTO));
}
```

Figure 4.2: Snippet from `UserController`: endpoint to modify user profile information.

4.2.3 `com.wineadvisor.wineadvisor.model`

The model package represents the core domain entities of our application, encapsulating the key business data and related operations. It forms the foundation of our business logic by defining the main objects we work with. In this package, each class directly corresponds to a document stored within the MongoDB collections. This design choice ensures a straightforward mapping between the application's data structures and the database schema, facilitating seamless data persistence and retrieval through Spring Boot's built-in support.

- `Review`: This class models the documents within the "reviews" collection. It contains the details of a user's review on a specific wine, including the reviewer's username, the wine reviewed, the review text, and the given rating.
- `Wine`: This class corresponds to the documents in the "wines" collection. It holds all relevant information about a wine, such as its name, vintage, style, associated winery, and other key attributes that define the product.
- `User`: This class represents documents from the "users" collection. It stores user-specific information like username, email, and preferences related to the WineAdvisor platform.
- `Winery`: This class models the documents within the "wineries" collection. It contains data about a winery, including its name, location and contact info.
- `Admin`: This class represents administrative users in the system. It contains details about users with elevated privileges who can manage platform settings and moderate content.
- `Country`: This class corresponds to the "countries" collection. It contains information about the countries related to wine production, including country name and possibly other geographic or regulatory data.
- `Region`: This class represents documents in the "regions" collection. It holds data about specific wine-producing regions within countries, often including associated wineries and local characteristics.
- `Style`: This class models the "styles" collection, capturing different wine styles or categories to classify wines by their flavor profiles or production methods.

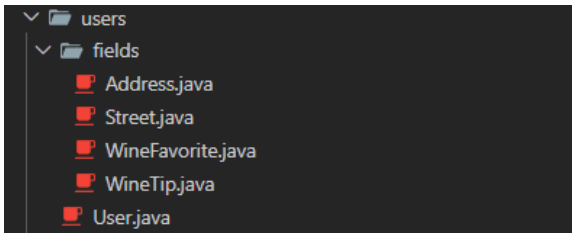


Figure 4.3: `User` class: folder structure.

```java
@Data
@NoArgsConstructor
@AllArgsConstructor
@Document(collection = "users")
@JsonInclude(JsonInclude.Include.NON_NULL)
public class User {
    @Id
    private ObjectId _id;

    private String gender;

    private Name name;

    private Address address;

    private String email;
    private String telephone;

    private Login login;

    private Registered registered;
    private Dob dob;

    private Picture picture;

    private ArrayList<ReviewEmbedded> reviews;
    private ArrayList<Long> likes;
    private ArrayList<Long> dislikes;

    @Field("wine_favorites")
    private ArrayList<WineFavorite> wineFavorites;

    @Field("wine_tips")
    private ArrayList<WineTip> wineTips;

    @Field("new_wine_tips")
    private ArrayList<WineTip> newWineTips;
```

Figure 4.4: `User` class: structure of the user model.

Inside the `model` package, there is also a sub-package named `analytics`. This folder contains classes that correspond to collections storing the results of various aggregation queries used for analytics purposes. The classes within this sub-package include:

- `TopWinesRatingsType`
- `TopWineriesRatings`
- `TopVintagesRatingsType`
- `TopVintagesQopType`
- `TopVintagesOurQopType` (which extends `TopVintagesQopType`)

4.2.4 com.wineadvisor.wineadvisor.service

This package implements the core business logic and service layers, processing data and applying application rules.

- AdminService
- AnalyticsService
- AuthenticationService
- CountryService
- RegionService
- IdCounterService - manages custom IDs for reviews and wines
- ReviewService
- StyleService
- UserService
- WineryService
- WineService

Figure 4.5: Snippet from UserService: service method for updating user information.

4.2.5 com.wineadvisor.wineadvisor.repository

The repository package serves as a dedicated layer for managing data access, isolating the underlying storage details from the rest of the application. It supports multiple databases, such as MongoDB and Neo4j, by offering tailored interfaces that leverage each database specific features. This structure promotes a clean separation of concerns and ensures efficient and consistent data operations across different storage technologies.

MongoDB Repositories

In our project, MongoDB repositories are interfaces that extend `MongoRepository`, allowing us to perform common database operations such as CRUD without writing boilerplate code. By annotating them with `@Repository`, Spring automatically generates the implementation at runtime, enabling seamless interaction with MongoDB collections through simple method definitions.

- `AdminRepository`
- `CountryRepository`
- `RegionRepository`
- `ReviewRepository`
- `StyleRepository`
- `UserRepository`
- `WineRepository`
- `WineryRepository`
- `TopWinesRatingsTypeRepository`
- `TopWineriesRatingsRepository`
- `TopVintagesRatingsTypeRepository`
- `TopVintagesQopTypeRepository`
- `TopVintagesOurQopTypeRepository`

```java
@Repository
public interface UserRepository extends MongoRepository<User, ObjectId> {
    Optional<User> findByLogin_Username(String username);
    Optional<User> findByEmail(String email);
    Optional<User> findByReviews_ReviewId(Long id);

    Page<User> findByName_FirstContainingIgnoreCaseAndName_LastContainingIgnoreCase(String firstName, String lastName, Pageable pageable);
    Page<User> findByName_FirstContainingIgnoreCase(String name, Pageable pageable);
    Page<User> findByName_LastContainingIgnoreCase(String last_name, Pageable pageable);

    ArrayList<User> findByReviews_WineId_Id(Long wineId);
    ArrayList<User> findByReviews_WineId_IdAndReviews_WineId_Year(Long wineId, Integer vintageYear);
    ArrayList<User> findByLikesOrDislikes(Long id, Long id2);
    ArrayList<User> findByAddress_Region(String region);
}
```

Figure 4.6: `UserRespository`

Neo4j Repositories

For Neo4j, the application uses repositories built with the Neo4jClient, allowing for fine-grained control over Cypher queries and data interactions. Instead of relying on auto-generated methods through Neo4jRepository, this approach enables the implementation of custom graph operations, optimized for specific logic such as limiting user reviews or managing dynamic relationships. This design ensures flexibility and full expressiveness in querying the graph database.

- `ReviewNeo4jRepository`
- `WineryNeo4jRepository`
- `WineryNeo4jRepository`

```java
@Repository
@RequiredArgsConstructor
public class UserNeo4jRepository {

    private final Neo4jClient neo4jClient;

    public void createUser(String username, String firstName, String lastName, String thumbnail) {
        Map<String, Object> params = new HashMap<>();
        params.put(key:"username", username);
        params.put(key:"firstName", firstName);
        params.put(key:"lastName", lastName);
        params.put(key:"thumbnail", thumbnail);

        neo4jClient.query("""
            CREATE (u:User {
                username: $username,
                firstName: $firstName,
                lastName: $lastName,
                thumbnail: $thumbnail
            })
        """)
        .bindAll(params)
        .run();
    }

    public Map<String, Object> findUserByUsername(String username) {
        return neo4jClient.query("""
                MATCH (u:User {username: $username})
                RETURN u.username AS username, u.firstName AS firstName, u.lastName AS lastName, u.thumbnail AS thumbnail
            """)
            .bind(username).to(name:"username")
            .fetch()
            .one()
            .orElse(other:null);
    }

    public void updateUser(String username, String firstName, String lastName, String thumbnail) {
        Map<String, Object> params = new HashMap<>();
        params.put(key:"username", username);
        params.put(key:"firstName", firstName);
        params.put(key:"lastName", lastName);
        params.put(key:"thumbnail", thumbnail);

        neo4jClient.query("""
                MATCH (u:User {username: $username})
                SET u.firstName = $firstName,
                    u.lastName = $lastName,
                    u.thumbnail = $thumbnail
            """)
            .bindAll(params)
            .run();
    }
```

Figure 4.7: Snippet from the `UserNeo4jRespository`

4.2.6 `com.wineadvisor.wineadvisor.DTO`

The DTO package contains the Data Transfer Objects used to encapsulate and exchanged data structure. Even though our project is backend-only, DTOs play a crucial role in decoupling the domain models from the exposed data, allowing for cleaner interfaces, improved maintainability, and better control over what is shared with the client. In particular, DTOs were used whenever the controller methods required a `@RequestBody` parameter, allowing for structured and validated input.

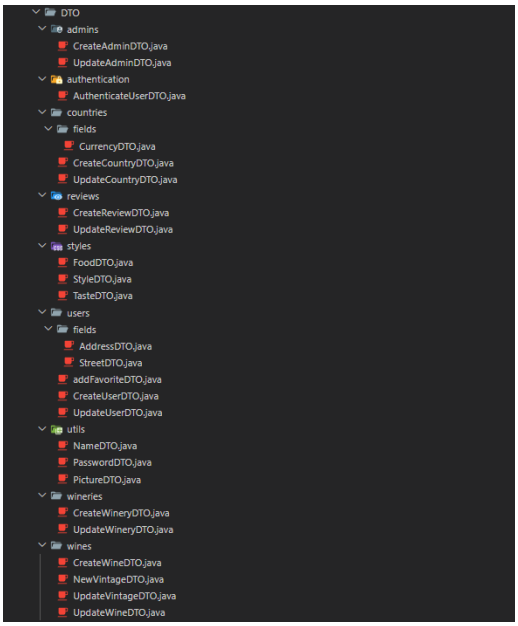


Figure 4.8: Overview of the DTO package organization.

```java
@Data
@JsonPropertyOrder({ "gender", "name", "address", "email", "telephone", "date of birth", "profile picture" })
public class UpdateUserDTO {
    @Pattern(regexp = "^(male|female|other)$", message = "Gender must be one among \"male\", \"female\" and \"other\" (or blank).")
    @Schema(description = "Gender of the updated user", example = "male")
    private String gender;

    @NotNull(message = "Name info cannot be null.")
    @Valid
    @Schema(name = "name", description = "Name info of the updated user")
    @JsonProperty("name")
    private NameDTO nameDTO;

    @NotNull(message = "Address info cannot be null.")
    @Valid
    @Schema(name = "address", description = "Home address of the updated user")
    @JsonProperty("address")
    private AddressDTO addressDTO;

    @NotBlank(message = "Email cannot be blank.")
    @Email(message = "Email must be a valid email address.")
    @Schema(description = "Email of the updated user", example = "mariorossi@example.com")
    private String email;

    @Pattern(regexp = "^\\+?[0-9\\s\\-()]+$", message = "Telephone must be a valid telephone number.")
    @Schema(description = "Telephone of the updated user", example = "+39 3331234567")
    private String telephone;

    @PastOrPresent(message = "Date must be in the past.")
    @Schema(name = "date of birth", description = "Date of birth of the updated user", example = "1970-01-01T00:00:00.000Z")
    @JsonProperty("date of birth")
    private Instant dob;

    @Schema(name = "profile picture", description = "Picture info of the updated user")
    @JsonProperty("profile picture")
    private PictureDTO pictureDTO;
```

Figure 4.9: `UpdateUserDTO`

4.2.7 `com.wineadvisor.wineadvisor.exception`

This package contains all the custom exception classes and centralized error-handling logic used in the application. It allows us to provide meaningful and consistent error responses when unexpected conditions occur during backend processing. By defining custom exceptions (e.g., `BadRequestException`, `AccessDeniedException`) and mapping them to appropriate HTTP status codes, we ensured better clarity and control over API behavior. Additionally, the `GlobalExceptionHandler` class uses `@RestControllerAdvice` to intercept and handle both custom and standard exceptions in a unified way, improving debuggability and user feedback.

- `AccessDeniedException`: Thrown when a user attempts to access a resource or perform an action without the necessary permissions. It returns an HTTP `403 FORBIDDEN` status.
- `BadRequestException`: Raised when a client request is invalid or malformed. It helps return an HTTP `400 BAD REQUEST` status.
- `DebugException`: A custom exception used for internal testing or debugging purposes. It always returns an HTTP `500 INTERNAL SERVER ERROR` with a fixed message.
- `ResourceAlreadyExistsException`: Used when trying to create a resource that already exists in the system. It results in an HTTP `409 CONFLICT` status.
- `ResourceNotFoundException`: Thrown when a requested resource cannot be found. This maps to an HTTP `404 NOT FOUND` status.

```java
@ResponseStatus(HttpStatus.NOT_FOUND)
public class ResourceNotFoundException extends RuntimeException {
    public ResourceNotFoundException(String message) {
        super(message);
    }
}
```

Figure 4.10: `ResourceNotFoundException`

- `GlobalExceptionHandler`: Centralizes exception handling for the entire application by mapping specific exceptions to meaningful HTTP responses. It ensures that client errors and server issues are returned with appropriate messages and status codes.

```java
@Hidden
@RestControllerAdvice
public class GlobalExceptionHandler {

    //////// AUTHENTICATION EXCEPTION ////////
    @ExceptionHandler(InternalAuthenticationServiceException.class)
    public ResponseEntity<?> handleInternalAuthenticationServiceException(InternalAuthenticationServiceException e) {
        System.out.println(x:"--- WRN: InternalAuthenticationServiceException thrown and intercepted.");
        System.err.println("--- ERR: " + e.getMessage() + "\n\n");
        return ResponseEntity.status(HttpStatus.UNAUTHORIZED).body("--- ERR: " + e.getMessage());
    }

    @ExceptionHandler(BadCredentialsException.class)
    public ResponseEntity<?> handleBadCredentialsException(BadCredentialsException e) {
        System.out.println(x:"--- WRN: BadCredentialsException thrown and intercepted.");
        System.err.println("--- ERR: " + e.getMessage() + "\n\n");
        return ResponseEntity.status(HttpStatus.UNAUTHORIZED).body("--- ERR: " + e.getMessage());
    }


    //////// AUTHORIZATION EXCEPTION ////////
    @ExceptionHandler(AuthorizationDeniedException.class)
    public ResponseEntity<?> handleAuthorizationDeniedException(AuthorizationDeniedException e) {
        System.out.println(x:"--- WRN: AuthorizationDeniedException thrown and intercepted.");
        System.err.println("--- ERR: " + e.getMessage() + ": user is trying to access to a resource which is not his.\n\n");
        return ResponseEntity.status(HttpStatus.FORBIDDEN).body("--- ERR: " + e.getMessage() + ": the resource you are trying to access is not yours.");
    }

    @ExceptionHandler(AccessDeniedException.class)
    public ResponseEntity<?> handleAccessDeniedException(AccessDeniedException e) {
        System.out.println(x:"--- WRN: AccessDeniedException thrown and intercepted.");
        System.err.println(x:"--- ERR: ACCESS DENIED: user is trying to access to a resource which is not his.\n\n");
        return ResponseEntity.status(HttpStatus.FORBIDDEN).body(body:"--- ERR: ACCESS DENIED: the resource you are trying to access is not yours.");
    }
```

Figure 4.11: Snippet from the `GlobalExceptionHandler`: each exception is intercepted and mapped to a meaningful HTTP status and message.

4.3 RESTful API Documentation

This section summarizes all endpoints exposed by the WineAdvisor application. The API documentation shown in following snapshots was generated using Swagger UI. The RESTful APIs are organized by collections, with each controller handling a specific entity of the application (e.g., *wines*, *wineries*, *users*). As shown in following sections, for each collection, all available endpoints are listed, along with an example of usage for one representative endpoint. These examples include both the request setup and the server response, as displayed in the Swagger UI.

4.3.1 `/api/wines`

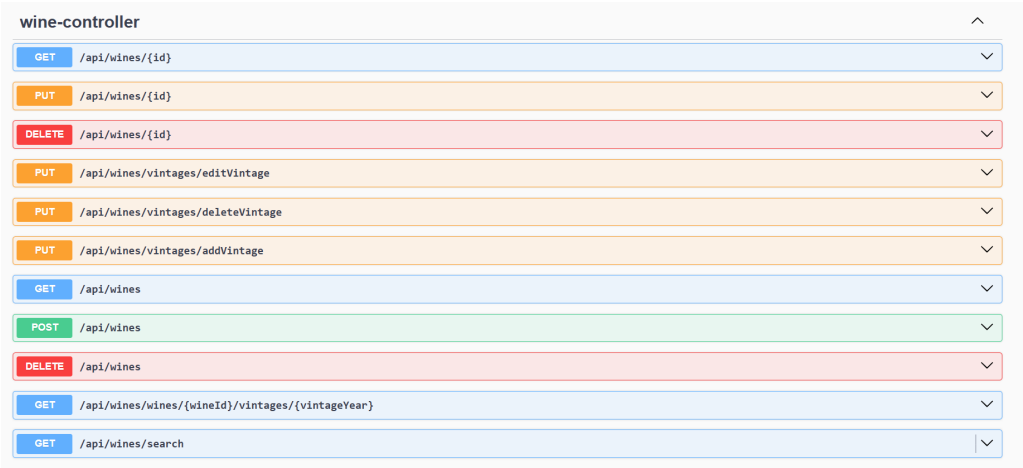


Figure 4.12: Overview of the available `/wines` endpoints.

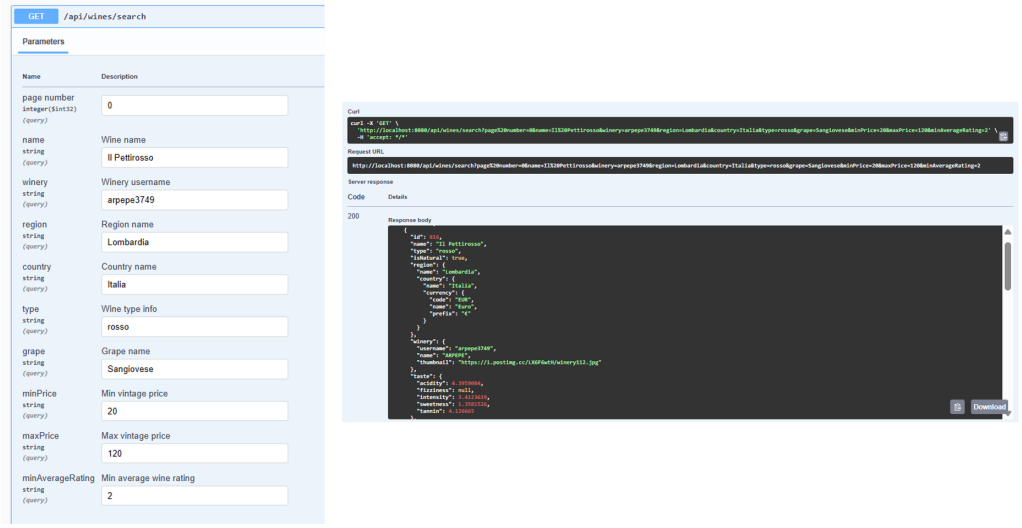


Figure 4.13: Example of request and response for the `/search GET` endpoint in the `/wines` endpoints.

4.3.2 /api/wineries

Figure 4.14: Overview of the available `/wineries` endpoints.

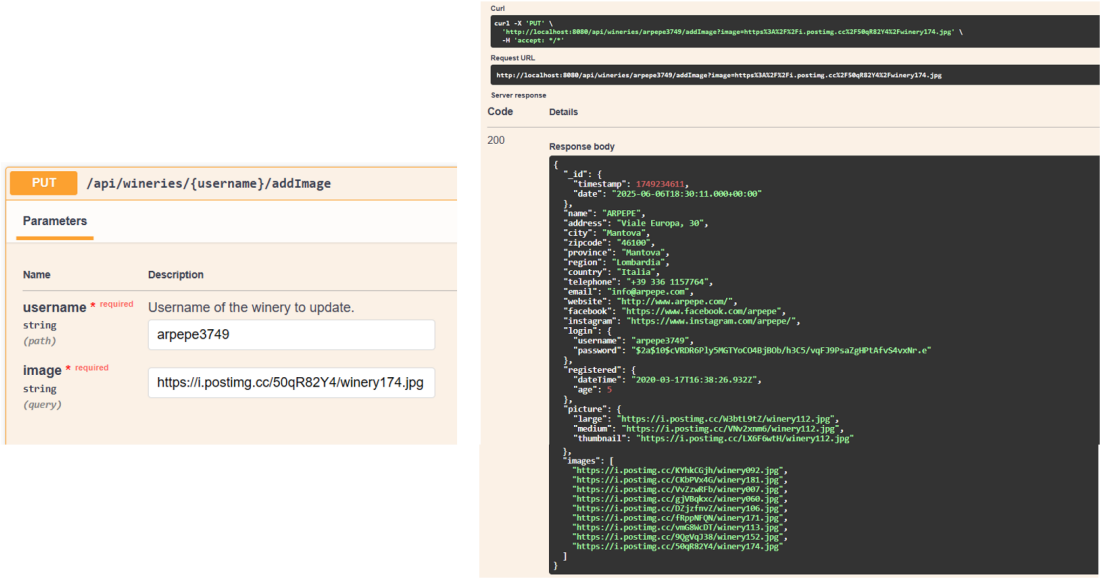


Figure 4.15: Example of request and response for the `/{username}/addImage PUT` endpoint in the `/wineries` endpoints.

4.3.3 /api/users

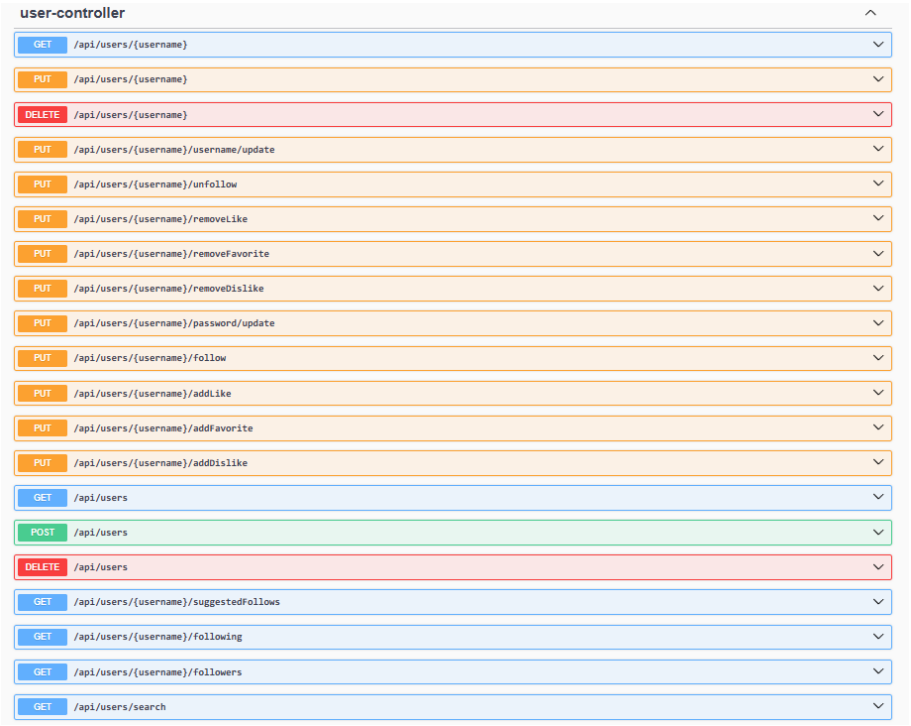


Figure 4.16: Overview of the available `/users` endpoints.

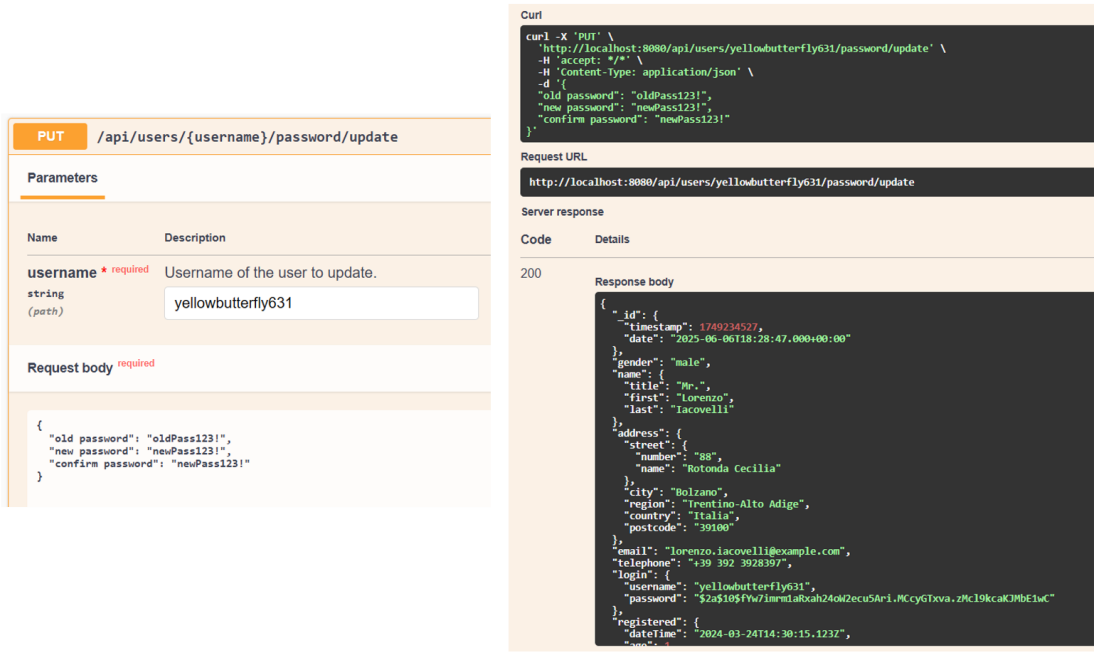


Figure 4.17: Example of request and response for the `/{username}/password/update PUT` endpoint in the `/users` controller.

4.3.4 /api/styles

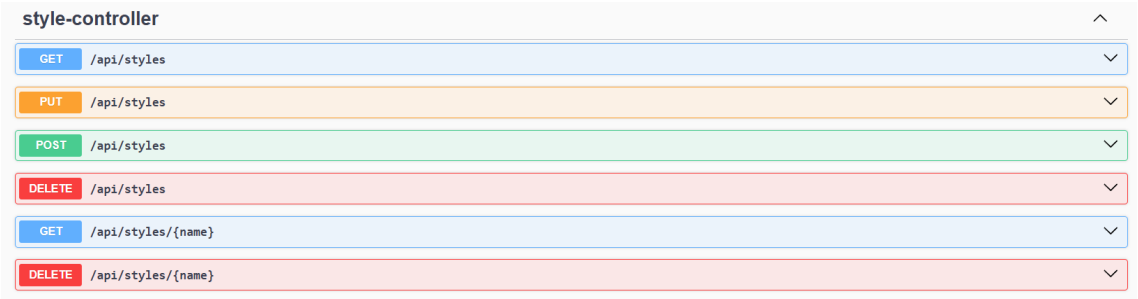


Figure 4.18: Overview of the available `/styles` endpoints.

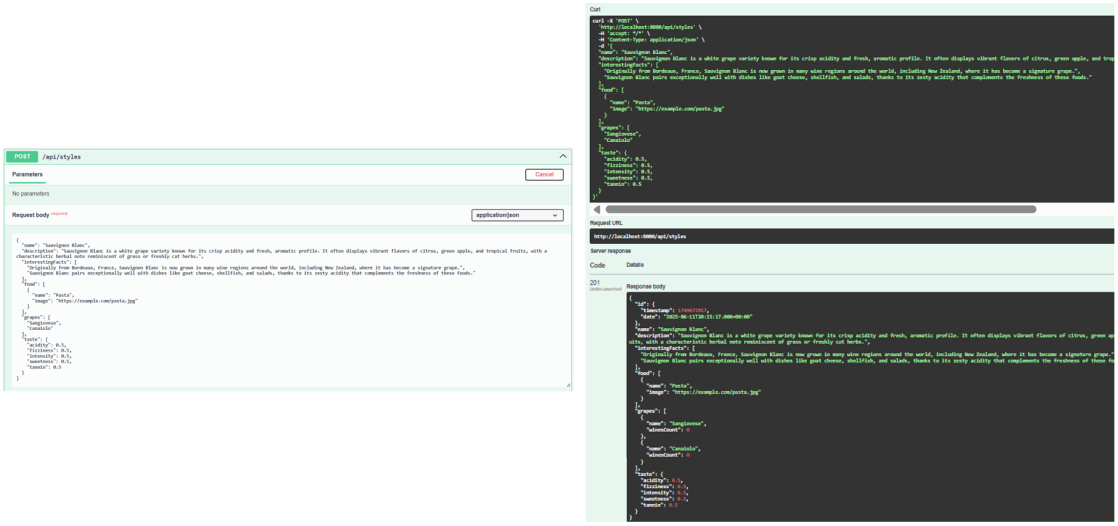


Figure 4.19: Example of request and response for the `POST` endpoint (used to create a new style) in the `/styles` controller.

4.3.5 /api/reviews

Figure 4.20: Overview of the available `/reviews` endpoints.

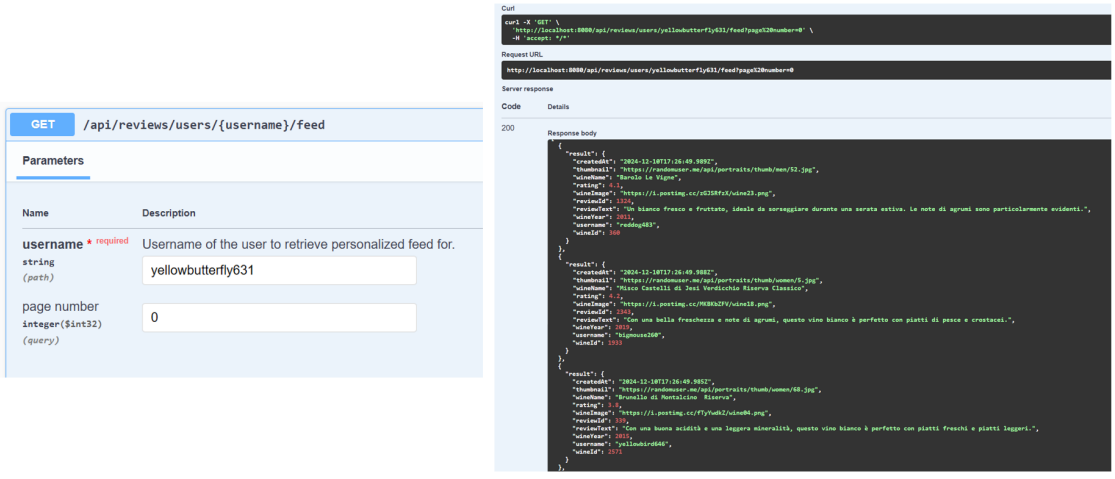


Figure 4.21: Example of request and response for the `/users/{username}/feed` `GET` endpoint in the `/reviews` controller.

4.3.6 /api/regions

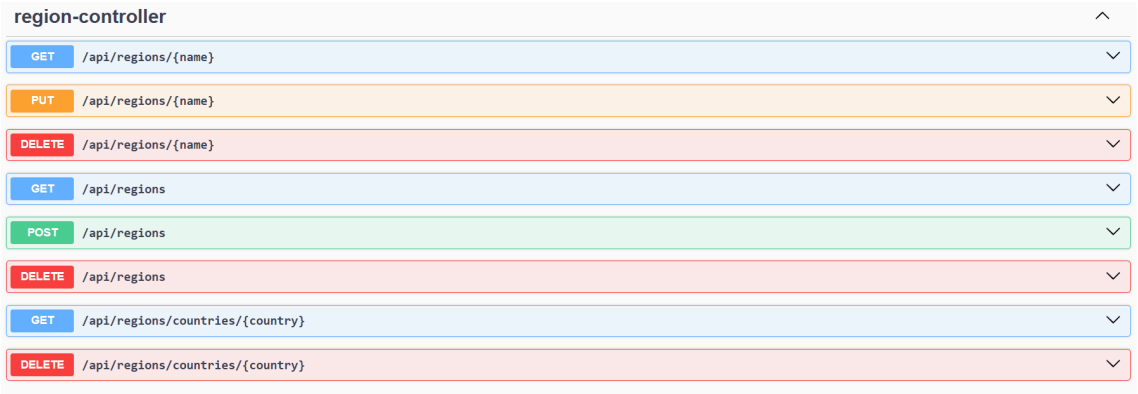


Figure 4.22: Overview of the available `/regions` endpoints.

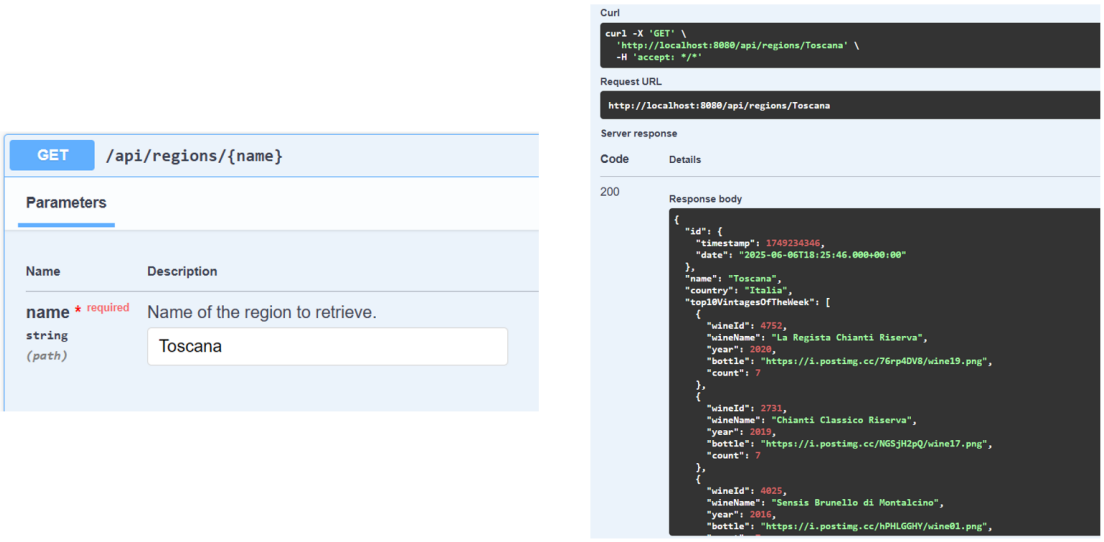


Figure 4.23: Example of request and response for the `/{name} GET` endpoint in the `/regions` controller.

4.3.7 /api/countries

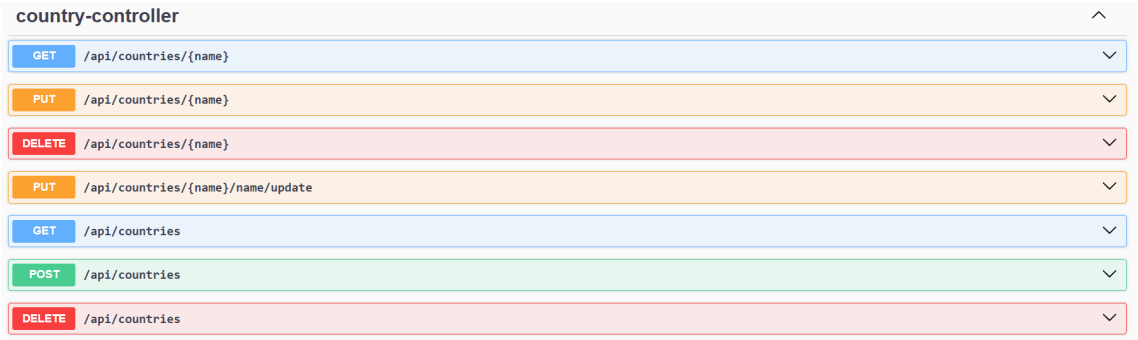


Figure 4.24: Overview of the available `/countries` endpoints.

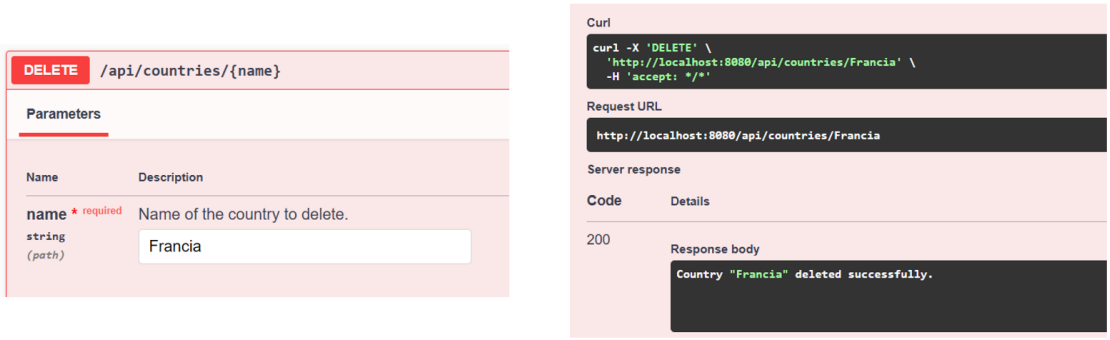


Figure 4.25: Example of request and response for the `/{name} DELETE` endpoint in the `/countries` controller.

4.3.8 /api/admins

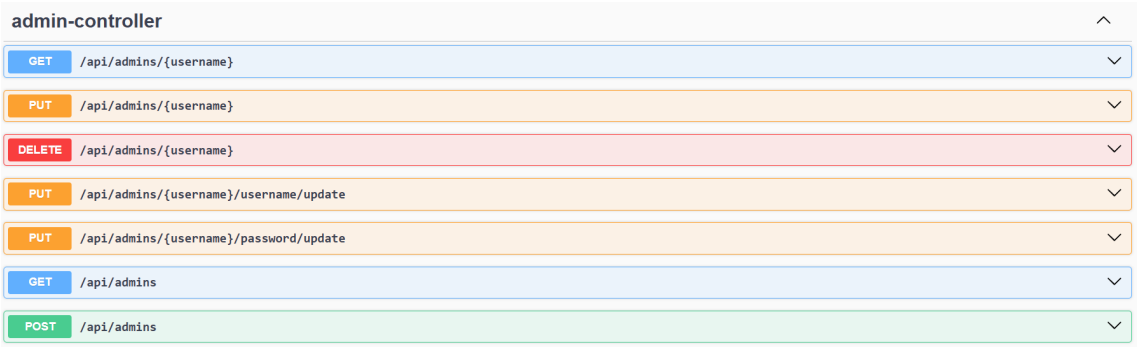


Figure 4.26: Overview of the available `/admins` endpoints.

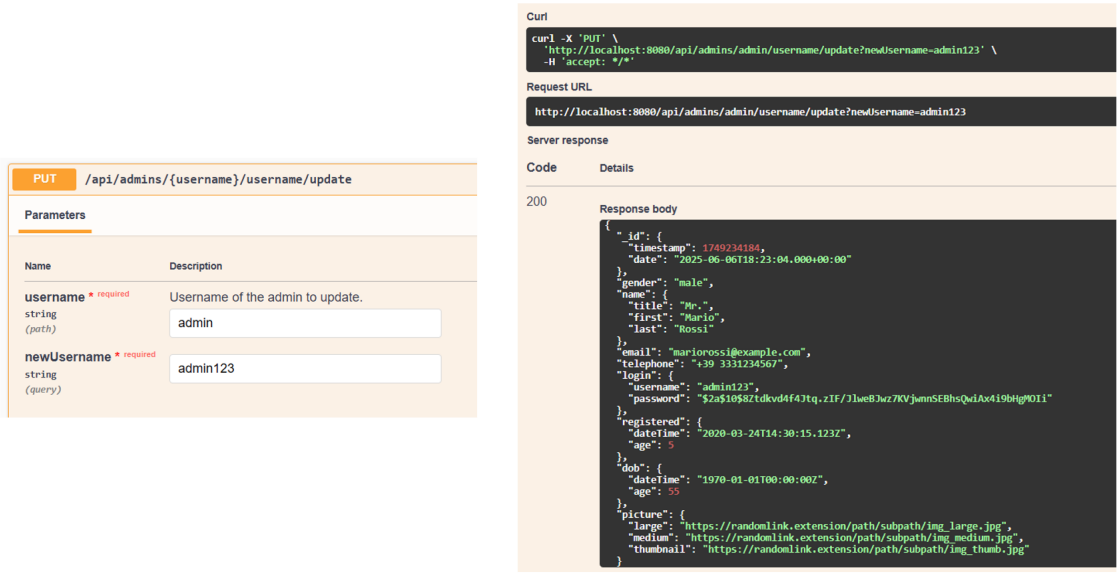


Figure 4.27: Example of request and response for the `/{username}/username/update PUT` endpoint in the `/admins` controller.

4.3.9 /api/analytics

Figure 4.28: Overview of the available `/analytics` endpoints.

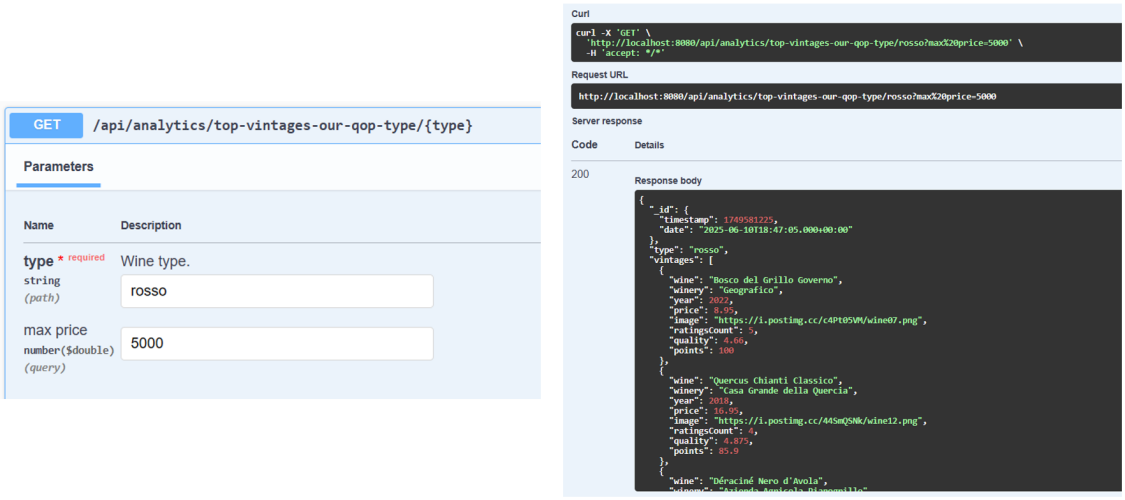


Figure 4.29: Example of request and response for the `/top-vintages-our-qop-type/{type} GET` endpoint in the `/analytics` controller.

4.3.10 `/api/authentication`

For this endpoint, the images of the requests and responses are provided using Postman, due to the fact that an authentication request needs to be non-authenticated in our application.

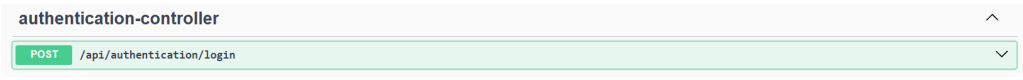


Figure 4.30: `/authentication` endpoint.

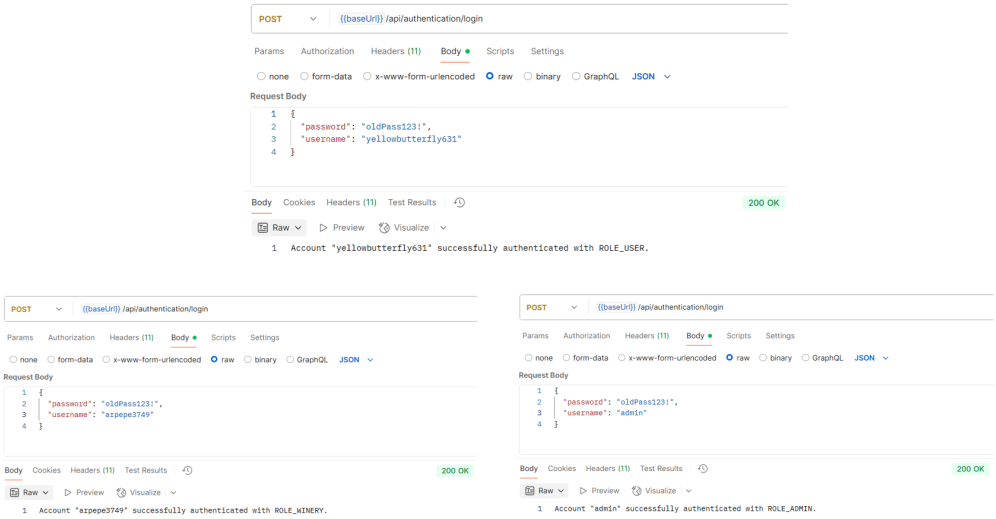


Figure 4.31: Example of 3 different Postman requests and responses for the `/login` `POST` endpoint in the `/authentication` controller.

4.4 MongoDB Operations

To provide a clear and organized structure, this section is divided into two parts: one dedicated to basic CRUD operations, and another focused on more complex queries such as aggregations and related analytics.

4.4.1 CRUD operations

CRUD operations have been implemented for each of the following collections: *users*, *reviews*, *wineries*, *wines*, *admins*, *countries*, *regions*, and *styles*. To avoid redundancy, we will not report all of them. Instead, we will use the *wines* collection as a representative example, providing code snapshots to illustrate the implementation.

Create a Wine

To create a new wine, the application performs the following steps, invoking the methods listed below.

- `public ResponseEntity<?> createWine(@RequestBody CreateWineDTO wine)`
- `public Wine addWine(CreateWineDTO createWineDTO, String wineryUsername)`
- `public Optional<Winery> findByLogin_Username(String username)`
- `public Optional<Country> findByName(String name)`
- `public Optional<Style> findByName(String name)`
- `public Wine save(Wine wine)`

The first method is an endpoint inside the `@RestController` and is responsible for handling the `POST` request when a winery user attempts to add a new wine. The method is secured with the `@Secured("ROLE_WINERY")` annotation to restrict access only to authenticated wineries. It extracts the username of the winery from the *Spring Security* context and delegates the creation logic to the service layer.

In the service method `addWine()`, several validation and composition steps are carried out before persisting the new wine into the database.

- The winery is retrieved by its username using `wineryRepository.findByLogin_Username()`. If not found, a `ResourceNotFoundException` is thrown.
- Similarly, the winery's country is retrieved from the `countries` collection.
- A new `Wine` object is instantiated and populated with fields from the `DTO` and additional embedded documents such as winery, style, region, and country.

- If the specified wine style exists in the database, its information is embedded inside the wine. Otherwise, the `style` field is left `null`.
- The `id` of the wine is generated using a custom `IdCounterService`.
- Initial statistics and timestamps are set, and the final object is saved to the *wines* collection via `wineRepository.save()`.

The following figures show some of the most relevant code snippets involved in the wine creation process.

```java
@PostMapping
@Secured( "ROLE_WINERY" )
public ResponseEntity<?> createWine(
        @NotNull(message = "New wine info cannot be null.") @Valid @RequestBody CreateWineDTO wine) {
    String wineryUsername = (String) SecurityContextHolder.getContext().getAuthentication().getName();
    Wine savedWine = wineService.addWine(wine, wineryUsername);
    return ResponseEntity.status(HttpStatus.CREATED).header(headerName:"Location", "/api/wines/" + savedWine.getId()).body(savedWine);
}
```

Figure 4.32: `@RestController` method handling the wine creation request.

```java
public Wine addWine(CreateWineDTO createWineDTO, String wineryUsername) throws ResourceNotFoundException {
    // Controllo che esista la winery
    Winery winery = wineryRepository.findByLogin_Username(wineryUsername).orElseThrow(
        () -> new ResourceNotFoundException("Winery not found with username: " + wineryUsername + ".")
    );

    // Controllo che esista il country nella collection country
    Country country = countryRepository.findByName(winery.getCountry()).orElseThrow(
        () -> new ResourceNotFoundException("Country not found with name: " + winery.getCountry() + ".")
    );

    Wine wine = new Wine();
    wine.setId(idCounterService.generateSequence(seqName:"wines"));
    wine.setName(createWineDTO.getName());
    wine.setType(createWineDTO.getType());
    wine.setIsNatural(createWineDTO.getIsNatural());

    wine.setTaste(taste:null);

    Optional<Style> style_to_find = styleRepository.findByName(createWineDTO.getStyle());
    if(style_to_find.isEmpty()) {
        wine.setStyle(style:null);
    } else {
        Style style = style_to_find.get();

        wine.setStyle(new StyleEmbedded());
        wine.getStyle().setName(style.getName());
        wine.getStyle().setDescription(style.getDescription());
        wine.getStyle().setInterestingFacts(style.getInterestingFacts());
        wine.getStyle().setFood(style.getFood());
        wine.getStyle().setGrapes(style.getGrapes());
    }
    wine.setVintages(new ArrayList<Vintage>());
    wine.setStatistics(new Statistics());
    wine.getStatistics().setRatingsAverage(ratingsAverage:0.0);
    wine.getStatistics().setRatingsCount((long) 0);

    wine.setWinery(new WineryEmbedded());
    wine.getWinery().setName(winery.getName());
    wine.getWinery().setUsername(winery.getLogin().getUsername());
    wine.getWinery().setThumbnail(winery.getPicture().getThumbnail());

    wine.setRegion(new RegionEmbedded());
    wine.getRegion().setName(winery.getRegion());
    wine.getRegion().setCountry(new CountryEmbedded());
    wine.getRegion().getCountry().setName(country.getName());
    wine.getRegion().getCountry().setCurrency(country.getCurrency());

    wine.setCreatedAt(Instant.now());

    return wineRepository.save(wine);
}
```

Figure 4.33: Service method that creates and saves a new wine.

Wine search

To search for wines based on multiple optional filters, the application performs the following steps, invoking the methods listed below.

- `public ResponseEntity<?> searchWines(...)`
- `public Page<Wine> searchWines(...)`
- `public Page<Wine> findByNameContainingIgnoreCaseAndWinery_UsernameContainingIgnoreCaseAndRegion_NameAndRegion_Country_NameAndTypeAndStyle_Grapes_NameAndStatistics_RatingsAverageGreaterThanEqualAndVintages_PriceBetween(...)`

The first method is a `GET` endpoint inside the `@RestController` that handles search requests sent by the client. It accepts a wide set of optional query parameters, including page number, wine name, winery username, region, country, type, grape variety, minimum price, maximum price and minimum average rating. If not provided, each parameter is initialized with a default value, such as empty strings for textual fields and default numeric bounds for price and rating. Basic validation is performed at the controller level: for example, numeric values such as prices and ratings must be non-negative, and the `minPrice` must not exceed the `maxPrice`. If any validation constraint is violated, a `BadRequestException` is thrown. Once validated and normalized, the controller delegates the request to the service layer.

The service method `searchWines(...)` builds a paginated query using the provided filters. Such method calls a derived query on the `WineRepository`, where all filters are combined using a single, compound method name to automatically generate the MongoDB query:

- `name` is matched as a case-insensitive sub-string;
- `winery` filters by the embedded field `winery.username`;
- `region` and `country` are matched exactly against `region.name` and `region.country.name`, respectively;
- `type` is matched exactly against the `type` field;
- `grape` filters on `style.grapes.name`;
- `minPrice` and `maxPrice` define an inclusive range on `vintages.price`;
- `minAverageRating` sets a lower bound on `statistics.ratingsAverage`.

Pagination is applied using a fixed-size `PageRequest`. After executing the query, the result page is validated using the `checkReturnedPage()` utility method:

- if no wines match the given filters, a `ResourceNotFoundException` is thrown;

- if the requested page index exceeds the number of available pages, a `BadRequestException` is thrown;
- if successful, the service returns a `Page<Wine>` object to the controller, which then wraps it in a `ResponseEntity` with status code `200 OK`.

The following figures show some of the most relevant code snippets involved in this process.

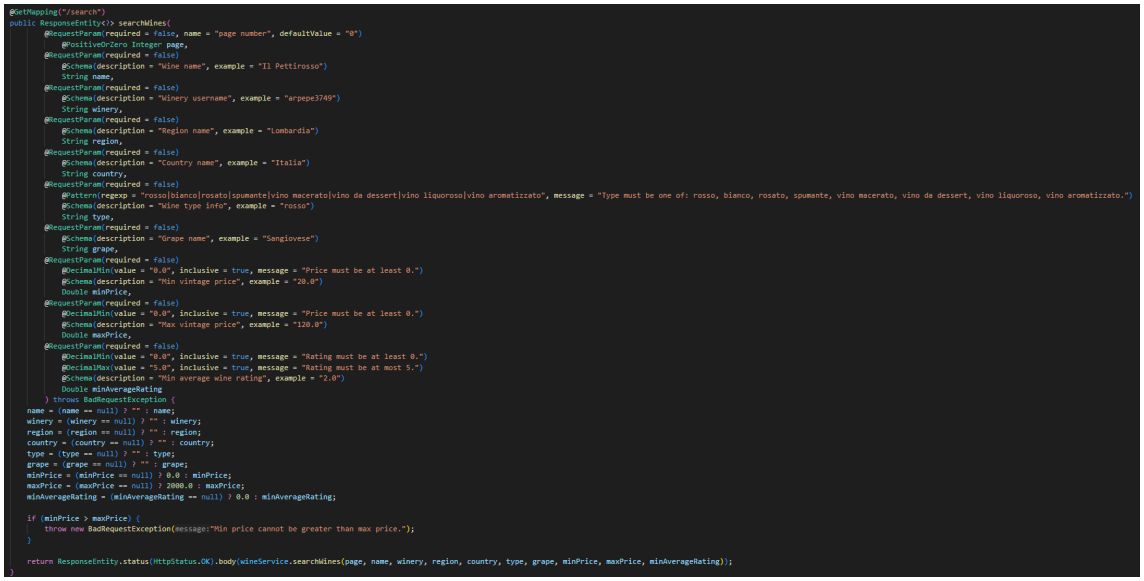


Figure 4.34: `@RestController` method handling the wines search request.

Figure 4.35: Service method that searches for wines.

```
private void checkReturnedPage(Page<Wine> page, String notFoundMessage) throws ResourceNotFoundException, BadRequestException {
    if (page.getTotalElements() == 0) {
        throw new ResourceNotFoundException(notFoundMessage);
    }
    if (page.getPageable().getPageNumber() >= page.getTotalPages()) {
        throw new BadRequestException(message:"Page requested too high.");
    }
}
```

Figure 4.36: Method used to validate the results of a paginated wine search, ensuring that the requested page exists and contains data.

Update a Wine

To update the details of an existing wine, the application performs the following steps, invoking the methods listed below.

- `public ResponseEntity<?> updateWine(@RequestBody @Valid UpdateWineDTO wine)`
- `public Wine updateWine(UpdateWineDTO updatedWine, String username)`
- `public Optional<Wine> findById(String id)`
- `public Optional<Wine> findByIdAndWinery_Username(String id, String username)`
- `public Optional<Style> findByName(String name)`
- `public ArrayList<User> findByReviews_WineId_Id(String id)`
- `public ArrayList<Review> findByWineId_Id(String id)`
- `public Wine save(Wine wine)`

The first method is a `PUT` endpoint defined in the `@RestController`, secured with the `@Secured("ROLE_WINERY")` annotation to allow access only to authenticated wineries. It extracts the username from the *Spring Security* context and delegates the update operation to the service layer.

In the service method `updateWine()`, the following steps are performed.

- The target wine is retrieved by its ID. If not found, a `ResourceNotFoundException` is thrown.
- A secondary check verifies that the wine belongs to the winery attempting the update by querying `findByIdAndWinery_Username()`. If the match fails, a `ResourceNotFoundException` is raised.
- The wine's main fields such as `name`, `type`, and `isNatural` are updated.
- If a valid style is found in the database, it is embedded into the wine object; otherwise, the `style` field is set to `null`.
- The updated wine name is propagated to all `ReviewEmbedded` objects in the *users* collection and to all reviews in the *reviews* collection to ensure data consistency.
- After all updates are applied, the wine is saved back into the *wines* collection using `wineRepository.save()`.

The update operation is marked with `@Transactional` to ensure atomicity. If any part of the process fails, the transaction is rolled back.

The following figures show some of the most relevant code snippets involved in the wine update process.

```java
@PutMapping
@Secured({ "ROLE_WINERY" })
public ResponseEntity<?> updateWine(
            @RequestBody @Valid UpdateWineDTO wine) {
    // Prendo username della winery che vuole modificare il vino
    String username = SecurityContextHolder.getContext().getAuthentication().getName();
    Wine updatedWine = wineService.updateWine(wine, username);
    return ResponseEntity.status(HttpStatus.OK).body(updatedWine);
}
```

Figure 4.37: `@RestController` method handling the wine update request.

```java
@Transactional
public Wine updateWine(UpdateWineDTO updatedWine, String username) throws ResourceNotFoundException {
    return wineRepository.findById(updatedWine.getWineId())
        .map(wine -> {
            // Controllo che il vino sia della winery username
            if(wineRepository.findByIdAndWinery_Username(updatedWine.getWineId(), username).isEmpty()){
                throw new ResourceNotFoundException("Winery " + username + " does not own wine with id " + updatedWine.getWineId() + ".");
            }

            wine.setName(updatedWine.getName());
            wine.setType(updatedWine.getType());
            wine.setIsNatural(updatedWine.getIsNatural());

            Optional<Style> style_to_find = styleRepository.findByName(updatedWine.getStyle());
            if(style_to_find.isEmpty()) {
                wine.setStyle(style:null);
            } else {
                Style style = style_to_find.get();

                wine.setStyle(new StyleEmbedded());
                wine.getStyle().setName(style.getName());
                wine.getStyle().setDescription(style.getDescription());
                wine.getStyle().setInterestingFacts(style.getInterestingFacts());
                wine.getStyle().setFood(style.getFood());
                wine.getStyle().setGrapes(style.getGrapes());
            }

            // Devo aggiornare il nome anche nelle reviews embedded nella collection users
            ArrayList<User> users = userRepository.findByReviews_WineId_Id(updatedWine.getWineId());
            if (!users.isEmpty()){
                for (User user : users){
                    for (ReviewEmbedded review : user.getReviews()){
                        if (review.getWineId().getId().equals(updatedWine.getWineId())){
                            review.getWineId().setName(updatedWine.getName());
                        }
                    }
                    userRepository.save(user);
                }
            }

            // Devo aggiornare il nome anche nelle reviews della collection reviews
            ArrayList<Review> reviews = reviewRepository.findByWineId_Id(updatedWine.getWineId());
            if (!reviews.isEmpty()){
                for (Review review : reviews){
                    review.getWineId().setName(updatedWine.getName());
                    reviewRepository.save(review);
                }
            }

            return wineRepository.save(wine);
        })
        .orElseThrow(
            () -> new ResourceNotFoundException("Wine with id " + updatedWine.getWineId() + " not found.")
        );
}
```

Figure 4.38: Service method that updates a wine.

Delete a Wine

To delete a wine, the following steps are performed by the application, invoking the methods listed below.

- `public ResponseEntity<?> deleteWine(@PathVariable Long id)`
- `public void deleteWineById(Long wineId, String username)`
- `public Optional<Wine> findById(Long id)`
- `public Optional<Wine> findByIdAndWinery_Username(Long id, String username)`
- `public ArrayList<Review> findByWineId_Id(Long id)`
- `public Iterable<User> findAll()`
- `public void deleteAllByWineId_Id(Long id)`
- `public void deleteById(Long id)`

The process starts from a `DELETE` endpoint accessible only to users with roles `ROLE_ADMIN` or `ROLE_WINERY`, secured using the `@Secured` annotation. The ID of the wine to be deleted is extracted from the URL path, and the authenticated user's username is obtained via the *Spring Security* context. The deletion is then delegated to the service layer.

In the `deleteWineById()` service method, the application performs the following operations.

- Checks whether the wine with the given ID exists in the database. If not, a `ResourceNotFoundException` is thrown.
- Verifies that the wine is owned by the winery making the request. If not, another `ResourceNotFoundException` is raised.
- Retrieves all reviews associated with the wine using `findByWineId_Id()`, and removes each review:
  - From the embedded list of reviews in each *user* document.
  - From the lists of liked and disliked reviews of *users*, by checking if the review ID matches.
  - If a user was modified, they are saved back to the database using `userRepository.save()`.
- Deletes all reviews of the wine from the *reviews* collection via `deleteAllByWineId_Id()`.
- Finally, the wine is removed from the *wines* collection using `deleteById()`.

This method is annotated with `@Transactional` to ensure that all deletions and updates are performed atomically. If any part of the process fails, the transaction is rolled back, preserving data consistency.

The code snippets below show the core components of the wine deletion functionality.

```java
@DeleteMapping("/{id}")
@Secured({ "ROLE_ADMIN", "ROLE_WINERY" })
public ResponseEntity<?> deleteWine(
        @PathVariable @NotNull(message = "ID cannot be null.") @Positive(message = "ID cannot be negative.") Long id) {
    // Prendo username della winery che vuole aggiungere la vintage al vino
    String username = SecurityContextHolder.getContext().getAuthentication().getName();
    wineService.deleteWineById(id, username);
    return ResponseEntity.status(HttpStatus.OK).body(body:"Wine deleted successfully.");
}
```

Figure 4.39: `@RestController` method handling the wine deletion request.

```java
@Transactional
public void deleteWineById(Long wineId, String username) throws ResourceNotFoundException {
    // Controllo che il vino specificato esista
    if (wineRepository.findById(wineId).isEmpty()){
        throw new ResourceNotFoundException("Wine with id " + wineId + " not found.");
    }

    // Controllo che il vino sia della winery username
    if(wineRepository.findByIdAndWinery_Username(wineId, username).isEmpty()){
        throw new ResourceNotFoundException("Winery " + username + " does not own wine with id " + wineId + ".");
    }

    // Se elimino un vino, devo eliminare anche tutte le recensioni fatte su quel vino
    // e togliere l'id di quelle recensioni dagli array "likes"/"dislikes" degli users.
    ArrayList<Review> reviews_to_delete = reviewRepository.findByWineId_Id(wineId);
    ArrayList<User> users = (ArrayList<User>) userRepository.findAll();
    for (User user : users){
        boolean modified = user.getReviews().removeIf(r -> r.getWineId().getId().equals(wineId));
        if (user.getLikes().removeIf(likeId -> reviews_to_delete.stream().anyMatch(r -> r.getId().equals(likeId)))) {
            modified = true;
        }
        if (user.getDislikes().removeIf(dislikeId -> reviews_to_delete.stream().anyMatch(r -> r.getId().equals(dislikeId)))) {
            modified = true;
        }
        if (modified) {
            userRepository.save(user);
        }
    }

    reviewRepository.deleteAllByWineId_Id(wineId);
    wineRepository.deleteById(wineId);
}
```

Figure 4.40: Service method that deletes a wine.

4.4.2 MongoDB Aggregations for Analytics

Before exploring the details of the *aggregation pipelines* implemented in the system, it is important to clarify how analytics are handled within the application. Rather than computing aggregations on demand, the system **periodically** executes predefined aggregation pipelines and stores their results in dedicated collections within the MongoDB database. This design choice improves **performance** and **scalability**, allowing clients to simply retrieve precomputed analytics through specific `GET` endpoints.

Each aggregation is therefore associated with a corresponding collection that is updated at regular intervals, ensuring that the data remains relevant and up-to-date without introducing overhead during query execution.

Daily Update of Vintage Statistics

This aggregation pipeline is scheduled to run daily at midnight. It updates the `statistics` field inside each `vintage` field within the documents of the *wines* collection. The statistics include the number of ratings (`ratings_count`) and the average rating (`ratings_average`) computed from the *reviews* collection. The pipeline performs the following steps:

1. `$unwind` the `vintages` array to process each vintage individually.
2. `$lookup` reviews that match both the wine's ID and the vintage year.
3. `$project` a new field `statistics` containing the computed values.

To ensure the performance of this pipeline, a compound index on the fields `wine_id.id` and `wine_id.year` is created in the *reviews* collection. The aggregation results are then used to update the corresponding vintage `statistics` field in the *wines* collection via individual update operations.

```
@Transactional
@Scheduled(cron = "0 0 0 * * ?")    // Ogni giorno a mezzanotte
protected void updateStatisticsPerVintage() {
    System.out.println(x:"--- INFO: Declaring aggregation pipeline stages for vintage statistics update.");
    List<Document> pipeline = Arrays.asList(
            new Document(key:"$unwind", value:"$vintages"),
            new Document(key:"$lookup",
                    new Document(key:"from", value:"reviews")
                            .append(key:"let",
                                    new Document(key:"wineId", value:"$_id")
                                            .append(key:"vintageYear", value:"$vintages.year"))
                            .append(key:"pipeline", Arrays.asList(new Document(key:"$match",
                                    new Document(key:"$expr",
                                            new Document(key:"$and", Arrays.asList(
                                                    new Document(key:"$eq", Arrays.asList(...a:"$wine_id.id", "$$wineId")),
                                                    new Document(key:"$eq",
                                                            Arrays.asList(...a:"$wine_id.year", "$$vintageYear")))))),
                                    new Document(key:"$project",
                                            new Document(key:"rating", value:1L))))
                            .append(key:"as", value:"matched_reviews")),
            new Document(key:"$project",
                    new Document(key:"_id", value:1L)
                            .append(key:"year", value:"$vintages.year")
                            .append(key:"statistics",
                                    new Document(key:"ratings_count",
                                            new Document(key:"$size", value:"$matched_reviews"))
                                            .append(key:"ratings_average",
                                                    new Document(key:"$cond", Arrays.asList(
                                                            new Document(key:"$gt",
                                                                    Arrays.asList(new Document(key:"$size",
                                                                            value:"$matched_reviews"), 0L)),
                                                            new Document(key:"$avg", value:"$matched_reviews.rating"),
                                                            new BsonNull())))))));
    // Assicuro che esistano gli indici FUNZIONALI alla pipeline.
    mongoTemplate
        .indexOps(collectionName:"reviews")
        .ensureIndex(
            new Index()
                .on(key:"wine_id.id", Sort.Direction.ASC)
                .on(key:"wine_id.year", Sort.Direction.ASC)
                .named(name:"index_for_vintage_statistics_update")
        );
    System.out.println(x:"--- INFO: Ensured index on fields \"wine_id.id\" and \"wine_id.year\" for collection \"reviews\".");
    System.out.println(x:"--- INFO: Executing aggregation pipeline for vintage statistics update...");
    AggregateIterable<Document> results = mongoTemplate.getCollection(collectionName:"wines").aggregate(pipeline).allowDiskUse(allowDiskUse:true);
    System.out.println(x:"--- INFO: Aggregation pipeline executed successfully.");
    System.out.println(x:"--- INFO: Processing results from aggregation pipeline...");
    for (Document doc : results) {
        Object idObj = doc.get(key:"_id");
        Object yearObj = doc.get(key:"year");
        Document stats = (Document) doc.get(key:"statistics");
        // Verifico che tutti i campi siano presenti
        if (idObj == null || yearObj == null || stats == null) {
            continue;
        }
        Long wineId = ((Number) idObj).longValue();
        int year = ((Number) yearObj).intValue();
        // Query per selezionare il vino e la vintage giusta
        Query query = new Query(Criteria.where(key:"_id").is(wineId)
            .and(key:"vintages.year").is(year));
        // Update per sovrascrivere il campo statistics nella vintage corretta
        Update update = new Update().set(key:"vintages.$.statistics", stats);
        mongoTemplate.updateFirst(query, update, collectionName:"wines");
    }
    System.out.println(x:"--- INFO: Vintage statistics updated successfully.\n\n");
```

Figure 4.41: `updateStatisticsPerVintage()`

Daily Update of Wine Statistics

This aggregation pipeline also runs daily at midnight and is responsible for updating the overall `statistics` field of each document of the *wines* collection (per wine, not per vintage). It calculates the total number of ratings and their average, based on all the reviews referring to that wine. The steps involved are:

1. `$lookup` reviews for each wine using the wine's ID.
2. `$project` the new `statistics` field for each wine.
3. `$merge` the updated statistics back into the *wines* collection, overwriting existing values when matched.

A single-field index on `wine_id.id` is ensured for the *reviews* collection to support efficient matching. Unlike the vintage update, this pipeline uses the `$merge` operator to apply bulk updates directly within the aggregation framework.

```
@Scheduled(cron = "0 0 0 * * ?")    // Ogni giorno a mezzanotte
protected void updateStatisticsPerWine() {
    System.out.println(x:"--- INFO: Declaring aggregation pipeline stages for wine statistics update.");
    List<Document> pipeline = Arrays.asList(
            new Document(key:"$project",
                    new Document(key:"_id", value:1L)),
            new Document(key:"$lookup",
                    new Document(key:"from", value:"reviews")
                            .append(key:"localField", value:"_id")
                            .append(key:"foreignField", value:"wine_id.id")
                            .append(key:"as", value:"matched_reviews")),
            new Document(key:"$project",
                    new Document(key:"_id", value:1L)
                            .append(key:"statistics",
                                    new Document(key:"ratings_count",
                                            new Document(key:"$size", value:"$matched_reviews"))
                                            .append(key:"ratings_average",
                                                    new Document(key:"$cond",
                                                            Arrays.asList(
                                                                    new Document(key:"$gt",
                                                                            Arrays.asList(new Document(key:"$size",
                                                                                    value:"$matched_reviews"), 0L)),
                                                                    new Document(key:"$avg", value:"$matched_reviews.rating"),
                                                                    new BsonNull()))))),
            new Document(key:"$merge",
                    new Document(key:"into", value:"wines")
                            .append(key:"on", value:"_id")
                            .append(key:"whenMatched", Arrays.asList(new Document(key:"$set",
                                    new Document(key:"statistics", value:"$$new.statistics"))))
                            .append(key:"whenNotMatched", value:"discard")));


    // Assicuro che esistano gli indici FUNZIONALI alla pipeline.
    mongoTemplate
        .indexOps(collectionName:"reviews")
        .ensureIndex(
            new Index()
                .on(key:"wine_id.id", Sort.Direction.ASC)
                .named(name:"index_for_wine_statistics_update")
        );
    System.out.println(x:"--- INFO: Ensured index on field \"wine_id.id\" for collection \"reviews\".");


    System.out.println(x:"--- INFO: Executing aggregation pipeline for wine statistics update...");
    mongoTemplate.getCollection(collectionName:"wines").aggregate(pipeline).allowDiskUse(allowDiskUse:true).toCollection();
    System.out.println(x:"--- INFO: Wine statistics updated successfully.\n\n");
}
```

Figure 4.42: `updateStatisticsPerWine()`

Daily Update of Wines Count per Grape in *Styles* and *Wines* Collections

This aggregation pipeline is scheduled to run daily at midnight and updates the `wines_count` field associated with each grape variety both within the nested `style.grapes` array in the *wines* collection and in the `grapes` array of the *styles* collection. The process consists of:

1. `$unwind` the `style.grapes` array in the *wines* collection to handle each grape variety separately.
2. `$group` by the grape name to count how many wines contain that grape (`wines_count`).

The aggregation results are iterated to:

- Update all matching grape entries in `style.grapes` of the *wines* collection by setting the new `wines_count`.
- Update the `grapes` array inside the *styles* collection analogously.

The updates leverage `arrayFilters` to target the correct grape entries within arrays, and `updateMulti` ensures multiple documents are updated where applicable. This approach guarantees that the grape wine counts remain consistent and up-to-date across related collections, facilitating accurate analytics and reporting.

```
@Transactional
@Scheduled(cron = "0 0 0 * * ?")    // Ogni giorno a mezzanotte
protected void updateWinesCountInGrapesPerWine() {
    System.out.println(x:"--- INFO: Declaring aggregation pipeline stages for grapes count update.");
    List<Document> pipeline = Arrays.asList(
            new Document(key:"$unwind",
                    new Document(key:"path", value:"$style.grapes")),
            new Document(key:"$group",
                    new Document(key:"_id", value:"$style.grapes.name")
                            .append(key:"wines_count",
                                    new Document(key:"$sum", value:1L))));


    System.out.println(x:"--- INFO: Executing aggregation pipeline for grapes count update...");
    AggregateIterable<Document> results = mongoTemplate.getCollection(collectionName:"wines").aggregate(pipeline).allowDiskUse(allowDiskUse:true);
    System.out.println(x:"--- INFO: Aggregation pipeline executed successfully.");


    System.out.println(x:"--- INFO: Processing results from aggregation pipeline...");
    for (Document doc : results) {
        String grapeName = doc.getString(key:"_id");
        Long winesCount = doc.getLong(key:"wines_count");

        if (grapeName == null || winesCount == null) {
            continue;
        }

        // Seleziona tutti i vini che contengono questa grape
        Query query_on_wines = new Query(Criteria.where(key:"style.grapes.name").is(grapeName));
        Query query_on_styles = new Query(Criteria.where(key:"grapes.name").is(grapeName));

        // Aggiorna ogni entry dentro style.grapes con quel nome
        Update update_on_wines = new Update().set(key:"style.grapes.$[g].wines_count", winesCount);
        Update update_on_styles = new Update().set(key:"grapes.$[g].wines_count", winesCount);

        // Definisce il filtro per gli array ("arrayFilters")
        update_on_wines.filterArray(identifier:"g.name", grapeName);
        update_on_styles.filterArray(identifier:"g.name", grapeName);

        // Applica l'update su tutti i documenti che contengono quella grape
        mongoTemplate.updateMulti(query_on_wines, update_on_wines, collectionName:"wines");
        mongoTemplate.updateMulti(query_on_styles, update_on_styles, collectionName:"styles");
    }

    System.out.println(x:"--- INFO: Grapes count updated successfully.\n\n");
}
```

Figure 4.43: `updateWinesCountInGrapesPerWine()`

Weekly Update of Top 10 Vintages per Region

This aggregation pipeline is scheduled to execute every Monday at 2:00 AM. It computes the **top 10 most reviewed vintages** for each region and updates the *regions* collection accordingly. A vintage is defined as a unique pair of wine and year. The pipeline operates on the *reviews* collection and proceeds through the following stages:

1. `$lookup` joins the *reviews* collection with the *wines* collection by matching the id field, enriching each review with the wine region.
2. `$unwind` flattens the joined results.
3. `$project` extracts only the relevant fields: wine information (ID, name, image, year) and the region.
4. `$group` clusters the data by wine ID and vintage year, computing the number of reviews for each combination.
5. `$sort` ranks the combinations by review count in descending order.
6. A second `$group` aggregates the data per region, pushing the vintage details into an array.
7. `$project` slices the array to retain only the top 10 vintages per region.
8. A final `$project` renames fields for clarity and consistency, preparing documents with fields `name` (region name) and `top_10_vintages_of_the_week`.
9. `$merge` writes the output into the *regions* collection, updating existing entries by matching on `name`, and discarding any unmatched records.

Additionally, the method ensures that a unique index on `name` in the *regions* collection is in place to support this pipeline.

This aggregation ensures that each document in the *regions* collection reflects the current top 10 most popular vintages based on user reviews, enabling efficient retrieval of high-level insights into regional trends in wine appreciation.

```java
@Scheduled(cron = "0 0 2 * * MON")      // Ogni lunedì alle 2 di notte
protected void updateTop10VintagesPerRegion() {
    System.out.println(x:"--- INFO: Declaring aggregation pipeline stages for collection \"regions\".");
    List<Document> pipeline = Arrays.asList(
            new Document(key:"$lookup",
                    new Document(key:"from", value:"wines")
                            .append(key:"localField", value:"wine_id.id")
                            .append(key:"foreignField", value:"_id")
                            .append(key:"as", value:"result")),
            new Document(key:"$unwind",
                    new Document(key:"path", value:"$result")),
            new Document(key:"$project",
                    new Document(key:"_id", value:0L)
                            .append(key:"wine_id", value:"$wine_id.id")
                            .append(key:"wine_name", value:"$wine_id.name")
                            .append(key:"wine_image", value:"$wine_id.image")
                            .append(key:"year", value:"$wine_id.year")
                            .append(key:"region", value:"$result.region.name")),
            new Document(key:"$group",
                    new Document(key:"_id",
                            new Document(key:"wine_id", value:"$wine_id")
                                    .append(key:"year", value:"$year"))
                            .append(key:"wine_name",
                                    new Document(key:"$first", value:"$wine_name"))
                            .append(key:"wine_image",
                                    new Document(key:"$first", value:"$wine_image"))
                            .append(key:"region",
                                    new Document(key:"$first", value:"$region"))
                            .append(key:"count",
                                    new Document(key:"$sum", value:1L))),
            new Document(key:"$sort",
                    new Document(key:"count", -1L)),
            new Document(key:"$group",
                    new Document(key:"_id", value:"$region")
                            .append(key:"vintages",
                                    new Document(key:"$push",
                                            new Document(key:"wine_id", value:"$_id.wine_id")
                                                    .append(key:"wine_name", value:"$wine_name")
                                                    .append(key:"year", value:"$_id.year")
                                                    .append(key:"bottle", value:"$wine_image")
                                                    .append(key:"count", value:"$count")))),
            new Document(key:"$project",
                    new Document(key:"_id", value:0L)
                            .append(key:"name", value:"$_id")
                            .append(key:"vintages",
                                    new Document(key:"$slice", Arrays.asList(...a:"$vintages", 10L)))),
            new Document(key:"$project", new Document(key:"_id", value:0)
                    .append(key:"name", value:1)
                    .append(key:"top_10_vintages_of_the_week", value:"$vintages")),
            new Document(key:"$merge", new Document(key:"into", value:"regions")
                    .append(key:"on", value:"name")
                    .append(key:"whenMatched", value:"merge")
                    .append(key:"whenNotMatched", value:"discard")));
    // Assicuro che esistano gli indici NECESSARI alla pipeline.
    mongoTemplate
        .indexOps(collectionName:"regions")
        .ensureIndex(
            new Index()
                .on(key:"name", Sort.Direction.ASC)
                .named(name:"on_name_UNIQUE")
                .unique()
        );
    System.out.println(x:"--- INFO: Ensured UNIQUE index on field \"name\" for collection \"regions\".");
    System.out.println(x:"--- INFO: Executing aggregation pipeline for collection \"regions\"...");
    mongoTemplate.getCollection(collectionName:"reviews").aggregate(pipeline).allowDiskUse(allowDiskUse:true).toCollection();
    System.out.println(x:"--- INFO: Collection \"regions\" updated successfully.\n\n");
}
```

Figure 4.44: `updateTop10VintagesPerRegion()`

Weekly Update of Top 100 Vintages per Country

This aggregation pipeline is scheduled to run every Monday at 2:00 AM. It computes the top 100 most reviewed wine vintages for each country and updates the *countries* collection accordingly. Each vintage is defined as a unique pair of wine and year. The pipeline works on the *reviews* collection and consists of the following stages:

1. `$lookup` joins reviews with wines data by matching wine ids, adding information about the wine country.
2. `$unwind` flattens the resulting array from the join.
3. `$project` retains only the relevant fields: wine ID, name, image, year, and country.
4. `$group` aggregates by wine ID and vintage year, counting the number of reviews for each unique combination.
5. `$sort` ranks these combinations by descending review count.
6. A second `$group` collects vintage information for each country into an array.
7. `$project` slices this array to retain only the top 100 vintages.
8. Another `$project` renames the fields to store the data as `name` (country) and `top_100_vintages_of_the_week`.
9. `$merge` writes the results into the *countries* collection, merging on the `name` field and discarding non-matching documents.

Additionally, the method ensures that a unique index on `name` in the *countries* collection is in place to support this pipeline.

This process ensures that each document in the *countries* collection accurately reflects the most popular wine vintages in that country based on user activity, enabling efficient retrieval of high-level insights into national trends in wine appreciation.

```java
@Scheduled(cron = "0 0 2 * * MON")      // Ogni lunedì alle 2 di notte
protected void updateTop100VintagesPerCountry() {
    System.out.println(x:"--- INFO: Declaring aggregation pipeline stages for collection \"countries\".");
    List<Document> pipeline = Arrays.asList(
            new Document(key:"$lookup",
                    new Document(key:"from", value:"wines")
                            .append(key:"localField", value:"wine_id.id")
                            .append(key:"foreignField", value:"_id")
                            .append(key:"as", value:"result")),

            new Document(key:"$unwind",
                    new Document(key:"path", value:"$result")),

            new Document(key:"$project",
                    new Document(key:"_id", value:0L)
                            .append(key:"wine_id", value:"$wine_id.id")
                            .append(key:"wine_name", value:"$wine_id.name")
                            .append(key:"wine_image", value:"$wine_id.image")
                            .append(key:"year", value:"$wine_id.year")
                            .append(key:"country", value:"$result.region.country.name")),

            new Document(key:"$group",
                    new Document(key:"_id",
                            new Document(key:"wine_id", value:"$wine_id")
                                    .append(key:"year", value:"$year"))
                            .append(key:"wine_name",
                                    new Document(key:"$first", value:"$wine_name"))
                            .append(key:"wine_image",
                                    new Document(key:"$first", value:"$wine_image"))
                            .append(key:"country",
                                    new Document(key:"$first", value:"$country"))
                            .append(key:"count",
                                    new Document(key:"$sum", value:1L))),

            new Document(key:"$sort",
                    new Document(key:"count", -1L)),

            new Document(key:"$group",
                    new Document(key:"_id", value:"$country")
                            .append(key:"vintages",
                                    new Document(key:"$push",
                                            new Document(key:"wine_id", value:"$_id.wine_id")
                                                    .append(key:"wine_name", value:"$wine_name")
                                                    .append(key:"year", value:"$_id.year")
                                                    .append(key:"bottle", value:"$wine_image")
                                                    .append(key:"count", value:"$count")))),

            new Document(key:"$project",
                    new Document(key:"_id", value:0L)
                            .append(key:"name", value:"$_id")
                            .append(key:"vintages",
                                    new Document(key:"$slice", Arrays.asList(...a:"$vintages", 100L)))),

            new Document(key:"$project", new Document(key:"_id", value:0)
                    .append(key:"name", value:1)
                    .append(key:"top_100_vintages_of_the_week", value:"$vintages")),

            new Document(key:"$merge", new Document(key:"into", value:"countries")
                    .append(key:"on", value:"name")
                    .append(key:"whenMatched", value:"merge")
                    .append(key:"whenNotMatched", value:"discard")));


    // Assicuro che esistano gli indici NECESSARI alla pipeline.
    mongoTemplate
        .indexOps(collectionName:"countries")
        .ensureIndex(
            new Index()
                .on(key:"name", Sort.Direction.ASC)
                .named(name:"on_name_UNIQUE")
                .unique()
        );
    System.out.println(x:"--- INFO: Ensured UNIQUE index on field \"name\" for collection \"countries\".");

    System.out.println(x:"--- INFO: Executing aggregation pipeline for collection \"countries\"...");
    mongoTemplate.getCollection(collectionName:"reviews").aggregate(pipeline).allowDiskUse(allowDiskUse:true).toCollection();

    System.out.println(x:"--- INFO: Collection \"countries\" updated successfully.\n\n");
}
```

Figure 4.45: `updateTop100VintagesPerCountry()`

Weekly Update of Wine Tips per User

In order to provide personalized and dynamic wine recommendations, we defined a scheduled aggregation that updates the `wine_tips` field in each document of the *users* collection every Monday at 3:00 AM.

1. The aggregation pipeline begins by projecting the necessary fields from the *users* collection: the user's username and the list of their own favorite wines (stored under `wine_favorites.id`).
2. A first `$lookup` is then performed on the *reviews* collection to extract the top 3 highest-rated wines reviewed by the user, sorted by rating and review timestamp.
3. These are combined with the user's favorite wines using a `$setUnion` operation to form a comprehensive set of relevant wines for the user.
4. The pipeline then unwinds this combined list and retrieves detailed information for each wine via a `$lookup` on the *wines* collection, focusing on extracting the wine's style (`style.name`).
5. For each user-style pair, a representative wine is selected using `$group` and `$first`.
6. Another `$lookup` is used to retrieve up to 25 highly rated wines of the same style.
7. These are sorted by descending average rating and rating count.
8. Then, they are enriched with relevant metadata such as name, type, average price, image, and creation date.
9. The resulting tips are then grouped per user and stored as a list in the `wine_tips` field.
10. To maintain a manageable size, this list is limited to 100 items per user using the `$slice` operator.
11. Finally, the results are merged back into the original *users* collection using a `$merge` stage that updates existing documents based on the `login.username` field.

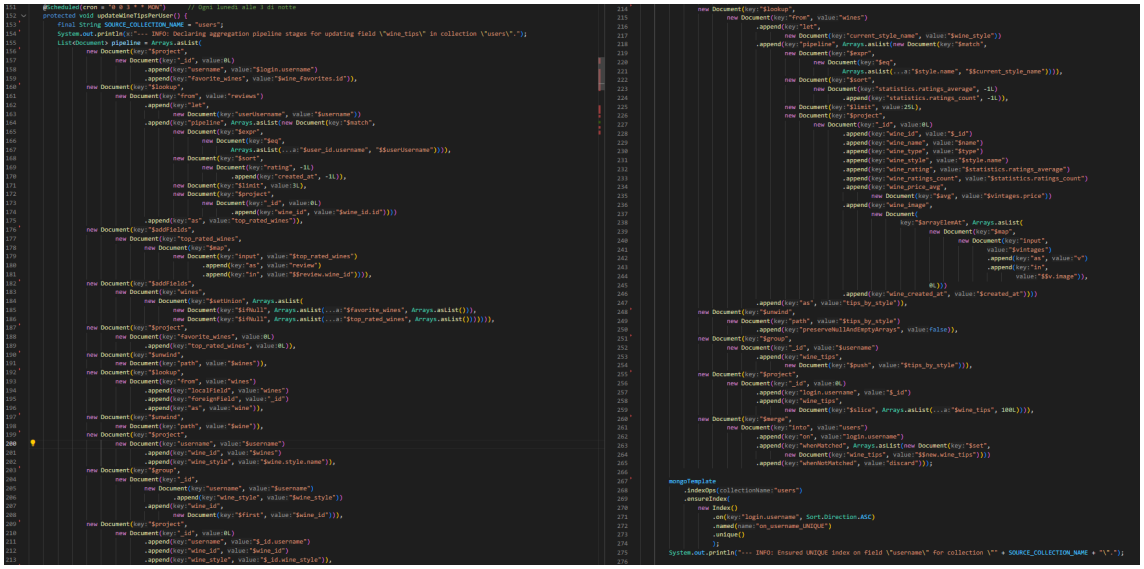


Figure 4.46: `updateWineTipsPerUser()`: part 1.

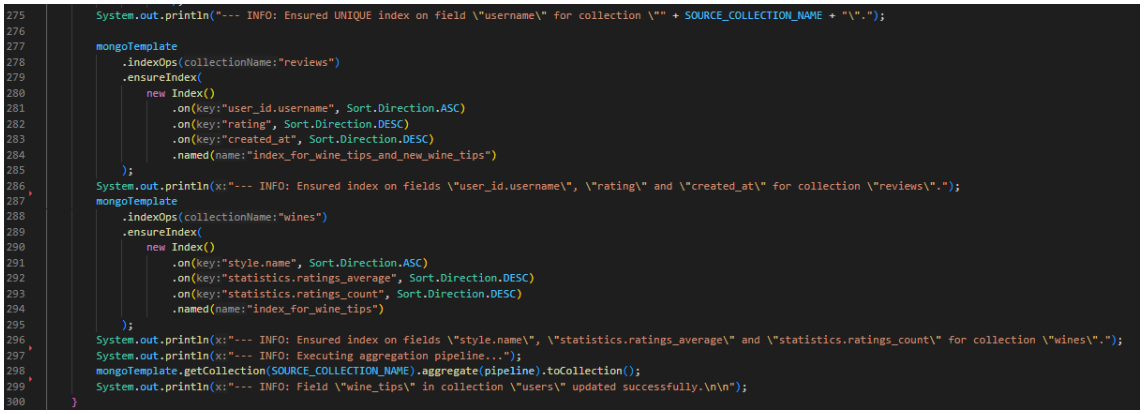


Figure 4.47: `updateWineTipsPerUser()`: part 2.

As we can see from the screenshots above, the aggregation ensures the presence of the following indexes for performance optimization:

- A unique index on `login.username` in the *users* collection.
- A compound index on `user_id.username`, `rating`, and `created_at` in the *reviews* collection.
- An compound on `style.name`, `statistics.ratings_average` and `statistics.ratings_count` in the *wines* collection.

Weekly Update of New Wine Tips per User

This aggregation pipeline runs every Monday at 3:00 AM and aims to update the `new_wine_tips` field in each document of the *users* collection. The goal is to generate personalized wine recommendations by combining user's favorite wines and their top-rated reviews, taking into account only new wines.

The pipeline proceeds as follows.

1. It starts by projecting each user's `username` and the IDs of their favorite wines.
2. Then, a lookup stage fetches the top 3 wines reviewed by the user, prioritizing those with the highest rating and most recent review date.
3. These two sets of wine IDs — favorites and top reviewed — are merged into a unique list to avoid duplicates.
4. For each wine in this combined list, the pipeline retrieves detailed information from the *wines* collection, including the wine's style.
5. Next, the documents are grouped by user and wine style, selecting one representative wine per style.
6. Another lookup stage uses each wine style to find up to 25 recently created wines (within the last 6 months), sorted by creation date, including relevant data such as name, type, average rating, price, and image.
7. The recommendations are then grouped per user, collecting up to 100 new wine tips.
8. Finally, the pipeline merges these results back into the *users* collection, updating the `new_wine_tips` field only for existing users.

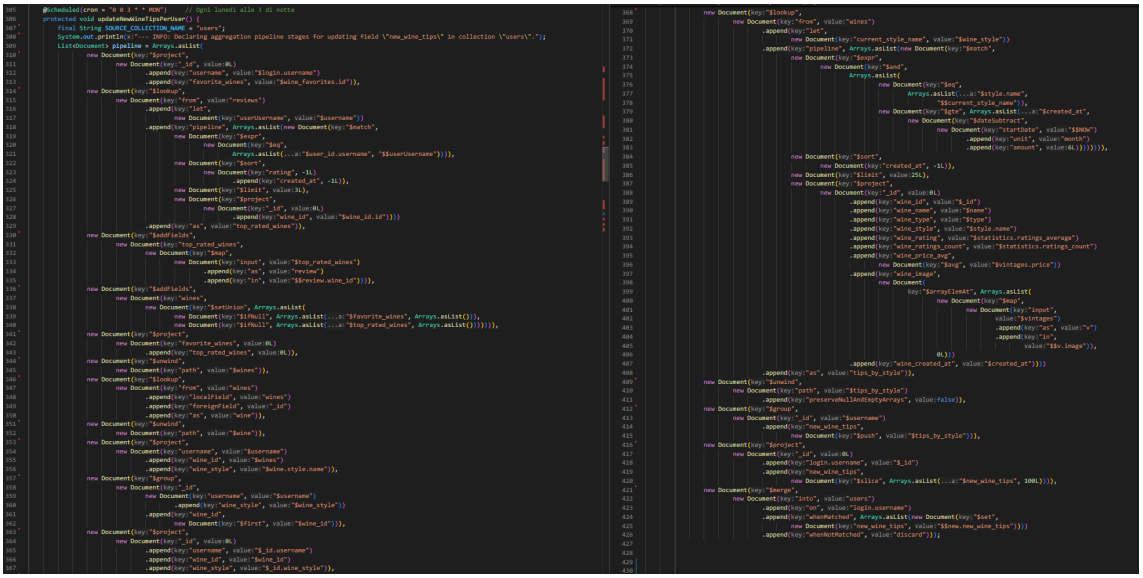


Figure 4.48: `updateNewWineTipsPerUser()`: part 1.

Figure 4.49: `updateNewWineTipsPerUser()`: part 2.

As we can see from the screenshots above, the aggregation ensures the presence of the following indexes for performance optimization:

- A unique index on `login.username` in the *users* collection.
- A compound index on `user_id.username`, `rating`, and `created_at` in the *reviews* collection.
- An compound on `style.name` and `created_at` in the *wines* collection.

Weekly Update of Top Vintages by Our Quality/Price Ratio per Wine Type

This aggregation pipeline is scheduled to run every Monday at 2:00 AM. It calculates the top vintages ranked by a custom quality/price (QoP) formula for each wine type (e.g., red, white, rosé) and updates the *top_vintages_by_our_qop_per_type* collection accordingly.

The pipeline operates on the *wines* collection and includes the following stages:

1. `$unwind` separates vintages embedded within each wine document.
2. `$project` retains necessary fields such as wine name, type, winery name, vintage year, image, ratings count, average quality rating, price, and computes the QoP score using the formula:

$$\text{QoP} = \frac{e^{\text{ratings\_average}} \times \sqrt{\text{ratings\_count}}}{\ln(\text{price})}$$

3. `$sort` orders vintages by QoP and ratings count in descending order.
4. `$group` aggregates vintages by wine type to determine the maximum QoP value per type and collects all vintages.
5. `$unwind` flattens vintages again to assign the max QoP for each type to every vintage.
6. `$addFields` adds a `points` field, representing each vintage's QoP as a percentage of the max QoP within its wine type, rounded to one decimal place.
7. `$group` regroups vintages by wine type, assembling the vintages with their respective points.
8. `$project` formats the output by excluding the internal ID, renaming the wine type, and limiting vintages to the top 1000 entries.
9. `$merge` updates the *top_vintages_by_our_qop_per_type* collection by replacing existing documents with matching wine types or inserting new ones.

Before running the pipeline, the method ensures that appropriate collections and indexes exist for efficient aggregation and merging. This process maintains an up-to-date materialized view of the highest-ranked vintages per wine type by our QoP formula, facilitating data-driven quality and price analysis for wine recommendations.

```java
private void createTopVintagesByOurQoPPerType() {
    if (!mongoTemplate.collectionExists(TOP_VINTAGES_OUR_QOP)) {
        mongoTemplate.createCollection(TOP_VINTAGES_OUR_QOP);
        System.out.println("--- INFO: Materialized view \"" + TOP_VINTAGES_OUR_QOP + "\" created successfully.");
    } else {
        System.out.println("--- INFO: Materialized view \"" + TOP_VINTAGES_OUR_QOP + "\" already exists.");
    }

    // Assicuro che esistano gli indici NECESSARI alla pipeline.
    mongoTemplate
        .indexOps(TOP_VINTAGES_OUR_QOP)
        .ensureIndex(
            new Index()
                .on(key:"type", Sort.Direction.ASC)
                .named(name:"on_type_UNIQUE")
                .unique()
        );
    System.out.println("--- INFO: Ensured UNIQUE index on field \"type\" for materialized view \"" + TOP_VINTAGES_OUR_QOP + "\".");
}
```

Figure 4.50: `createTopVintagesByOurQoPPerType()`

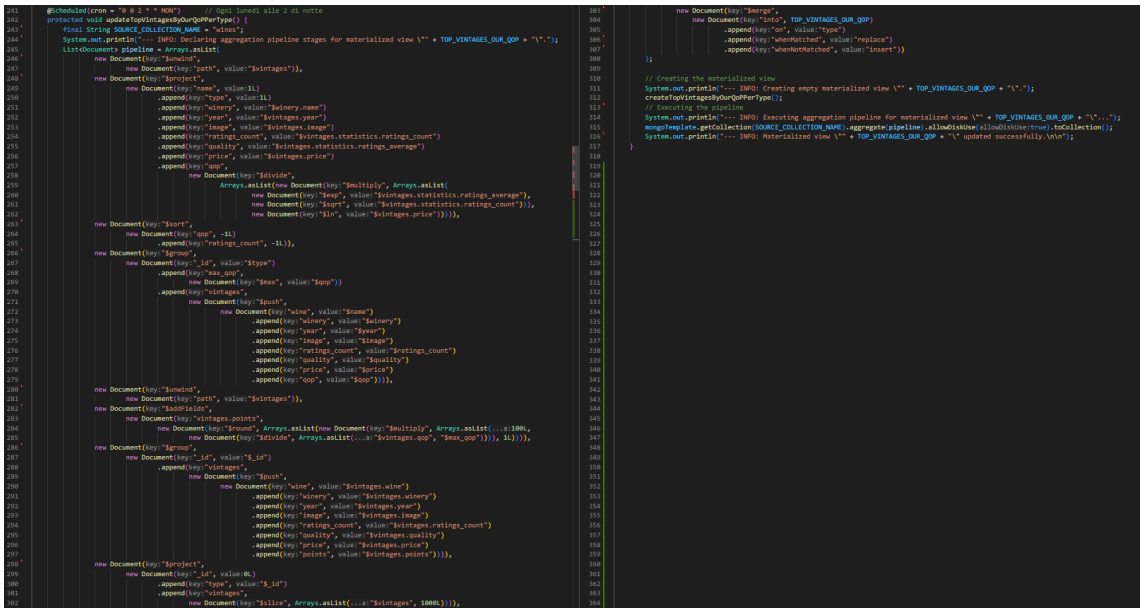


Figure 4.51: `updateTopVintagesByOurQoPPerType()`

Weekly Update of Top Vintages by Base Quality/Price Ratio per Wine Type

This aggregation pipeline is scheduled to run every Monday at 2:00 AM. It calculates the top vintages ranked by a base quality/price (QoP) formula for each wine type (e.g., red, white, rosé) and updates the *top_vintages_by_qop_per_type* collection accordingly.

The pipeline operates on the *wines* collection and includes the following stages:

1. `$unwind` separates vintages embedded within each wine document.
2. `$project` retains necessary fields such as wine name, type, winery name, vintage year, image, ratings count, average quality rating, price, and computes the QoP score using the formula:

$$\text{QoP} = \frac{\text{ratings\_average}}{\text{price}}$$

3. `$sort` orders vintages by QoP and ratings count in descending order.
4. `$group` aggregates vintages by wine type to determine the maximum QoP value per type and collects all vintages.
5. `$unwind` flattens vintages again to assign the max QoP for each type to every vintage.
6. `$addFields` adds a `points` field, representing each vintage's QoP as a percentage of the max QoP within its wine type, rounded to one decimal place.
7. `$group` regroups vintages by wine type, assembling the vintages with their respective points.
8. `$project` formats the output by excluding the internal ID, renaming the wine type, and limiting vintages to the top 1000 entries.
9. `$merge` updates the *top_vintages_by_qop_per_type* collection by replacing existing documents with matching wine types or inserting new ones.

Before running the pipeline, the method ensures that the appropriate collection and indexes exist for efficient aggregation and merging. This process maintains an up-to-date materialized view of the highest-ranked vintages per wine type by base QoP formula, enabling data-driven quality and price analysis for wine recommendations.

```java
private void createTopVintagesByQoPPerType() {
    if (!mongoTemplate.collectionExists(TOP_VINTAGES_QOP)) {
        mongoTemplate.createCollection(TOP_VINTAGES_QOP);
        System.out.println("--- INFO: Materialized view \"" + TOP_VINTAGES_QOP + "\" created successfully.");
    } else {
        System.out.println("--- INFO: Materialized view \"" + TOP_VINTAGES_QOP + "\" already exists.");
    }

    // Assicuro che esistano gli indici NECESSARI alla pipeline.
    mongoTemplate
        .indexOps(TOP_VINTAGES_QOP)
        .ensureIndex(
            new Index()
                .on(key:"type", Sort.Direction.ASC)
                .named(name:"on_type_UNIQUE")
                .unique()
        );
    System.out.println("--- INFO: Ensured UNIQUE index on field \"type\" for materialized view \"" + TOP_VINTAGES_QOP + "\".");
}
```

Figure 4.52: `createTopVintagesByQoPPerType()`

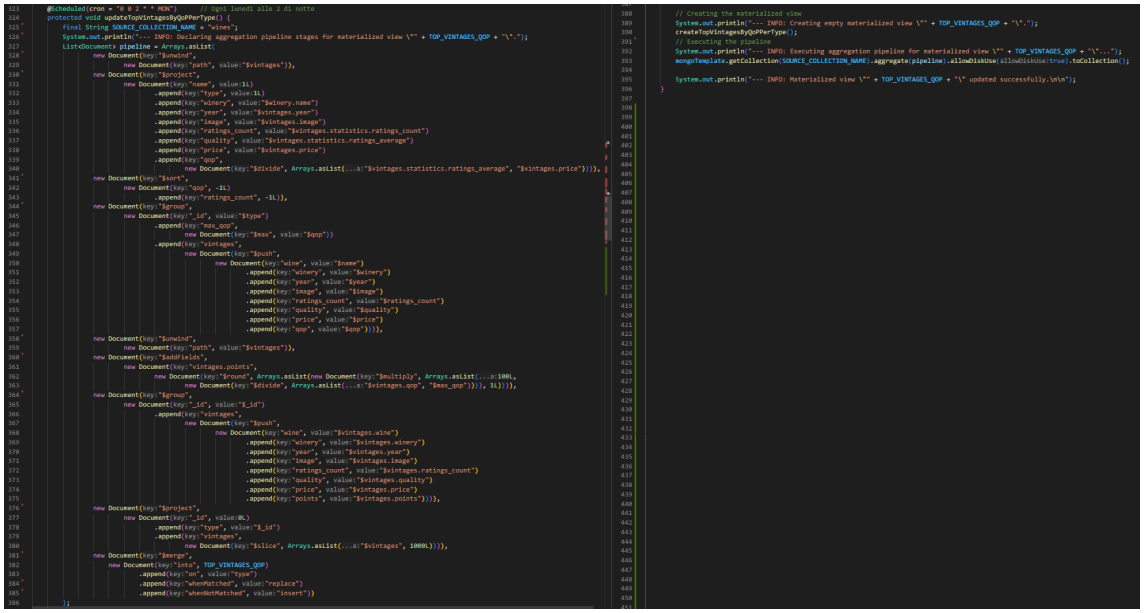


Figure 4.53: `updateTopVintagesByQoPPerType()`

Weekly Update of Top Vintages by Average Rating per Wine Type

This aggregation pipeline is scheduled to run every Monday at 2:00 AM. It calculates a ranking of the top vintages based on their average ratings for each wine type (e.g., red, white, rosé) and updates the *top_vintages_by_ratings_per_type* collection accordingly.

The pipeline operates on the *wines* collection and includes the following stages:

1. `$unwind` separates vintages embedded within each wine document.
2. `$project` retains necessary fields such as wine name, type, winery name, vintage year, price, image, average rating, and ratings count.
3. `$sort` orders vintages by average rating and ratings count in descending order.
4. `$group` aggregates vintages by wine type, collecting all vintages to form the top vintages list per type.
5. `$project` formats the output by excluding the internal ID, renaming the wine type, and limiting vintages to the top 1000 entries.
6. `$merge` updates the *top_vintages_by_ratings_per_type* collection by replacing existing documents with matching wine types or inserting new ones.

Before running the pipeline, the method ensures that the appropriate collection and indexes exist for efficient aggregation and merging. This process maintains an up-to-date materialized view of the highest-ranked vintages per wine type by rating average, enabling data-driven quality and price analysis for wine recommendations.

```
private void createTopVintagesByRatingsPerType() {
    if (!mongoTemplate.collectionExists(TOP_VINTAGES_RATINGS)) {
        mongoTemplate.createCollection(TOP_VINTAGES_RATINGS);
        System.out.println("--- INFO: Materialized view \"" + TOP_VINTAGES_RATINGS + "\" created successfully.");
    } else {
        System.out.println("--- INFO: Materialized view \"" + TOP_VINTAGES_RATINGS + "\" already exists.");
    }

    // Assicuro che esistano gli indici NECESSARI alla pipeline.
    mongoTemplate
        .indexOps(TOP_VINTAGES_RATINGS)
        .ensureIndex(
            new Index()
                .on(key:"type", Sort.Direction.ASC)
                .named(name:"on_type_UNIQUE")
                .unique()
        );
    System.out.println("--- INFO: Ensured UNIQUE index on field \"type\" for materialized view \"" + TOP_VINTAGES_RATINGS + "\".");
}
```

Figure 4.54: `createTopVintagesByRatingsPerType()`

```java
@Scheduled(cron = "0 0 2 * * MON")      // Ogni lunedì alle 2 di notte
protected void updateTopVintagesByRatingsPerType() {
    final String SOURCE_COLLECTION_NAME = "wines";
    System.out.println("--- INFO: Declaring aggregation pipeline stages for materialized view \"" + TOP_VINTAGES_RATINGS + "\".");
    List<Document> pipeline = Arrays.asList(
            new Document(key:"$unwind",
                    new Document(key:"path", value:"$vintages")),
            new Document(key:"$project",
                    new Document(key:"name", value:1L)
                            .append(key:"type", value:1L)
                            .append(key:"winery", value:"$winery.name")
                            .append(key:"year", value:"$vintages.year")
                            .append(key:"price", value:"$vintages.price")
                            .append(key:"image", value:"$vintages.image")
                            .append(key:"ratings_average", value:"$vintages.statistics.ratings_average")
                            .append(key:"ratings_count", value:"$vintages.statistics.ratings_count")),
            new Document(key:"$sort",
                    new Document(key:"ratings_average", -1L)
                            .append(key:"ratings_count", -1L)),
            new Document(key:"$group",
                    new Document(key:"_id", value:"$type")
                            .append(key:"vintages",
                                    new Document(key:"$push",
                                            new Document(key:"wine", value:"$name")
                                                    .append(key:"winery", value:"$winery")
                                                    .append(key:"year", value:"$year")
                                                    .append(key:"price", value:"$price")
                                                    .append(key:"image", value:"$image")
                                                    .append(key:"ratings_average", value:"$ratings_average")
                                                    .append(key:"ratings_count", value:"$ratings_count")))),
            new Document(key:"$project",
                    new Document(key:"_id", value:0L)
                            .append(key:"type", value:"$_id")
                            .append(key:"vintages",
                                    new Document(key:"$slice", Arrays.asList(...a:"$vintages", 1000L)))),
            new Document(key:"$merge",
                    new Document(key:"into", TOP_VINTAGES_RATINGS)
                            .append(key:"on", value:"type")
                            .append(key:"whenMatched", value:"replace")
                            .append(key:"whenNotMatched", value:"insert"))
    );

    // Creating the materialized view
    System.out.println("--- INFO: Creating empty materialized view \"" + TOP_VINTAGES_RATINGS + "\".");
    createTopVintagesByRatingsPerType();

    // Executing the pipeline
    System.out.println("--- INFO: Executing aggregation pipeline for materialized view \"" + TOP_VINTAGES_RATINGS + "\"...");
    mongoTemplate.getCollection(SOURCE_COLLECTION_NAME).aggregate(pipeline).allowDiskUse(allowDiskUse:true).toCollection();

    System.out.println("--- INFO: Materialized view \"" + TOP_VINTAGES_RATINGS + "\" updated successfully.\n\n");
}
```

Figure 4.55: `updateTopVintagesByRatingsPerType()`

Weekly Update of Top Wines by Average Rating per Wine Type

This aggregation pipeline is executed every Monday at 2:00 AM. It generates a ranking of the top wines based on their overall average rating, categorized by wine type (e.g., red, white, rosé). The final result is stored in the *top_wines_by_ratings_per_type* materialized view.

The aggregation is performed on the *wines* collection and includes the following stages:

1. `$project` extracts the wine name, type, winery name, and overall average statistics. It also computes the average price across vintages (rounded to two decimal places), and selects the first available image from the list of vintages.

2. `$sort` orders wines by descending average rating and ratings count, giving priority to highly rated and frequently reviewed wines.

3. `$group` aggregates wines by type and collects the relevant fields into a list named `wines` for each category.

4. `$project` prepares the final output by renaming the wine type field and limiting the number of top wines to 1000 per type.

5. `$merge` updates the *top_wines_by_ratings_per_type* materialized view, replacing existing documents where the type matches or inserting new entries otherwise.

Before running the pipeline, the method ensures that the appropriate collection and indexes exist for efficient aggregation and merging. This process maintains an up-to-date materialized view of the highest-ranked wines per wine type by rating average, enabling data-driven quality and price analysis for wine recommendations.

```
private void createTopWinesByRatingsPerType() {
    if (!mongoTemplate.collectionExists(TOP_WINES_RATINGS)) {
        mongoTemplate.createCollection(TOP_WINES_RATINGS);
        System.out.println("--- INFO: Materialized view \"" + TOP_WINES_RATINGS + "\" created successfully.");
    } else {
        System.out.println("--- INFO: Materialized view \"" + TOP_WINES_RATINGS + "\" already exists.");
    }

    // Assicuro che esistano gli indici NECESSARI alla pipeline.
    mongoTemplate
        .indexOps(TOP_WINES_RATINGS)
        .ensureIndex(
            new Index()
                .on(key:"type", Sort.Direction.ASC)
                .named(name:"on_type_UNIQUE")
                .unique()
        );
    System.out.println("--- INFO: Ensured UNIQUE index on field \"type\" for materialized view \"" + TOP_WINES_RATINGS + "\".");
}
```

Figure 4.56: `createTopWinesByRatingsPerType()`

```java
@Scheduled(cron = "0 0 2 * * MON")      // Ogni lunedì alle 2 di notte
protected void updateTopWinesByRatingsPerType() {
    final String SOURCE_COLLECTION_NAME = "wines";
    System.out.println("--- INFO: Declaring aggregation pipeline stages for materialized view \"" + TOP_WINES_RATINGS + "\".");
    List<Document> pipeline = Arrays.asList(
            new Document(key:"$project",
                    new Document(key:"name", value:1L)
                            .append(key:"type", value:1L)
                            .append(key:"winery", value:"$winery.name")
                            .append(key:"prices_average",   // dato che i prezzi sono diversi per ogni vintage, calcolo la media dei prezzi delle annate
                                    new Document(key:"$round",
                                            Arrays.asList(new Document(key:"$avg", value:"$vintages.price"), 2L)))
                            .append(key:"image",
                                    new Document(key:"$arrayElemAt", Arrays.asList(...a:"$vintages.image", 0L)))
                            .append(key:"ratings_average", value:"$statistics.ratings_average")
                            .append(key:"ratings_count", value:"$statistics.ratings_count")),
            new Document(key:"$sort",
                    new Document(key:"ratings_average", -1L)
                            .append(key:"ratings_count", -1L)),
            new Document(key:"$group",
                    new Document(key:"_id", value:"$type")
                            .append(key:"wines",
                                    new Document(key:"$push",
                                            new Document(key:"wine", value:"$name")
                                                    .append(key:"winery", value:"$winery")
                                                    .append(key:"image", value:"$image")
                                                    .append(key:"prices_average", value:"$prices_average")
                                                    .append(key:"ratings_average", value:"$ratings_average")
                                                    .append(key:"ratings_count", value:"$ratings_count")))),
            new Document(key:"$project",
                    new Document(key:"_id", value:0L)
                            .append(key:"type", value:"$_id")
                            .append(key:"wines",
                                    new Document(key:"$slice", Arrays.asList(...a:"$wines", 1000L)))),
            new Document(key:"$merge",
                    new Document(key:"into", TOP_WINES_RATINGS)
                            .append(key:"on", value:"type")
                            .append(key:"whenMatched", value:"replace")
                            .append(key:"whenNotMatched", value:"insert"))
    );

    // Creating the materialized view
    System.out.println("--- INFO: Creating empty materialized view \"" + TOP_WINES_RATINGS + "\"...");
    createTopWinesByRatingsPerType();

    // Executing the pipeline
    System.out.println("--- INFO: Executing aggregation pipeline for materialized view \"" + TOP_WINES_RATINGS + "\"...");
    mongoTemplate.getCollection(SOURCE_COLLECTION_NAME).aggregate(pipeline).allowDiskUse(allowDiskUse:true).toCollection();

    System.out.println("--- INFO: Materialized view \"" + TOP_WINES_RATINGS + "\" updated successfully.\n\n");
}
```

Figure 4.57: `updateTopWinesByRatingsPerType()`

Weekly Update of Top Wineries by Average Wine Ratings

This aggregation pipeline is executed every Monday at 2:00 AM. It generates a unified ranking of the top wineries based on the average rating of all their wines, regardless of wine type. The final result is stored in the *top_wineries_by_wines_ratings* materialized view.

The aggregation is performed on the *wines* collection and includes the following stages:

1. `$group` aggregates all wines by the winery's username, calculating the average rating and the total number of ratings across all wines produced by each winery.

2. `$lookup` joins the result with the *wineries* collection to retrieve the full winery information using the username as a key.

3. `$unwind` flattens the resulting array from the previous join to access the embedded winery document.

4. `$sort` orders wineries by descending average rating and ratings count, prioritizing both quality and popularity.

5. `$project` reshapes the document to include only relevant fields such as winery name, location, thumbnail image, and the aggregated statistics, rounding the average rating to one decimal place.

6. `$merge` writes the final output to the *top_wineries_by_wines_ratings* materialized view, replacing existing entries or inserting new ones based on the winery username.

Before running the pipeline, the method ensures that the appropriate collection and indexes exist for efficient aggregation and merging. This process maintains an up-to-date materialized view of the highest-ranked wineries by wine rating average, allowing users to easily discover the most highly rated producers.

```java
private void createTopWineriesByWineRatings() {
    if (!mongoTemplate.collectionExists(TOP_WINERIES_RATINGS)) {
        mongoTemplate.createCollection(TOP_WINERIES_RATINGS);
        System.out.println("--- INFO: Materialized view \"" + TOP_WINERIES_RATINGS + "\" created successfully.");
    } else {
        System.out.println("--- INFO: Materialized view \"" + TOP_WINERIES_RATINGS + "\" already exists.");
    }

    // Assicuro che esistano gli indici NECESSARI alla pipeline.
    mongoTemplate
        .indexOps(TOP_WINERIES_RATINGS)
        .ensureIndex(
            new Index()
                .on(key:"winery_username", Sort.Direction.ASC)
                .named(name:"on_winery_username_UNIQUE")
                .unique()
        );
    System.out.println("--- INFO: Ensured UNIQUE index on field \"winery_username\" for materialized view \"" + TOP_WINERIES_RATINGS + "\".");

    // Assicuro che esistano gli indici FUNZIONALI alla pipeline.
    mongoTemplate
        .indexOps(collectionName:"wineries")
        .ensureIndex(
            new Index()
                .on(key:"login.username", Sort.Direction.ASC)
                .named(name:"on_username_UNIQUE")
                .unique()
        );
    System.out.println(x:"--- INFO: Ensured UNIQUE index on field \"login.username\" for collection \"wineries\".");
}
```

Figure 4.58: `createTopWineriesByWineRatings()`

```java
@Scheduled(cron = "0 0 2 * * MON")      // Ogni lunedì alle 2 di notte
protected void updateTopWineriesByWineRatings() {
    final String SOURCE_COLLECTION_NAME = "wines";
    System.out.println("--- INFO: Declaring aggregation pipeline stages for materialized view \"" + TOP_WINERIES_RATINGS + "\".");
    List<Document> pipeline = Arrays.asList(
            new Document(key:"$group",
                    new Document(key:"_id", value:"$winery.username")
                            .append(key:"ratings_average",
                                    new Document(key:"$avg", value:"$statistics.ratings_average"))
                            .append(key:"ratings_count",
                                    new Document(key:"$sum", value:"$statistics.ratings_count"))),
            new Document(key:"$lookup",
                    new Document(key:"from", value:"wineries")
                            .append(key:"localField", value:"_id")
                            .append(key:"foreignField", value:"login.username")
                            .append(key:"as", value:"winery")),
            new Document(key:"$unwind",
                    new Document(key:"path", value:"$winery")),
            new Document(key:"$sort",
                    new Document(key:"ratings_average", -1L)
                            .append(key:"ratings_count", -1L)),
            new Document(key:"$project",
                    new Document(key:"_id", value:0L)
                            .append(key:"winery_username", value:"$winery.login.username")
                            .append(key:"thumbnail", value:"$winery.picture.thumbnail")
                            .append(key:"winery", value:"$winery.name")
                            .append(key:"region", value:"$winery.region")
                            .append(key:"country", value:"$winery.country")
                            .append(key:"ratings_average",
                                    new Document(key:"$round", Arrays.asList(...a:"$ratings_average", 1L)))
                            .append(key:"ratings_count", value:"$ratings_count")),
            new Document(key:"$merge",
                    new Document(key:"into", TOP_WINERIES_RATINGS)
                            .append(key:"on", value:"winery_username")
                            .append(key:"whenMatched", value:"replace")
                            .append(key:"whenNotMatched", value:"insert"))
    );

    // Creating the materialized view
    System.out.println("--- INFO: Creating empty materialized view \"" + TOP_WINERIES_RATINGS + "\".");
    createTopWineriesByWineRatings();

    // Executing the pipeline
    System.out.println("--- INFO: Executing aggregation pipeline for materialized view \"" + TOP_WINERIES_RATINGS + "\"...");
    mongoTemplate.getCollection(SOURCE_COLLECTION_NAME).aggregate(pipeline).allowDiskUse(allowDiskUse:true).toCollection();

    System.out.println("--- INFO: Materialized view \"" + TOP_WINERIES_RATINGS + "\" updated successfully.\n\n");
}
```

Figure 4.59: `updateTopWineriesByWineRatings()`

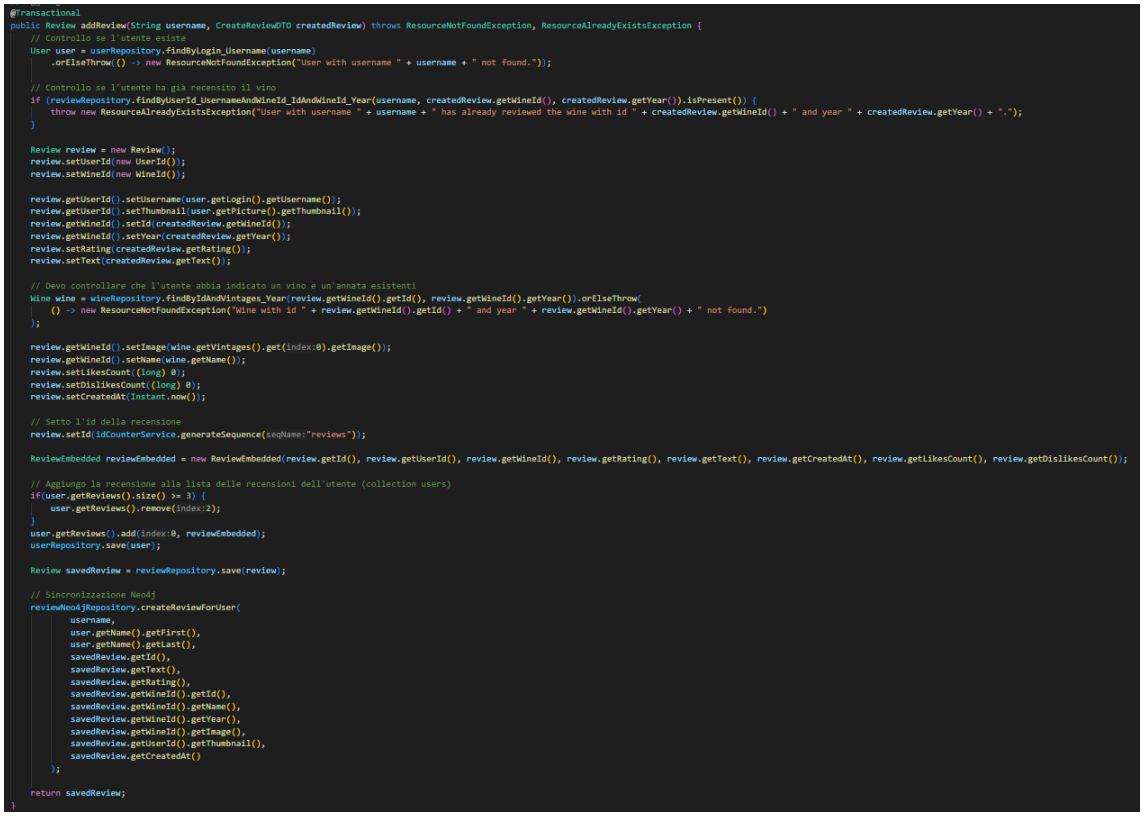
4.5 Neo4j Graph Queries

4.5.1 Synchronization Between MongoDB and Neo4j

CRUD operations have been implemented for each of the following entities in Neo4j: users, wineries, and reviews. To avoid redundancy, we will not detail all of them. It is important to note that these Neo4j operations are tightly synchronized with the corresponding MongoDB actions.

With regard to *reviews* collection:

- When a new review is **added** in MongoDB, it is immediately created in the Neo4j graph.
- When a review is **updated** in MongoDB, the change is promptly reflected in Neo4j, provided the review exists in the graph.
- When a review is **deleted** from MongoDB, it is also removed from the Neo4j graph if present.



```
@Transactional
public Review addReview(String username, CreateReviewDTO createdReview) throws ResourceNotFoundException, ResourceAlreadyExistsException {
    // Controllo se l'utente esiste
    User user = userRepository.findByLogin_Username(username)
        .orElseThrow(() -> new ResourceNotFoundException("User with username " + username + " not found."));

    // Controllo se l'utente ha già recensito il vino
    if (reviewRepository.findByUserId_UsernameAndWineId_IdAndWineId_Year(username, createdReview.getWineId(), createdReview.getYear()).isPresent()) {
        throw new ResourceAlreadyExistsException("User with username " + username + " has already reviewed the wine with id " + createdReview.getWineId() + " and year " + createdReview.getYear() + ".");
    }

    Review review = new Review();
    review.setUserId(new UserId());
    review.setWineId(new WineId());

    review.getUserId().setUsername(user.getLogin().getUsername());
    review.getUserId().setThumbnail(user.getPicture().getThumbnail());
    review.getWineId().setId(createdReview.getWineId());
    review.getWineId().setYear(createdReview.getYear());
    review.setRating(createdReview.getRating());
    review.setText(createdReview.getText());

    // Devo controllare che l'utente abbia indicato un vino e un'annata esistenti
    Wine wine = wineRepository.findByIdAndVintages_Year(review.getWineId().getId(), review.getWineId().getYear()).orElseThrow(
        () -> new ResourceNotFoundException("Wine with id " + review.getWineId().getId() + " and year " + review.getWineId().getYear() + " not found.")
    );

    review.getWineId().setImage(wine.getVintages().get(index:0).getImage());
    review.getWineId().setName(wine.getName());
    review.setLikesCount((long) 0);
    review.setDislikesCount((long) 0);
    review.setCreatedAt(Instant.now());

    // Setto l'id della recensione
    review.setId(idCounterService.generateSequence(seqName:"reviews"));

    ReviewEmbedded reviewEmbedded = new ReviewEmbedded(review.getId(), review.getUserId(), review.getWineId(), review.getRating(), review.getText(), review.getCreatedAt(), review.getLikesCount(), review.getDislikesCount());

    // Aggiungo la recensione alla lista delle recensioni dell'utente (collection users)
    if(user.getReviews().size() >= 3) {
        user.getReviews().remove(index:2);
    }
    user.getReviews().add(index:0, reviewEmbedded);
    userRepository.save(user);

    Review savedReview = reviewRepository.save(review);

    // Sincronizzazione Neo4j
    reviewNeo4jRepository.createReviewForUser(
            username,
            user.getName().getFirst(),
            user.getName().getLast(),
            savedReview.getId(),
            savedReview.getText(),
            savedReview.getRating(),
            savedReview.getWineId().getId(),
            savedReview.getWineId().getName(),
            savedReview.getWineId().getYear(),
            savedReview.getWineId().getImage(),
            savedReview.getUserId().getThumbnail(),
            savedReview.getCreatedAt()
        );

    return savedReview;
}
```

Figure 4.60: `addReview()`: an example method to demonstrate that Neo4j is updated immediately after MongoDB.

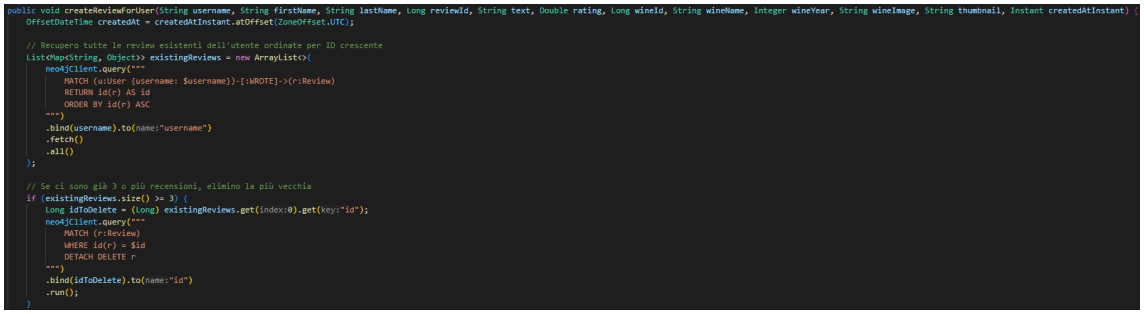


Figure 4.61: `createReviewForUser()` in `ReviewNeo4jRepository`.

The same synchronization logic applies to *users* and *wineries* collections, ensuring data consistency across both databases.

With regard to *users* collection:

- When a new user is **added** in MongoDB, it is immediately created in the Neo4j graph.
- When a user is **updated** in MongoDB, the change is promptly reflected in Neo4j.
- When a user is **deleted** from MongoDB, it is also removed from the Neo4j graph if present.

With regard to *wineries* collection:

- When a new winery is **added** in MongoDB, it is immediately created in the Neo4j graph.
- When a winery is **updated** in MongoDB, the change is promptly reflected in Neo4j.
- When a winery is **deleted** from MongoDB, it is also removed from the Neo4j graph if present.

4.5.2 Relevant Operations

User's feed creation

To retrieve the personalized feed, the application calls in cascade the following methods.

- `public ResponseEntity<?> getUserFeed(`
  `@PathVariable String username,`
  `@RequestParam(defaultValue = "0") Integer page)`
- `public List<Map<String, Object> > getUserFeed(`
  `String username,`
  `Integer page)`

- `public List<Map<String, Object> > getUserFeed(`
  `String username,`
  `Integer skip,`
  `Integer limit)`

The first method is a `@RestController` endpoint and is responsible for handling a GET request to the `/api/reviews/users/{username}/feed` path. It is secured so that only the user themselves or an administrator can access it.

The controller delegates the logic to the service layer, which calculates the correct offset (`skip`) based on the `page` argument and `PAGE_SIZE` constant, and then calls the repository method to fetch the data.

The third method interacts directly with the Neo4j graph. It performs the following operations.

- Finds all the users followed by the given user.
- For each followed user, retrieves the most recent reviews (up to three).
- Collects the necessary review details (e.g., wine name, rating, date).
- Orders the results by date in descending order and applies pagination using `skip` and `limit`.

This mechanism allows the system to provide each user with a feed that contains the latest reviews written by the users they follow, maintaining a social and personalized experience.

The following figures show some of the code snippets involved in the process.

```
@GetMapping("/users/{username}/feed")
@Secured({ "ROLE_USER", "ROLE_ADMIN" })
@PreAuthorize("#username == authentication.principal.username or hasRole('ROLE_ADMIN')")
public ResponseEntity<?> getUserFeed(
        @Parameter(description = "Username of the user to retrieve personalized feed for.", schema = @Schema(example = "yellowbutterfly631")) @NotBlank(message = "Username cannot be blank.") @PathVariable String username,
        @RequestParam(required = false, name = "page number", defaultValue = "0") Integer page) {
    return ResponseEntity.status(HttpStatus.OK).body(reviewService.getUserFeed(username, page));
}
```

Figure 4.62: `@RestController` method handling the get user's feed request.

```
public List<Map<String, Object>> getUserFeed(String username, Integer page) {
    Integer skip = page * PAGE_SIZE;
    return reviewNeo4jRepository.getUserFeed(username, skip, PAGE_SIZE);
}
```

Figure 4.63: Service method to create user's feed.

```
public List<Map<String, Object>> getUserFeed(String username, Integer skip, Integer limit) {
    String query = """
        MATCH (me:User {username: $username})-[:FOLLOWS]->(friend:User)
        MATCH (friend)-[:WROTE]->(review:Review)
        WITH friend, review
        ORDER BY review.createdAt DESC
        WITH friend, collect(review)[0..3] AS recentReviews
        UNWIND recentReviews AS review
        RETURN {
            reviewId: review.reviewId,
            username: friend.username,
            thumbnail: friend.thumbnail,
            wineId: review.wineId,
            wineName: review.wineName,
            wineYear: review.wineYear,
            wineImage: review.wineImage,
            rating: review.rating,
            reviewText: review.text,
            createdAt: review.createdAt
        } AS result
        ORDER BY result.createdAt DESC
        SKIP $skip
        LIMIT $limit
    """;

    return new ArrayList<>(
        neo4jClient.query(query)
            .bindAll(
                Map.of(
                    k1:"username", username,
                    k2:"skip", skip,
                    k3:"limit", limit
                )
            )
            .fetch()
            .all()
    );
}
```

Figure 4.64: Neo4j query (in `ReviewNeo4jRepository`) that retrieves up to three of the most recent reviews per followed user, sorted by creation date and paginated to build the personalized feed.

Get follow suggestions

To retrieve follow suggestions, the application calls in cascade the following methods.

- `public ResponseEntity<?> getUserSuggestedFollows(`
  `@PathVariable String username)`
- `public List<Map<String, Object> > getUserSuggestedFollows(`
  `String username)`
- `public List<Map<String, Object> > getUserSuggestedFollows(`
  `String username)`
- `public List<Map<String, Object> > getRandomFollows(`
  `String username,`
  `int limit)`

The first method is a `@RestController` endpoint that handles a GET request to the `/api/users/{username}/suggestedFollows` path. It is protected so that only the user or an administrator can request suggestions.

The controller delegates the logic to the service layer, which calls the main repository method to retrieve follow suggestions. If fewer than 10 suggestions are returned, the service fetches additional random suggestions using a fallback method.

The third method queries the Neo4j graph to identify second-degree connections (i.e., "friends of friends") who are not already followed by the user. It performs the following operations.

- Finds all the users followed by the target user.
- Looks for users or wineries followed by those friends but not yet followed by the user.
- Excludes the user themselves from the suggestions.
- Returns up to 10 suggestions, ordered by the number of mutual followers.

If the number of suggestions is insufficient, the fourth method is used to:

- find random users or wineries not already followed by the user;
- filter out the user themselves;
- return a random subset up to the remaining number of slots needed.

This strategy ensures that the user is offered relevant and varied suggestions to expand their network, enhancing the social aspect of the application.

The following figures show some of the code snippets involved in the process.

```
@GetMapping("/{username}/suggestedFollows")
@Secured({"ROLE_ADMIN", "ROLE_USER" })
@PreAuthorize("#username == authentication.principal.username or hasRole('ROLE_ADMIN')")
public ResponseEntity<?> getUserSuggestedFollows(
        @Parameter(description = "Username of the user for whom to retrieve suggested follows.", schema = @Schema(example = "yellowbutterfly631")) @PathVariable String username) {
    return ResponseEntity.status(HttpStatus.OK).body(userService.getUserSuggestedFollows(username));
}
```

Figure 4.65: `@RestController` method handling the get follow suggestions request.

```
public List<Map<String, Object>> getUserSuggestedFollows(String username) {
    List<Map<String, Object>> suggestions = userNeo4jRepository.getUserSuggestedFollows(username);
    if (suggestions.size() < 10) {
        int remaining = 10 - suggestions.size();
        List<Map<String, Object>> randoms = userNeo4jRepository.getRandomFollows(username, remaining);
        suggestions.addAll(randoms);
    }

    return suggestions;
}
```

Figure 4.66: Service method to get follow suggestions.

```
public List<Map<String, Object>> getUserSuggestedFollows(String username) {
    String query = """
        MATCH (me:User {username: $username})-[:FOLLOWS]->(friend:User)
        MATCH (friend)-[:FOLLOWS]->(suggested)
        WHERE NOT (me)-[:FOLLOWS]->(suggested) AND suggested <> me
            AND (suggested:User OR suggested:Winery)
        RETURN suggested.username AS username,
            suggested.thumbnail AS thumbnail,
            COUNT(*) AS mutualFollows
        ORDER BY mutualFollows DESC
        LIMIT 10
    """;

    List<Map<String, Object>> result = neo4jClient.query(query)
    .bind(username).to(name:"username")
    .fetch()
    .all()
    .stream()
    .map(record -> {
        Map<String, Object> r = new HashMap<>();
        r.put(key:"username", (String) record.get(key:"username"));
        r.put(key:"thumbnail", (String) record.get(key:"thumbnail"));
        r.put(key:"mutualFollows", ((Number) record.get(key:"mutualFollows")).intValue());
        return r;
    })
    .collect(Collectors.toList());

    System.out.println(">>> SUGGESTED: " + result);
    return result;
}
```

Figure 4.67: `getUserSuggestedFollows()` in `UserNeo4jRepository`

```
public List<Map<String, Object>> getRandomFollows(String username, int limit) {
    String query = """
        MATCH (s)
        WHERE (s:User OR s:Winery)
        MATCH (me:User {username: $username})
        WHERE NOT (me)-[:FOLLOWS]->(s) AND s.username <> $username
        AND s.username <> $username
        RETURN s.username AS username,
            s.thumbnail AS thumbnail
        ORDER BY rand()
        LIMIT $limit
    """;

    return neo4jClient.query(query)
        .bind(username).to(name:"username")
        .bind(limit).to(name:"limit")
        .fetch()
        .all()
        .stream()
        .map(record -> {
            Map<String, Object> r = new HashMap<>();
            r.put(key:"username", (String) record.get(key:"username"));
            r.put(key:"thumbnail", (String) record.get(key:"thumbnail"));
            return r;
        })
        .collect(Collectors.toList());
}
```

Figure 4.68: `getRandomFollows()` in `UserNeo4jRepository`

Follow a user

To follow another user or winery, the application must call the following methods.

- `public ResponseEntity<?> follow(`
  `@PathVariable String username,`
  `@RequestParam String targetUsername)`
- `public void follow(`
  `String followerUsername,`
  `String targetUsername)`
- `public void follow(`
  `String followerUsername,`
  `String targetUsername,`
  `boolean isWinery)`

The first method is a `@RestController` endpoint that handles a PUT request to the `/api/users/{username}/follow` path. It is secured so that only the user themselves can perform this operation, ensuring proper access control.

The controller delegates the operation to the service layer, which determines whether the target is a user or a winery by querying the appropriate repository. If the target is valid, the service invokes the correct database logic. Otherwise, it throws a `ResourceNotFoundException`.

The third method interacts directly with the Neo4j graph. It performs the following operations.

- Matches the follower node in the graph.
- Matches the target node, which can be either a user or a winery depending on the `isWinery` flag.
- Creates a `FOLLOWS` relationship between the two nodes using the `MERGE` clause to avoid duplicates.

This structure ensures that users can establish social connections within the platform in a consistent and secure manner, enabling the application to maintain accurate and meaningful relationship data in the graph database.

The following figures show some of the code snippets involved in the process.

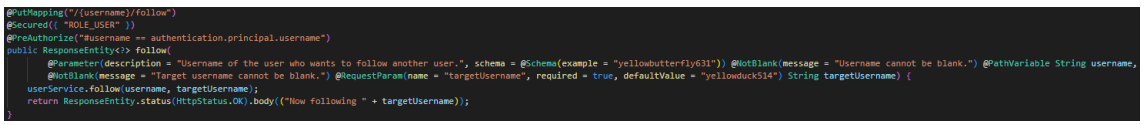


Figure 4.69: `@RestController` method handling the follow request.

```java
public void follow(String followerUsername, String targetUsername) throws ResourceNotFoundException {
    if (userRepository.existsByLogin_Username(targetUsername)) {
        userNeo4jRepository.follow(followerUsername, targetUsername, isWinery:false);
    } else if (wineryRepository.existsByLogin_Username(targetUsername)) {
        userNeo4jRepository.follow(followerUsername, targetUsername, isWinery:true);
    } else {
        throw new ResourceNotFoundException("No user or winery found with username: " + targetUsername);
    }
}
```

Figure 4.70: Service method to follow a user.

```java
public void follow(String followerUsername, String targetUsername, boolean isWinery) {
    String query = isWinery
        ? """
            MATCH (f:User {username: $follower})
            MATCH (w:Winery {username: $target})
            MERGE (f)-[:FOLLOWS]->(w)
          """
        : """
            MATCH (f:User {username: $follower})
            MATCH (t:User {username: $target})
            MERGE (f)-[:FOLLOWS]->(t)
          """;

    neo4jClient.query(query)
        .bind(followerUsername).to(name:"follower")
        .bind(targetUsername).to(name:"target")
        .run();
}
```

Figure 4.71: `follow()` method in `UserNeo4jRepository`

Unfollow a User

To unfollow another user or winery, the application must call the following methods.

- `public ResponseEntity<?> unfollow(`
  `@PathVariable String username,`
  `@RequestParam String targetUsername)`
- `public void unfollow(`
  `String followerUsername,`
  `String targetUsername)`
- `public void unfollow(`
  `String followerUsername,`
  `String targetUsername)`

The first method is a `@RestController` endpoint that handles a PUT request to the path `/api/users/{username}/unfollow`. It is secured using `@Secured` and

`@PreAuthorize` annotations to ensure that only the authenticated user can perform the operation, enforcing proper access control.

The controller delegates the request to the service layer, which performs validation by checking whether the target user or winery exists in the respective repositories. If the target is not found, a `ResourceNotFoundException` is thrown. Otherwise, the service invokes the Neo4j layer to update the graph.

The third method interacts directly with the Neo4j database. It performs the following operations.

- Matches the user node corresponding to the follower.
- Matches the target node (either user or winery) using the given username.
- Deletes the `FOLLOWS` relationship between the two nodes, effectively removing the connection from the graph.

This structure ensures that unfollowing is handled securely and efficiently at all layers of the application. It maintains the consistency of the social graph, allowing users to manage their connections while keeping the relationship data up to date.

```
@PutMapping("/{username}/unfollow")
@Secured({ "ROLE_USER" })
@PreAuthorize("#username == authentication.principal.username")
public ResponseEntity<?> unfollow(
        @Parameter(description = "Username of the user who wants to unfollow another user.", schema = @Schema(example = "yellowbutterfly631")) @NotBlank(message = "Username cannot be blank.") @PathVariable String username,
        @NotBlank(message = "Target username cannot be blank.") @RequestParam(name = "targetUsername", required = true, defaultValue = "yellowduck514") String targetUsername) {
    userService.unfollow(username, targetUsername);
    return ResponseEntity.status(HttpStatus.OK).body(("Unfollowed " + targetUsername));
}
```

Figure 4.72: `@RestController` method handling the unfollow request.

```
public void unfollow(String followerUsername, String targetUsername) throws ResourceNotFoundException {
    if (!userRepository.existsByLogin_Username(targetUsername) &&
        !wineryRepository.existsByLogin_Username(targetUsername)) {
        throw new ResourceNotFoundException("No user or winery found with username: " + targetUsername);
    }

    userNeo4jRepository.unfollow(followerUsername, targetUsername);
}
```

Figure 4.73: Service method to unfollow a user.

```
public void unfollow(String followerUsername, String targetUsername) {
    neo4jClient.query("""
        MATCH (f:User {username: $follower})-[rel:FOLLOWS]->(t)
        WHERE t.username = $target
        DELETE rel
    """)
    .bind(followerUsername).to(name:"follower")
    .bind(targetUsername).to(name:"target")
    .run();
}
```

Figure 4.74: `unfollow()` method in `UserNeo4jRepository`

4.6 Admin Operations

Admin operations on the *reviews* collection

An admin (`ROLE_ADMIN`) has access to following operations on *reviews* collection:

- Retrieve all reviews in the system, paginated (`GET /api/reviews`).
- Retrieve a personalized review feed for any user (`GET /api/reviews/feed`).
- Delete all reviews in the system (`DELETE /api/reviews`).
- Delete a specific review by its ID (`DELETE /api/reviews/{id}`).
- Delete all reviews written by a specific user (`DELETE /api/reviews/user/{username}`).
- Delete all reviews associated with a specific wine (`DELETE /api/reviews/wines/{wineId}`).
- Delete all reviews associated with a specific vintage of a wine (`DELETE /api/reviews/wines/{wineId}/vintages{vintageYear}`).

Admin operations on the *wines* collection

An admin (`ROLE_ADMIN`) has access to following operations on *wines* collection:

- Retrieve all wines in the system, paginated (`GET /api/wines`).
- Delete all wines in the system (`DELETE /api/wines`).
- Delete a specific wine by its ID (`DELETE /api/wines/{id}`).

Admin operations on the *users* collection

An admin (`ROLE_ADMIN`) has access to following operations on *users* collection:

- Retrieve all users in the system, paginated (`GET /api/users`).
- Search users by first name, last name, or both (`GET /api/users/search`).
- Retrieve a specific user by their username (`GET /api/users/{username}`).
- Retrieve suggested follows for a specific user (`GET /api/users/suggestions?username={username}`).
- Update any user's information (`PUT /api/users/{username}`).
- Update any user's username (`PUT /api/users/{username}/username/update?newUsername={newUsername}`).
- Delete all users in the system (`DELETE /api/users`).
- Delete a specific user by their username (`DELETE /api/users/{username}`).

Admin operations on the *wineries* collection

An admin (`ROLE_ADMIN`) has access to following operations on *wineries* collection:

- Retrieve all wineries in the system, paginated (`GET /api/wineries`).
- Search wineries by name, paginated (`GET /api/wineries/search?name={name}`).
- Retrieve a specific winery by username (`GET /api/wineries/{username}`).
- Update a winery's data (`PUT /api/wineries/{username}`).
- Update a winery's username (`PUT /api/wineries/{username}/username/update`).
- Add an image to a winery profile (`PUT /api/wineries/{username}/addImage`).
- Remove an image from a winery profile (`PUT /api/wineries/{username}/removeImage`).
- Delete all wineries in the system (`DELETE /api/wineries`).
- Delete a specific winery by username (`DELETE /api/wineries/{username}`).

Admin operations on the *countries* collection

An admin (`ROLE_ADMIN`) has access to following operations on *countries* collection:

- Create a new country (`POST /api/countries`).
- Retrieve all countries in the system, paginated (`GET /api/countries`).
- Retrieve a specific country by name (`GET /api/countries/{name}`).
- Update a country's data (`PUT /api/countries/{name}`).
- Update a country's name (`PUT /api/countries/{name}/name/update`).
- Delete all countries in the system (`DELETE /api/countries`).
- Delete a specific country by name (`DELETE /api/countries/{name}`).

Admin operations on the *regions* collection

An admin (`ROLE_ADMIN`) has access to following operations on *regions* collection:

- Create a new region associated with a country (`POST /api/regions`).
- Retrieve all regions in the system, paginated (`GET /api/regions`).
- Retrieve a specific region by name (`GET /api/regions/{name}`).
- Retrieve all regions in a specific country, paginated (`GET /api/regions/countries/{country}`).
- Update a region's associated country (`PUT /api/regions`).
- Delete all regions in the system (`DELETE /api/regions`).
- Delete a specific region by name (`DELETE /api/regions/{name}`).
- Delete all regions in a specific country (`DELETE /api/regions/countries/{country}`).

Admin operations on the *styles* collection

An admin (`ROLE_ADMIN`) has access to following operations on *styles* collection:

- Create a new style (`POST /api/styles`).
- Retrieve all styles in the system, paginated (`GET /api/styles`).
- Retrieve a specific style by name (`GET /api/styles/{name}`).
- Update an existing style (`PUT /api/styles`).
- Delete all styles in the system (`DELETE /api/styles`).
- Delete a specific style by name (`DELETE /api/styles/{name}`).

Admin operations on analytics collections

An admin (`ROLE_ADMIN`) has access to following operations on analytics collections:

- Delete all documents in *top-vintages-our-qop-type* collection (`DELETE /api/analytics/top-vintages-our-qop-type`).
- Delete a specific document by type in *top-vintages-our-qop-type* collection (`DELETE /api/analytics/top-vintages-our-qop-type/{type}`).
- Delete all documents in *top-vintages-qop-type* collection (`DELETE /api/analytics/top-vintages-qop-type`).

- Delete a specific document by type in *top-vintages-qop-type* collection (`DELETE /api/analytics/top-vintages-qop-type/{type}`).
- Delete all documents in *top-vintages-ratings-type* collection (`DELETE /api/analytics/top-vintages-ratings-type`).
- Delete a specific document by type in *top-vintages-ratings-type* collection (`DELETE /api/analytics/top-vintages-ratings-type/{type}`).
- Delete all documents in *top-wines-ratings-type* collection (`DELETE /api/analytics/top-wines-ratings-type`).
- Delete a specific document by type in *top-wines-ratings-type* collection (`DELETE /api/analytics/top-wines-ratings-type/{type}`).
- Delete all documents in *top-wineries-ratings* collection (`DELETE /api/analytics/top-wineries-ratings`).
- Delete a specific document by winery username in *top-wineries-ratings* collection (`DELETE /api/analytics/top-wineries-ratings/{winery_username}`).

Admin operations on the *admins* collection

An admin (`ROLE_ADMIN`) has access to following operations on *admins* collection:

- Create a new admin (`POST /api/admins`).
- Retrieve all admins, paginated (`GET /api/admins`).
- Retrieve a specific admin by username (`GET /api/admins/{username}`).
- Update the admin's own data (`PUT /api/admins/{username}`).
- Update the admin's own username (`PUT /api/admins/{username}/username/update`).
- Update the admin's own password (`PUT /api/admins/{username}/password/update`).
- Delete the admin's own account (`DELETE /api/admins/{username}`).

Note: In order to have at least one admin account in the system, the previous `DELETE` endpoint will return an error if the authenticated admin is the only one in the *admins* collection.

Chapter 5

Database Choices

5.1 Introduction to VM Organization

Before exploring the specifics of this chapter, it is helpful to present an info-graphic that depicts the overall structure of the *Virtual Machines* (*VMs*) available to us, accompanied by explanations of the processes operating on each one.

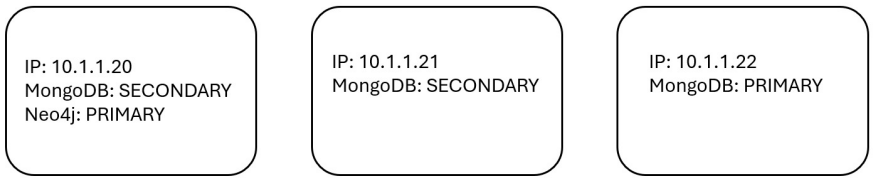


Figure 5.1: An overview of our VMs and their roles in the application.

5.2 Replication

5.2.1 MongoDB Replication

To meet the *non-functional requirement* of **high availability**, we set up a *replica set* using the VMs available to us.

The organization of the replica set can be summarized as follows.

- A **primary** node with the IP address 10.1.1.22, responsible for receiving all write requests (and possibly read requests).
- Two **secondary** nodes with IP addresses 10.1.1.21 and 10.1.1.20, storing copies of the primary and receiving read requests.

We have used the next priorities to set up the replica set:

- 10.1.1.22: priority=5 (primary in normal conditions).
- 10.1.1.21: priority=2 (likely to become primary if needed).
- 10.1.1.20: priority=1 (last choice for primary).

We have set the JAVA application to contact the cluster with next settings:

```
mongodb://10.1.1.22:27018,10.1.1.21:27018,10.1.1.20:27018/
  ?replicaSet=wineadvisor_repl_set&retryWrites=true&w=1
                &readPreference=nearest
```

5.2.2 Considerations on CAP Theorem

Looking at the requirements, it turns out that a **strict consistency** between replicas is **not necessary** at all: our application stays in the AP side of the CAP triangle. This allows us to set the following parameters:

- `w=1` (a write operation is considered successful when it is completed just on the primary node, in order to have a server response time as short as possible);
- `readPreference=nearest` (a read operation is directed from the client to the node with the lowest latency).

5.2.2.1 AP side

As stated before, in the architectural design of our application, we have deliberately chosen to prioritize **Availability** and **Partition Tolerance** (**AP**). This decision is critical for an application like ours, which heavily relies on social networking features.

By focusing on AP, we ensure that our system remains operational and accessible even in the event of network failures or data partitioning issues. This approach is vital to maintain uninterrupted user engagement with the social aspects of the application. We do this by replicating data across multiple nodes, thus ensuring availability even if some nodes are unreachable. While this might lead to temporary inconsistencies, it's a trade-off we accept to guarantee constant availability, which is crucial for user interactions within our social platform.

5.2.2.2 Failures recovery

The configuration of the secondary nodes is aligned with our considerations: in the event of primary node's failure, the secondary node at `10.1.1.21`, with a priority of 2, is more likely to take over as the new primary node. This ensures continuous service delivery, even during primary node down times.

The `10.1.1.21` has been chosen over the `10.1.1.20` because the latter hosts the Neo4j graph (and process): in this way, we achieve better load balancing and reduce as much as possible the problem of the *single point of failure*.

5.3 Sharding

At present, the system does not require sharding, due to the small dataset size and good performance metrics during development and testing. However, since the application behaves basically as a social network (with users following others and generating content) it is reasonable to anticipate that, in large-scale production environments, sharding may become beneficial to ensure the following properties.

- **Scalability**: distribute data across multiple servers placed in different "regions" (with a concept of region that may change following the growth and the distribution of the user base) to handle growing user volumes.
- **Performance**: improve response times by reducing query bottlenecks.
- **Maintainability**: better distribute workload and isolate data domains for easier updates and monitoring.

5.3.1 Possible Sharding Strategies

In this context, the following strategies are under consideration for potential future scaling.

- **User-based sharding**
  Data might be partitioned based on user identity (e.g., by username): this approach could be effective because most operations are scoped to individual users, such as personal reviews, likes, and follows, but brings as downsides an additional overhead when dealing (in a single operation) with resources owned by users in different servers.
- **Geographic sharding**
  Data might be partitioned based on the country or region of the users who own it: this is well-suited if the application will grow with a user base distributed in distinct countries and can help reduce latency by locating data closer to the user.
- **Feature-based sharding**
  Data might be partitioned separating it in "functional domains" and distributing it in different shards (for example, storing reviews in one shard, users in another, and wineries in a third): this approach helps optimize domain-specific queries and isolate workloads but generates an overhead when dealing with operations that use more than one collection (in general, more than one "functional domain").
- **Hybrid sharding**
  Data might be partitioned using a combination of multiple strategies, such as applying user-based sharding within geographic partitions: while this adds

complexity, it allows for more balanced and flexible scaling based on real-world usage patterns.

5.3.2 Recommended Sharding Strategy

In order to choose the best among the previous strategies, we need to consider some of the main characteristics of our application.

- Each user should be involved in an operation with another user in a social network, with no restrictions linked with their alphabetic order or other characteristics linked with their names or identity.
- Many operations in our application involve more than one collection (some of them even more than two) and this cannot be changed due to the embedding of documents, which is an intrinsic property of a document database.
- The presence of the graph database for the social network features of the application leads to the copy of information between the two database solutions, with three collections that are completely or partially replicated on the graph. We should also take into account the possibility that this number should grow in the future by adding new social network features to the application.

Considering these three main reasons, we discard the first and the third solution (among the ones presented in the previous paragraph).

Considering then that our application already supports users (both basic users and wineries) from different regions and countries and avoiding the complexity of the fourth solution (the hybrid one), which is unlikely to be strictly necessary in the initial growing phase of our application, the recommended sharding solution is the **geographic sharding** in a first growing phase of the application, partitioning data based on the country of the user who own it. Possible sub-sharding could be done based even on the region of the users.

5.4 Indexes

In this section we discuss about the indexes we have created in MongoDB in the replica set `wineadvisor_repl_set`, reachable on the virtual machines with addresses `10.1.1.22`, `10.1.1.21` and `10.1.1.20` at the port `27018`.

We can identify two different groups of indexes, among those we provide on the virtual cluster: **functional** indexes and **performance** indexes.

5.4.1 Neo4j indexes considerations

Before going on, we must specify that **we do not need** mandatory indexes for our Neo4j graph to be fully functional.

Moreover, graph queries are already really fast compared to equal queries developed on the document database: even the heavier graph query we have (the retrieval of the user's feed) expires in a time which is not perceptible by the user, so the need for performance indexes on graph decays. They might become necessary just in case of a very large user base (millions), but their implementation was not requested for this project.

Any further consideration for performance indexes **on graph** has not been done.

All the following considerations regard indexes on MongoDB.

5.4.2 Functional indexes

The **functional indexes** are mandatory for taking the application into a fully functional state. Most notably, we need them for executing a `$merge` as last stage of some aggregation pipelines: this stage automatically inserts the results of the pipeline into the respective collection. In this way, we avoid returning the entire result set of the pipeline to the JAVA application and then processing it a row at a time, which would be much more heavier and slow than the solution we actually implemented.

These indexes are mandatory because the `$merge` stage fails without them: they index the unique identifier of the documents in the collection we want to merge the result set in (e.g., if the result set is composed of a document for each country, the index should be created on the field `name` of the collection *country*, which is the unique identifier of the country documents across all the collections of our database).

Due to the fact that we cannot distribute our application without implementing these indexes, a review of the performance improvement they provide cannot be done at all.

In the following bullet list, all mandatory indexes are reported with structure:

<name_of_the_collection>:
`index(list(<indexed_field_path> <INDEX_DIRECTION>): <index_name>)`,
used in `<javaFunctionExecutingPipelineWhichUsesIndex>`.

- *top_vintages_by_our_qop_per_type*: `index(type ASC: on_type_UNIQUE)`, used in `updateTopVintagesByOurQoPPerType`
- *top_vintages_by_qop_per_type*: `index(type ASC: on_type_UNIQUE)`, used in `updateTopVintagesByOurQoPPerType`
- *top_vintages_by_ratings_per_type*: `index(type ASC: on_type_UNIQUE)`, used in `updateTopVintagesByRatingsPerType`
- *top_wines_by_ratings_per_type*: `index(type ASC: on_type_UNIQUE)`, used in `updateTopWinesByRatingsPerType`
- *top_wineries_by_wines_ratings*: `index(winery_username ASC: on_winery_username_UNIQUE)`, used in `updateTopWineriesByWineRatings`

- *regions*: `index(name ASC: on_name_UNIQUE)`, used in `updateTop10VintagesPerRegion`
- *countries*: `index(name ASC: on_name_UNIQUE)`, used in `updateTop100VintagesPerCountry`

5.4.3 Performance indexes

In opposition to mandatory indexes, **performance indexes** have been deployed in the replica set just to make operations faster.

In the following paragraphs, we report an analysis of the performance of the database **with and without** performance indexes. Some of the indexes are very similar one another because they are used by operations (queries or pipelines) which are themselves very similar: these leads to a performance improvement which is practically the same. For these indexes, just one for each "group" is reported below, with, in addition, the indication for the indexes that are similar to it.

5.4.3.1 reviews: `index(wine_id.id ASC)`

We need this index for the aggregation pipeline that updates the statistics on wines (fields `ratings_average` and `ratings_count`), in JAVA function `updateStatisticsPerWine`.

This function is executed once a week, but, as shown below, it is quite intensive, due to the presence of a lookup on collection *wines*. This heaviness is obviously destined to increase with the growth of the user base (along with the database). For these reasons, a performance improvement should be provided.

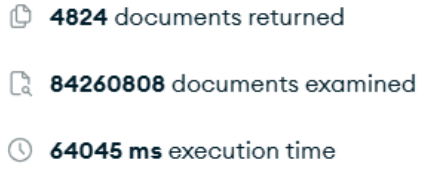
The index takes down to 1 the number of collection scans necessary and reduces the execution time massively (more than 200 times).


```
"totalDocsExamined": 84260808,
"totalKeysExamined": 0,
"collectionScans": 4825,
"collectionSeeks": 0,
"indexScans": 0,
"indexSeeks": 0,
"indexesUsed": [],
```

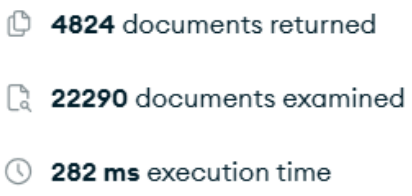

(a) Collection scans without index.


```
"totalDocsExamined": 22290,
"totalKeysExamined": 17466,
"collectionScans": 1,
"collectionSeeks": 17466,
"indexScans": 0,
"indexSeeks": 4824,
"indexesUsed": [
  "index_for_wine_statistics_update"
],
```


(b) Collection scans with index.



(a) Time used without index.



(b) Time used with index.

5.4.3.2 reviews: `index(user_id.username ASC, rating DESC, created_at DESC)`

We need this index for the aggregation pipelines that update the wine tips for each user (fields `wine_tips` e `new_wine_tips`), in JAVA functions `updateWineTipsPerUser` and `updateNewWineTipsPerUser`.

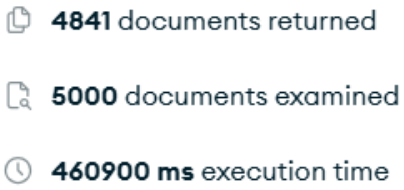
We take into account the first function: it is executed once a week, but, as shown below, it is quite intensive, due to the presence of three lookups: the first on collection *reviews*, the second on collection *wines* and the third again on collection *wines*. This heaviness is obviously destined to increase with the growth of the user base (along with the database). For these reasons, a performance improvement should be provided. The current index is necessary to optimize the performances of the first lookup.

Below, we show, for simplicity, the stats obtained from using/removing the first and the third index together. The second index cannot be removed because it is the auto-generated `_id` index on collection *wines*.

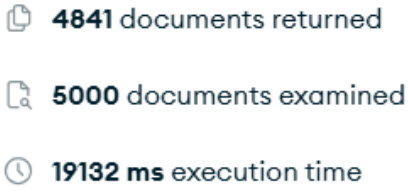
The indexes take down to 0 the number of collection scans necessary and reduces execution time massively (more than 95 times).

(a) Collection scans without index.

(b) Collection scans with index.



(a) Time used without index.



(b) Time used with index.

Function `updateNewWineTipsPerUser`

This function is really similar to function `updateWineTipsPerUser`, so much that in both of them we use the same first and second index. The third index is different, but in both function the performance improvement with their own index on the third lookup is exactly the same.

Screenshot of improvements are not reported for this pipeline they are exactly the same shown above.

5.4.3.3 wines: `index(style.name ASC, statistics.ratings_average DESC, statistics.ratings_count DESC)`

We need this index for the third lookup of the aggregation pipeline of the JAVA function `updateWineTipsPerUser` (only in this one). No additional considerations following those at the previous paragraph need to be done.

Below we show the stats of the collection scans of this lookup, which are taken down to 500. In this case, the improvement is clear but not perfect: MongoDB cannot reach less than this number of scans due to the presence of some sparse `null` values in the lookup fields at this stage of the pipeline. The stats regarding the execution time are provided in the previous sub-paragraph.


```
{
  "$lookup": {…},
  "totalDocsExamined": 57434544,
  "totalKeysExamined": 0,
  "collectionScans": 11906,
  "indexesUsed": [],
  "nReturned": 294178,
  "executionTimeMillisEstimate": 459125
},
```


(a) Collection scans without index.


```
{
  "$lookup": {…},
  "totalDocsExamined": 2693703,
  "totalKeysExamined": 281703,
  "collectionScans": 500,
  "indexesUsed": ["index_for_wine_tips"],
  "nReturned": 294178,
  "executionTimeMillisEstimate": 18072
},
```


(b) Collection scans with index.

5.4.3.4 wines: `index(style.name ASC, created_at DESC)`

We need this index for the third lookup of the aggregation pipeline of the JAVA function `updateNewWineTipsPerUser` (only in this one). As stated before, the index brings to this pipeline the exact same improving of the index at the previous sub-paragraph.

5.4.3.5 users: `index(login.username ASC UNIQUE)`

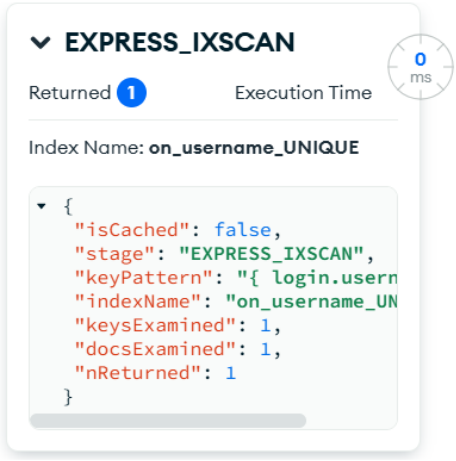
We need this index for the query that search a user with a given username. It needs to be unique because we use this field as substitute of the auto-generated `_id` field.

The heaviness of the query is not great in a single execution, but, looking at the volumes table we reported in this documentation, we should assume that it will be one of the most frequently executed: it is the query executed whenever a user opens a profile (both his or another profile). As a consequence, the total load that it would bring to the system is not negligible.

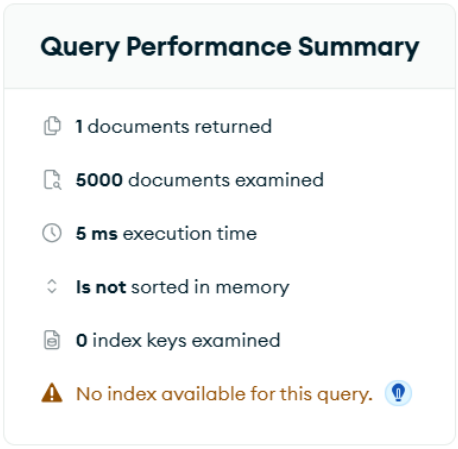
We tried both an `ASC` and `TEXT` index and they turned out to bring the same improvement to the query. No differences have been noticed also between an `ASC` and a `DESC` index. For simplicity, we choose to use an `ASC` index.

As shown below, the improvement is great, scanning just the necessary document and taking down to 0ms the execution time.

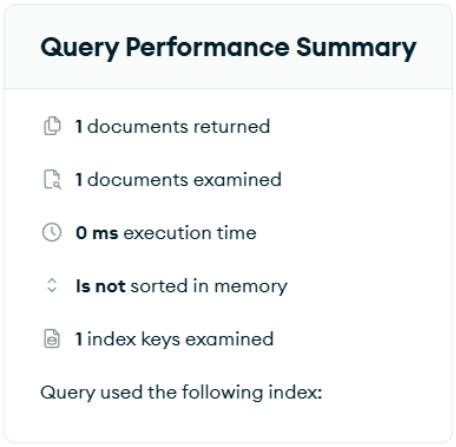
(a) Collection scans without index.



(b) Collection scans with index.



(a) Time used without index.



(b) Time used with index.

5.4.3.6 wineries: `index(login.username ASC UNIQUE)`

We need this index for the exact same reasons explained at the previous paragraph: the query which uses the current index is executed whenever a user opens a winery profile.

The considerations and the performance improvement we gave in previous sub-paragraph are exactly the same also for the current index.

5.4.3.7 styles: `index(name ASC UNIQUE)`

We need this index for the exact same reasons explained at the previous paragraph: the query which uses the current index is executed whenever a user opens the info page of a wine style.

Although the key of the field is different, considerations and performance improvement we reported in previous sub-paragraph are exactly the same also for the

current index.

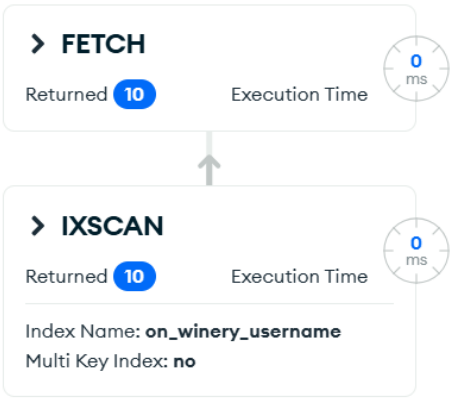
5.4.3.8 wines: `index(winery.username ASC)`

We need this index for the query that retrieves all the wines by a given winery.

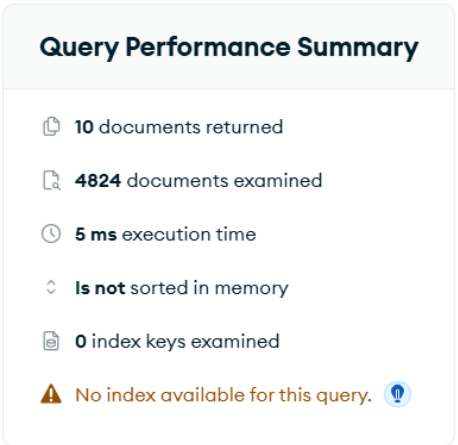
The heaviness of the query is not great in a single execution, but, looking at the volumes table we reported in this documentation, we should assume that it will be very frequently executed: it is the query executed whenever a user clicks on the button "Show all wines", hypothetically placed in the profile of a winery. As a consequence, the total load that it would bring to the system is not negligible.

As shown below, the improvement is great, scanning just the necessary documents and taking down to 0ms the execution time.

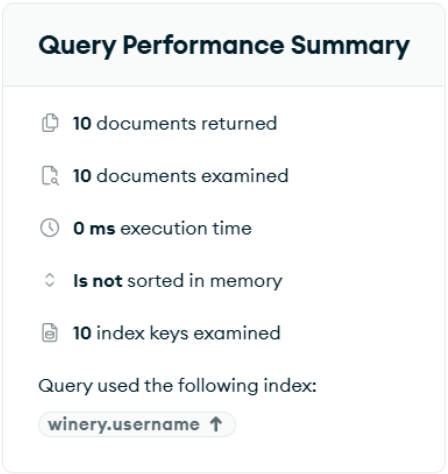
(a) Collection scans without index.



(b) Collection scans with index.



(a) Time used without index.



(b) Time used with index.

5.5 Discarded indexes

During the implementation of the project we tried to add also other indexes which turned out to be useless or even worsen. Examples are the indexes for the **wine search** function and the **user search** function.

5.5.1 Wine search

This function is the most complicated read function of the application: it allows to search a wine based on many fields (e.g. wine name, winery name, region, statistics, price...). The presence of many fields is not a problem for an index. The main issue with indexes for this query comes along with the way we manage the first two search fields: *wine name* and *winery name*.

These two fields are textual fields, then, to get a match, we had to choose among three possibilities.

- Searching for **perfect matches**: although this way was the most effective in terms of speed leveraging an index (`ASC` or `DESC`), it deletes any chance of searching sub-strings, which is very common need when searching wines or wineries names (not always you remember the entire name of wine or winery).

- Using `$text` **operator** and creating a `TEXT` **index**: although this solution is not effective as the first one in terms of speed but anyway quite effective, it matches only strings that start with the given sub-string (e.g. sub-string "Pettiros" will not match with string "Il Pettirosso"), it tokenizes the string just with spaces (e.g. sub-string "Vino" matches with string "Vino Nobile" but sub-string "Vino Nob" does not), it does a case-sensitive research (e.g. sub-string "grandi annate sangiovese" will not match string "Grandi Annate Sangiovese") and it treats in special ways recurrent words such as articles or prepositions (e.g. we can have unexpected behaviors with a string such as "Il Pettirosso").

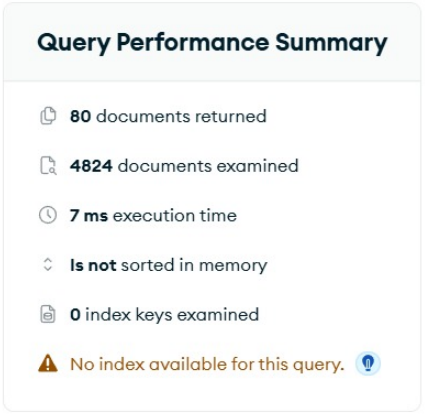
- Using `$regex` **operator**: this operator allows "the perfect search" on a text field, solving every problem we highlighted in previous points, but takes with it the consequence that it does not use any type of index (nor `TEXT`, neither `ASC` or `DESC`) in many situations (we figured out that the index is used only when you provide the entire wine or winery name to search).

After a careful examination of the functional and non-functional requirements, we opted for the **third solution**, in order to give the user a tool as flexible as possible to search for wines. We considered that forcing the user to search a wine with all the restrictions stated at points one and two would influence too much the user experience of the application.

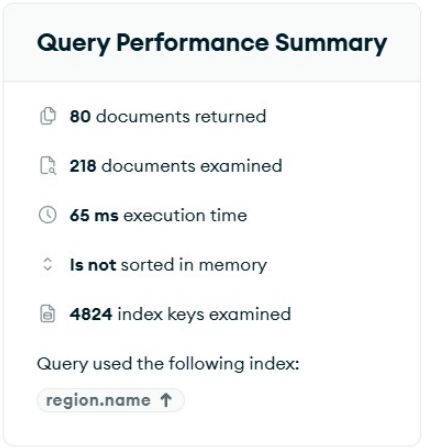
Moreover, in many commercial applications that need a search tool as ours, the implementation of a cheap "perfect search" is usually done with external tools that were out of our scope.

We also tried to implement an index considering the other fields of the query (both single or multiple indexes) but no other solution showed improvements in execution time. The latter even increased in basically every index we tried. The reason of this increase in time might be that MongoDB, in any case, examines many documents despite the presence (and the examination) of many index keys. This behavior, with our database dimension, takes up the execution time more than a single collection scan where MongoDB examines every document one time. The trend may change with consistent bigger amount of wines in the collection, but we can just assume it without experimentally prove it.

Below we report some tests we have done with different indexes, including with no index and with the complete one.

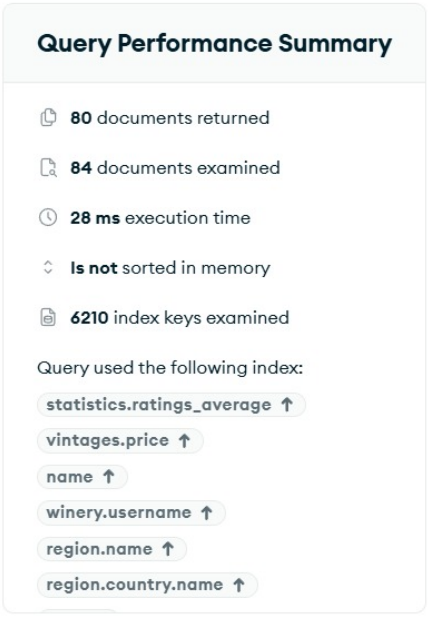


(a) Query stats without index.



(b) Query stats with partial index.

(a) Query stats with partial index.



(b) Query stats with complete index.

In the end, we chose last solution (the use of *regular expressions*): looking at the stats, a single search of a wine without indexes is not expensive or slow on his own at all. The operation would lead to a loss in the application performances just during really high traffic moments (peak time) with high number of users who request send request to the same endpoint at the same time. This can of course happen, but not with the current volumes of our database.

For these reasons, no index has been eventually provided to improve *wine search* performances.

5.5.2 User search

The user search function uses the first name and the last name fields to retrieve one or more users (field `name.first` and field `name.last`). No other field is taken into account. Both fields are text fields and all the considerations done in previous paragraph (*wine search*) about the fields `name` and `winery.username` of collection `wines` can be taken also to current fields and to the *user search*.

For these reason, no index has been eventually provided to improve *user search* performances.

Chapter 6

Future Implementations

6.1 Mobile Frontend Integration

A mobile application for Android and iOS could be developed to make WineAdvisor easily accessible to users. It would enable wine lovers to rate, review, and explore wines on the go, improving the user experience and engagement.

6.2 Gamification Elements

To enhance user engagement and make the platform more dynamic and enjoyable, future development could include the integration of gamification features. These may involve awarding badges, points, or ranks based on user activity, such as writing reviews, rating wines, attending events, or discovering rare and exclusive labels. For example, users could earn badges like "Wine Explorer" for reviewing wines from multiple regions, "Tasting Enthusiast" for attending verified events, or "Vintage Hunter" for reviewing older vintages. A points-based system could be introduced to unlock levels or rewards, encouraging continuous interaction with the platform. Gamification not only motivates users to stay active but also adds a layer of recognition and achievement to their experience. Furthermore, it fosters a sense of community and friendly competition among users, especially if accompanied by leaderboards or seasonal challenges. This approach can significantly boost user retention and make WineAdvisor a more social and interactive environment.

6.3 Event and Tastings Management

A future development could focus on enabling wineries to organize, publish, and manage wine-related events such as tastings, vineyard tours, or special product launches. This feature would allow wineries to promote their offerings directly within the platform and build a stronger connection with their audience. The backend could support RSVP management, including capacity limits, waiting lists, and confirmation messages. In addition, geo-localization features could help users discover events

happening nearby, enhancing the sense of community among wine lovers. After attending an event, users could leave reviews, share photos, and rate the experience, enriching the social dimension of the application. This implementation would position WineAdvisor not only as a wine review platform but also as a dynamic event hub, helping wineries engage with customers in a more personal and interactive way.

6.4 AI-Based Review Summarization

Incorporate Natural Language Processing (NLP) techniques to summarize large sets of reviews into concise overviews. This feature would help users quickly understand the general opinion about a wine.

6.5 Wine Expert Profile

A dedicated profile type for certified wine experts or sommeliers could be introduced. These users would have access to specialized features, such as expert-level reviews, verified badges, and the ability to curate wine lists or offer tasting notes. Their contributions would be highlighted throughout the platform, increasing user trust and helping guide less experienced users. Additionally, wineries could request expert reviews to promote their products with credibility.